CITY OF BOULDER, COLORADO
Finance Department / Purchasing Division
1136 Alpine
P.O. Box 791
Boulder, Colorado 80306
Telephone: 303-441-1857
Email: kasparekk@bouldercolorado.gov

REQUEST FOR BIDS

RFB NO. 44-2026

Streetlight Materials & Installation

ISSUE DATE: October 21, 2026

DUE DATE: 4:00 PM, Friday, November 20, 2026

Bids to be electronically saved to the Bidnet website PRIOR TO Proposal Due Date and Time above.

CONTACT:
Cory Benefiel
Traffic Signal Maintenance Supervisor
303-413-7103
BenefielC@bouldercolorado.gov

CITY OF BOULDER
NOTICE OF REQUEST FOR BIDS
RFB NO. 44-2026
STREETLIGHT MATERIALS & INSTALLATION

The City of Boulder ("City"), by and through its Transportation & Mobility department, is seeking bids for streetlight materials and installation. The award from this RFB will be for the initial year, plus four additional years at discretion of both parties. There will be a non-mandatory prebid meeting held on November 2, 2026, at 10:30 am. Details can be found in the RFB document.

Bidnet – Required. Solicitation documents will be issued electronically only. A copy of this Request for Bids (RFB) must be obtained only from the Rocky Mountain E-Purchasing System (Bidnet) at: www.bidnetdirect.com/colorado. Any modifications to this RFB or addenda pertaining to this RFB will be published to the above-mentioned website, and all bidders are responsible to periodically check the website for relevant updates prior to the submittal of a proposal.

Response to RFB. The City will only accept electronic submissions saved to the Bidnet website in response to this RFB. Electronic submissions are required to be considered for an award. To submit your RFB, please visit . Please authorize receiving notices through Bidnet relating to this procurement.

Insurance. This solicitation includes desired contractual insurance provisions. The City is open to alternative insurance policies and considerations. Proposers should not avoid proposing merely due to insufficient insurance. The City is open to alternative policy limits or language, subject to ultimate approval of the City's Risk Manager.

Deadline. The submission deadline is November 20**, 2026, at 4:00 p.m**. Mountain Time. This deadline is mandatory. You must save your submission to the Bidnet website before the deadline. Failure to do so may result in an inability to submit a response that is automatically implemented in the software. Except as set forth herein, no exceptions will be allowed. The City specifically reserves the right to reject late submissions.

Technical Submission Issues. In extraordinary circumstances, the City may permit late submissions if the City, in its sole discretion, determines the bidder has shown good cause due to technical issues. If you experience problems submitting your electronic response, please contact the Bidnet Direct technical support team (800-835-4603) ***prior*** to the submission deadline and document the technical errors and your efforts to resolve the same before the submission deadline. **Bidders are encouraged to document that their Bidnet submission was saved in case technical issues arise or are later discovered**. Your documentation may include e-mailing your submission to woodss@bouldercolrado.gov provided such submission is emailed by the deadline or immediately upon discovery of the issue. An emailed submission does not substitute the requirement for an electronic submission using Bidnet. By this paragraph, the City does not guarantee it will accept submissions that experience technical issues.

Costs. Proposals shall be prepared at the bidder's sole expense.

Public Record. Bid submissions are public records and subject disclosure under the Colorado Open Records Act (CORA).

Confidential Data. All submitted materials are presumed to be public records. However, confidential data, if unambiguously identified as such, will be held confidential upon request if: (1) it is clearly and unambiguously identified, (2) the request is made as part of the bid, and (3) if the City Attorney's Office determines the data meets the requirements of the CORA. The City's Access to Public Records Policy, which includes policies about release of solicitation records, can be found on the City's Central Records web page under the heading Additional Resources.

Preference. Preference is given to labor, materials, supplies, or provisions produced, manufactured, or grown in the state of Colorado, quality and price being equal to articles or services offered by competitors outside of Colorado.

Digital Accessibility. If the work or services described herein include information technology or digital content that must be made "accessible," the successful bidder must agree, if requested by the City, to undertake such reasonable efforts that may be needed of the successful bidder to adapt and convert work product to ensure it is accessible within the meaning of §§ 24-85-101, *et seq*., C.R.S., and its implementing regulations. For contracts which the provision of information technology comprises a primary or material component of the service (such as software and web services) the City anticipates requiring that the successful bidder agrees to ensure legal compliance, subject to limited exceptions.

Socially Responsible Procurement. The City is committed to socially responsible procurement and promoting social equity through its contracts. The City works to ensure open and fair procurements, competitive and fair pricing, environmentally sustainable solutions, and encourage members of the underserved business community to participate in our competitive solicitations.

Attention is called to the fact that **not less than the minimum salaries and wages** as set forth in the Contract Documents must be paid on this project, and that the Contractor must ensure that employees and applicants for employment are not discriminated against because of their race, color, religion, sex, or national origin (Davis Bacon Wages). If the bidder anticipates hiring employees to work on this job, the bidder should contact the local manpower office for qualified candidates.

Equal Opportunity. The City does not discriminate on the grounds of race, creed, ancestry, religion, religious expression, color, sexual orientation, gender identity, gender expression, genetic characteristics, marital status, age, sex, mental or physical disability, or national origin in consideration of an award. Disadvantaged business enterprises are afforded a full opportunity to submit proposals.

City's Reserved Rights. For avoidance of doubt, the City reserves the following rights:

- **Right to Supplement or Modify**. At any time prior to award, the City reserves the right to supplement or modify its solicitation with additional or different information, questions, contract language, or any other materials the City deems prudent to include in the supplementation or modification. Such supplementation or modification includes the right to modify the bid schedule at any time. Supplementation or modification will likely occur through Bidnet but may occur by email contact to bidders if the deadline has passed.

- **Right to Reject**. The City reserves the right to reject any bid that fails to meet the requirements described herein, the requirements or criteria in the City's Purchasing Ordinance, or in the City's Purchasing Policy Manual, as such ordinance or policy manual may be amended from time to time, or any other pertinent standard or criteria established by City. The City reserves the right to require new bids or reject any bid if the City believes it would not be in the best interest of the project or services set forth in this RFB to make an award to a given bidder. This right to reject includes the right to reject any bidder who appears on one or more lists designating the bidder as a debarred or suspended entity or individual for other public procurements.

- **Right to Waive Irregularities**. The City may waive, or decline to waive, any informality or irregularity contained in any bid.

- **Right to Revoke or Cancel Award**. At any time prior to the successful bidder and the City entering into a written and signed contract, the City reserves the right to revoke or cancel the award. If, for any reason, the City and a successful bidder are unable to come to terms of agreement, the City may, in its sole discretion, either award to the next highest scoring bidder or cancel this solicitation.

- **Right to Negotiate Changes in Scope or Services**. The City may negotiate changes to the services or work offered by the successful bidder as part of the final contracting process.

- **Right to Make Partial Award**. The City may award the contract by sections or in portions and reserves the right to adopt all or part of a bid.

Verbal Communications. Any verbal communications bidders may receive about this solicitation are non-binding upon the City. Vendors should rely only upon written statements issued by the authorized City representatives listed in the Instructions to Bidders and Bid Schedule section of this RFB or a member of the City's Purchasing Division.

Unauthorized Contacts. A bidder's unauthorized contacts with City staff about this solicitation may result in disqualification of the bidder.

Contract Terms. All awards are subject to the awarded bidder and the City successfully negotiating and entering into a written and signed contract in a form approved by the City Attorney's Office. The City's award to a successful bidder does not constitute consent by the City to (1) any revised terms or conditions contained in exceptions, or (2) any terms or conditions contained in the bid of the successful bidder.

In some instances, contract terms are resolved by appending this solicitation and the successful bid to the contract template attached hereto. This is common for the City's construction and public works contracts. However, most contracts necessitate development of a scope of work that further refines and clarifies the services. Bid language may merit modification to contract terms, even where a bidder has not made exceptions. While the City's position is that, generally, failure to make exceptions results in a waiver of the ability to later claim those exceptions, the City reserves the right to modify the contract as the City and the successful bidder deem appropriate.

In the event of any conflict between this RFB and the contract entered into by and between the City and the successful bidder, the terms and conditions of the contract shall control.

Bidder's Right to Withdraw. In the event the selected bidder and the City are unable to come to an agreement on final terms for the contract, and provided such inability is due to either (1) the bidder's exceptions or the City's required changes in response thereto, or (2) the City's required revisions to the version of the contract posted with this solicitation (as modified by an addendum), the selected bidder shall have the right to withdraw its bid without penalty or loss of any bid bond required by this solicitation.

Waiver. All bidders agree that damage or claim shall not be brought by any bidder because of rejections, and the bidders shall not seek any recourse of any kind against the City because of such rejections. The filing of any bid in response to this RFB shall constitute a binding agreement of the bidder to these conditions.

Federal, State, or Other Third-Party Approval. Bidders should be aware that some awards may require approval (or non-objection) by one or more third parties. Common examples of this include projects that use state or federal funding and for which, accordingly, state or federal agencies must approve the award (e.g., CDOT approval required for certain transportation projects). The City specifically reserves the right to revoke awards if a state, federal, or other entity providing financing or with oversight rights/responsibilities will not approve of or object to the award, or if the selected bidder is not otherwise fully authorized to perform the services under such state, federal, or third-party entity.

CONSENT TO PROTEST & APPEAL PROCESS AND REMEDIES

Protests, appeals and remedies under this solicitation are governed by Colorado Revised Statutes (C.R.S.) Sections 24-109-101.1 through 108, C.R.S. (protests and other pre-litigation resolution of controversies) and Sections 24-109-201 through 206, C.R.S., (appeals), Section 24-109-301, C.R.S. (interest accruing on costs), Sections 24-109-501 through 505, C.R.S. (remedies), as amended. Such sections apply in their entirety, except for the City's purposes, (1) all references to the "state" shall be to the "city", (2) the City's assigned purchasing agent renders the City's initial written decision, and (3) the City's Purchasing Manager renders the final agency determination upon any appeal therefrom, as well as any other functions and decisions of the "executive director" referenced in those sections.

As described in the statutes referenced, protests and appeals must be made in writing within ten (10) business days. Protests must be timely submitted as e-mail attachments to the purchasing agent for this solicitation at the following e-mail address:

Shay Woods
woodss@bouldercolorado.gov

Any subsequent appeals from the purchasing agents' initial determination must be timely submitted within ten (10) business days of the initial decision as e-mail attachments to the Purchasing Manager at the following e-mail address:

Michael Van Ausdal
Vanausdalm@bouldercolorado.gov

In addition, this solicitation is subject to limitations of remedies set forth in Sections 24-109-501 through 505, C.R.S., including, specifically, Section 24-109-505, C.R.S. Those remedies are limited to, in summary, the authority of the City to ratify, terminate, or cancel the solicitation, award, or contract, and, if ratification occurs, a successful aggrieved party's limitation of remedies to its reasonable costs incurred in connection with the solicitation, including bid preparation costs, plus interest on those costs.

By responding to this solicitation, bidders agree to the protest and appeal processes, requirements, and time limitations, as well as the limited remedies, set forth in the statutes above and as described herein.

Dated this October 21, 2026

City of Boulder, Colorado
A Municipal Corporation

INSTRUCTIONS FOR BIDDERS AND BID SCHEDULE

1. Bidders responding to this RFB must submit their bids as specified in this solicitation.
2. One electronic copy of the bid must be submitted online through the Rocky Mountain E-Purchasing System (www.bidnetdirect.com/colorado) and must be addressed to the Project Manager listed below. **Bidders are strongly encouraged to familiarize themselves with the Bidnet website in advance of the proposal deadline.**
3. Bids must be dated and signed by a duly authorized partner or corporate officer, with that person's name and title clearly identified. All bid terms, conditions, contents, fees, and charges shall be guaranteed by the bidder for a minimum of forty-five (45) days from the submission deadline. The City may, in its sole discretion, release any bid and return the bid bond prior to the end of such guarantee period.
4. Bidders who have questions concerning the scope of work or services or other requirements set forth in this RFB shall contact the following City representative:

   Project Manager
   Cory Benefiel
   Traffic Signal Maintenance Supervisor
   City of Boulder
   Transportation and Mobility
   303-413-7103
   BenefielC@bouldercolorado.gov

5. The above referenced individual is the only representative of the City with authority to provide any information, clarification, or interpretation regarding the plans, specifications, and any other contract documents or requirements. Contact with any other employee of the City of Boulder regarding this project is not authorized. Any information obtained from those other than an authorized City representative shall be considered invalid in the preparation of a proposal for this project.
6. Bidders who request clarification of the solicitation's requirements may submit written questions to the project manager at the email address listed above by 5:00 PM (Mountain Time) on November 6, 2026. All questions and answers will be posted as an addendum on Bidnet. Bidders should rely only on written statements issued by the project manager.
7. **Before submitting a bid, each bidder shall:**

   - Examine the contract documents thoroughly.
   - Become familiar with federal, state, and local laws, ordinances, rules, and regulations that may in any manner affect cost, progress, or performance of the work.
   - Study and carefully correlate the bidder's observations with the contract documents.
   - Notify the project manager of all conflicts, errors, omissions, ambiguities, or discrepancies in or among the contract documents.

8. All bids timely submitted shall become the property of the City and shall be retained in accordance with the City's records retention schedule. The City is a Colorado governmental entity and therefore, all information included in proposals and other written information submitted by the bidder to the City is subject to the provisions of the CORA. Bidders should expect that their proposal may be viewed by the general public and competitors once submitted to the City, subject to the City's Access to Public Records Policy. Merely marking information as "confidential," "proprietary," or otherwise stating a written intent to protect the information from disclosure is not necessarily sufficient to prevent disclosure under CORA.
9. **Bid Schedule (all times Mountain Time)**

RFB Issued	October 21, 2026
Non-Mandatory Prebid Meeting	November 2, 2026, @ 10:30am
Written Questions due	November 6, 2026, at 5:00 pm
Responses to Written Questions Posted	November 13, 2026
Bids Due	November 20, 2026, at 4:00 pm
Review of Bids/Successful Bidder Selected	November 30, 2026

**Pursuant to Section 2-8-4(a)(3) of the Boulder Revised Code 1981, the successful bidder is required to sign a contract with the City within 15 days of the successful bidder's acceptance of the award by the City. Failure to do so will result in the bidder's forfeiture of its bid bond.*

Interview – Optional. The City may elect to conduct interviews with one or more bidders. The City may do so to better understand the bidders, their teams, their approaches, products, and solutions, and any other information relevant to the work or project described in this solicitation. In some instances, the City may request that the interviews include a presentation by the interviewed bidders. In these instances, bidders may prepare PowerPoint or other written material that may assist the City in understanding and evaluating their bid. Bidders cannot plan on having an interview and, accordingly, are strongly encouraged to include all information necessary for the City's evaluation in their bid documents. For the avoidance of doubt, the City reserves the right to include information provided by bidders during interviews in the contract documents.

10. **Non Mandatory Prebid Meeting**

The City will hold a prebid meeting on November 2, 2026, at 10:30 am. This will be held virtually via Microsoft Teams. To receive an invitation to the meeting, please email Cory Benefiel at benefielc@bouldercolorado.gov.

11. **Required Documents for Bid Submission**.

To be considered, all bids shall include the following completed forms:

- **City of Boulder Bid Proposal**. Please include the separate excel spreadsheet with your submission.
- **City of Boulder Bid Bond** in the amount of 5% of the total bid (form attached).
- **City of Boulder Bid Form** (attached).
- **City of Boulder Acceptance of Terms and Conditions** (attached).
- **Cover or Introductory Letter**
    - Include the following elements of information in the letter as a minimum:
        - Company Project Team and Qualifications
        - Company Capabilities and Experience (Include description of staff and fleet availability)

In addition, the City requests submittal of the following documents at the time of bid submission, and the City will require such documents be completed and submitted in acceptable form to the City prior to contracting:

- **City of Boulder Non-Collusion Certificate** (attached).
- **City of Boulder Contractor Qualification Form** (attached).
- **City of Boulder Schedule of Manufacturers and Suppliers** (attached).
- **City of Boulder Schedule of Proposed Subcontractors** (attached).

- **Certificate of Good Standing from the Colorado Secretary of State**. If your company is not registered with the Colorado Secretary of State, include a statement addressing why it is not required to register or affirming that, prior to entering into a contract with the City, your company will register with the Colorado Secretary of State if the City awards your company the contract.

In addition to the forms attached to this RFB, all above-listed documents, except the Bid Proposal and Certificate of Good Standing, have been included as a separate PDF file in the Bidnet documents section labeled Required Bid Documents. You will also find the Microsoft Excel spreadsheet for pricing in the same documents section in Bidnet. The spreadsheet is labeled Streetlight Work Bid Pay Items General for 2027.
For avoidance of doubt, the City reserves the right to request that bidders correct minor irregularities or issues in the forms submitted, though bidders should not expect to have the opportunity to correct such irregularities or issues. With respect to the City's contracting terms and conditions, the City reserves the right to negotiate additional or different changes to contract language that the City deems necessary or prudent to account for bidder exceptions.

EVALUATION CRITERIA

The award will be provided to the lowest bid, submitted by a responsible and qualified bidder. Bids the City determines fail to respond to a material component of this Solicitation may be rejected. The lowest bid will include consideration of initial cost and, if applicable, life-cycle cost, including, without limitation, maintenance cost, over the normal lifetime of the product and energy-efficiency in consumption of non-renewal fuels (if applicable). Bidders are only responsible and qualified if the City is satisfied it meets the below responsibility criteria:

- The bidder's integrity and financial responsibility.
- The bidder's skill, relevant technical expertise, and ability to complete the contract promptly and satisfactorily.
    a. The bidder is required to be have a current City of Boulder Contractor License upon completion of contracting process.
    b. The bidder is recommended to be a Xcel Energy approved contractor
- Whether the bidder maintains a permanent place of business.
- Whether the bidder has adequate plant, equipment, and support services to perform the contract.
    a. The bidder shall maintain sufficient equipment, personnel, and resources to respond to emergency service requests and be on site and prepared to commence work within one (1) to one and a half (1½) hours of notification by the City.
- Whether the bidder has previously performed similar work satisfactorily.
- Whether the bidder is likely to be engaged in work that may impair the ability to finance the work covered by the bid or provide equipment for its proper execution.
- Whether the bidder proposes a reasonable approach to achieve the objectives sought.
- Whether there have been or are any claims that raise a substantial question about the bidder's ability to perform the contract.

The City applies the above criteria liberally, and may eliminate bidders based upon one or more of these criteria. Bidders must propose high-quality approaches, competitive and reasonable pricing, and provide adequate certainty surrounding the project or risk elimination from consideration. The City may conduct such investigations as the City deems necessary to assist in the evaluation of any bid and to establish the responsibility, qualifications, and financial ability of any bidder or their proposed subcontractors and other persons and organizations to do the work in accordance with the contract to City's satisfaction.
Further, the City may consider the qualifications and experience of subcontractors, suppliers, and other persons and

organizations proposed for those portions of the work as to which the identity of subcontractors, suppliers, and other persons and organizations must be submitted. The City also may consider the operating costs, maintenance requirements, performance data and guarantees of major items of materials and equipment proposed for incorporation in the work when such data is required to be submitted prior to the contract award.

ALL AWARDS ARE SUBJECT TO CONTRACT EXECUTION. If for any reason, the negotiations with the apparent lowest bidder who satisfies all applicable requirements are unsuccessful, the City may terminate those negotiations and begin negotiations with a different bidder, or cancel and repost the solicitation in order to contract with the bidder that best meets the needs of the City.

BID BOND

Note: Complete and return this form with your bid.

RFB Title: Streetlight Materials & Installation
RFB Number: 44-2026

Witness:

_____, as principal, hereinafter called the "bidding contractor," and _____, as "surety," a corporation properly organized under the laws of Colorado or authorized to do business in Colorado, hereby obligate themselves to the City of Boulder, as obligee, hereinafter called the "City," in the penal sum of _____ dollars ($_____), which penal sum shall secure the obligation of the bidding contractor to enter into contract with the City for the work associated with

_____

_____

_____

Project Number _____, if the bidding contractor is so selected by the City.

If the City accepts the bid of the bidding contractor, and if the bidding contractor properly executes a contract with the City to perform the work covered by the award, then this obligation shall be null and void.

In the alternative, if the bidding contractor refuses to enter into contract with the City for work described in the bid, but rather pays to the city the difference between the bidding contractor's stated bid amount and the actual amount bid by an alternate bidding contractor, who is accepted by the City, then this obligation shall be null and void.

EXECUTED on this _____ day of _____, 20____.

_____
(Contractor)

By: _____
(President)

_____
(Surety Company)

By: _____
(Attorney-in-Fact)

BID FORM

Note: Complete and return this form with your bid.

RFB Title: Streetlight Materials & Installation
RFB Number: 44-2026

The undersigned bidder does hereby declare and stipulate that the Bid Proposal is made in good faith, without collusion or connection with any other person or persons bidding for the same work, and that it is made in pursuance or and subject to all the terms and conditions of the bid documents all of which have been examined by the undersigned.

Bidders shall submit prices as provided in the Bid Proposal. Should an error occur in the extension of price, the unit price shall govern.

The City of Boulder ("City") reserves the right to make the award on the basis of the bid deemed most favorable to the City, to waive any informalities or to reject any or all bids.

The bidder shall complete the attached Contractor Qualification Form, with the bid proposal. Failure to provide this information may be sufficient cause for rejection of the bid.

The bidder shall complete the attached Proposed Subcontractors Form, with the bid proposal. Failure to provide this information may be sufficient cause for rejection of the bid.

The bidder shall complete the attached Schedule of Manufacturers and Suppliers Form, with the bid proposal. Failure to provide this information may be sufficient cause for rejection of the bid.

Addendum No.(s)_____ have been received.

Attached is an unconditional letter of credit, money order, certified check, or bid bond payable to the City of Boulder in an amount not less than five percent (5%) of the total amount of the bid.

If this bid is accepted, the bidder agrees to sign the contract agreement without qualifications and to furnish the performance bond, labor and material bond and the required evidence of insurance within ten (10) calendar days after notification of award.

All proposals shall be submitted on the bid proposal. The lowest responsible bid will be accepted; provided, however, that the City shall have the right to reject any or all bids and to waive any informalities or irregularities contained in said bid. The quantities are an estimate only and may be altered to meet actual City requirements.

Submit firm unit prices for the type of construction listed. Overhead shall not be priced and bid separately but shall be included in each bid item as required. All prices are for furnishing and installing the described item.

The bidder proposes to perform all work and provide all labor, materials, and equipment in full accordance with the bidding documents.

Confidential data, if identified as such, will be held confidential upon request, if the request is made as part of the bid and the City Attorney's Office determines the data meets the minimum requirements of the Colorado Public Records Act.

ATTEST:

_____ _____
Corporation

_____
President
Date_____
Address_____
City_____ State_____
Zip Code_____
Phone number_____

Email_____

_____
ATTEST: Company Name

_____ _____
Company Partner
Date_____
Address_____
City_____ State_____
Zip Code_____
Phone number_____

Email_____

ATTEST: _____
Company Name

_____

_____
Individual

Date_____
Address_____
City_____ State_____
Zip Code_____
Phone number_____

Email_____

Tax Identification Number_____

It is imperative that the Business and/or vendor name (surname for an individual) and TIN number

combination (Employer Identification Number for a Business or Social Security Number for an individual) match exactly.

ACCEPTANCE OF TERMS AND CONDITIONS

Note: Complete and return this form with your bid.

RFB Title: Streetlight Materials & Installation
RFB Number: 44-2026
The undersigned, as an authorized agent of the bidder, hereby certifies that the bidder:

1. Has received addendums;
2. Is familiar with all instructions, terms and conditions, and specifications stated in this RFB;
3. Is qualified to perform the work and services outlined in this RFB; and
4. Has reviewed the form of agreement attached to this RFB.

Each bidder must indicate exceptions that it takes to any terms and conditions listed in the form of agreement attached to this RFB, as well as the RFB itself. The City reserves the right to not consider exceptions that are not identified on this form. Proposals which take exception to the specifications, terms, or conditions of this RFB or offer substitutions shall explicitly state the exception(s), reasons(s) therefore, and language substitute(s) (if any) in this section of the proposal response. Failure to take exception(s) shall mean that the bidder accepts the conditions, terms, and specifications of the RFB, including the attached form of agreement.

Check the appropriate box below and complete this section as applicable:

☐ Bidder takes no exception to the specifications, terms, and conditions of this RFB.

☐ Bidder takes exception to the following terms and conditions in this RFB:

_____

_____

_____

_____

_____

_____

_____

Signed:

By: _____

Title: _____
Date: _____
For: _____

NON-COLLUSION CERTIFICATE

Note: Return this page with your bid.

RFB Title: Streetlight Materials & Installation
RFB Number: 44-2026

The undersigned, being duly sworn, deposes and says that the person, firm, association, partnership or corporation herein named, has not, either directly or indirectly, entered into any agreement, participated in any collusion, or otherwise taken any action in restraint of free competitive bidding in the preparation and submission of a proposal to the City of Boulder for consideration in the award of the project described in this solicitation.

_____

(Firm Name)

By: _____
(Authorized Signature)

Printed Name: _____

Title: _____

Email: _____

CONTRACTOR QUALIFICATION FORM

Note: Complete and return this form with your bid.

RFB Title: Streetlight Materials & Installation
RFB Number: 44-2026

I. General Information

Your Company:

Name: _____

Address: _____

How long have you been in business? _____

How long have you been in business under the present business name? _____ _____

What former name(s), if any, has your company operated under in the past five (5) years?

What is the largest contract you have been awarded in the past twelve (12) months?

_____

_____

II. Information: Previous Construction Contract

Provide the following information for three (3) contracts you have performed, or are currently performing, within the past five (5) years. At least two (2) contracts must be completed. If you have performed any contract with the City of Boulder, or County of Boulder, or state of Colorado, include information about at least one (1) such contract.

Contract 1

Owner _____
Name

_____ _____
Address Phone

Date Begun _____ Date Completed _____ Dollar Amount _____

Description of Work _____

Architect _____
Name

_____ _____
Address Phone

Surety for Performance Bond: _____

Address Phone

Surety for Labor and Material Bond: _____

Address Phone

Insurance Company: _ Name

Address Phone

Major Subcontractor: _ Name

Address Phone

Major Supplier: _ Name

Address Phone

Are there any lawsuits, claims, or disputes resulting from this project? If so, please describe:

_____

* * * * *

Contract 2

Owner _____
Name

Address Phone

Date Begun _____ Date Completed _____ Dollar Amount _____

Description of Work _____

Architect _____
Name

Address Phone

Surety for Performance Bond: _____

Address Phone

Address Phone

Surety for Labor and Material Bond: _____

_____ _____
Address Phone

Insurance Company: _ Name

_____ _____
Address Phone

Major Subcontractor: _ Name

_____ _____
Address Phone

Major Supplier: _ Name

_____ _____
Address Phone

Are there any lawsuits, claims, or disputes resulting from this project? If so, please describe:

_____

* * * * *

Contract 3

Owner _____
Name

_____ _____
Address Phone

Date Begun _____ Date Completed _____ Dollar Amount _____

Description of Work _____

Architect _____
Name

_____ _____
Address Phone

Surety for Performance Bond: _____

_____ _____
Address Phone

Surety for Labor and Material Bond: _____

_____ _____
Address Phone

Insurance Company: _ Name

_____ _____
Address Phone

Major Subcontractor: _ Name

_____ _____
Address Phone

Major Supplier: _ Name

_____ _____
Address Phone

Are there any lawsuits, claims, or disputes resulting from this project? If so, please describe:

_____

III. Describe the work you normally perform with your own forces:

_____

_____

_____

_____

IV. Have you, or any officer or partner of your organization, when working for another organization, failed to complete any work awarded to you or such officer or partner? If so, attach a separate sheet of explanation.

V. Are there any claims or law suits against your organization, against predecessor organizations of your organization, or against any officer or partner who currently works for your organization which have been brought during the previous five years? If so, attach a separate sheet of explanation.

VI. Resumes

Provide resumes for at least two (2) persons with your company. One resume should be for an officer of the company: president, vice-president, etc. The second resume should be for an operations manager of the company who has had experience directing a job as a contractor or superintendent. An operations manager may also be an officer of the company.

Officer: _____
Name

Operations Manager: __ Name

VII. Trade References:

1.

2.

3.

4.

5.

VIII. Bank References:

1.

2.

3.

4.

5.

IX. Name of Bonding Company and name and address of agent:

X. Attach a financial statement, audited if available, (consolidated if the contractor is part of an affiliated group), including contractor's latest balance sheet and income statement showing the following items:

A. Current Assets (*e.g.*, cash, joint venture accounts, accounts receivable, notes receivable, accrued income, deposits, materials inventory, and prepaid expenses):

B. Net Fixed Assets:

C. Other Assets:

D. Current Liabilities (*e.g.*, accounts payable, notes payable, accrued expenses, provision for income taxes, advances, accrued salaries, and accrued payroll taxes):

E. Other Liabilities (*e.g.*, capital, capital stock, authorized and outstanding shares par values, earned surplus, and retained earnings):

Name of firm preparing financial statement and date thereof:

_____

Is this financial statement for the identical organization named in Part I on page one?

_____

If not, explain the relationship and financial responsibility of the organization whose financial statement is provided (*e.g.*, parent-subsidiary).

Will this organization act as guarantor of the contract for construction?

SCHEDULE OF MANUFACTURERS AND SUPPLIERS

Note: Complete and return this form with your bid.
Note: The City reserves the right to reject the use of any particular manufacturer.

RFB Title: Streetlight Materials & Installation
RFB Number: 44-2026

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

PROPOSED SUBCONTRACTORS

Note: Complete and return this form with your bid.
Note: The City reserves the right to reject the use of any particular subcontractor.

RFB Title: Streetlight Materials & Installation
RFB Number: 44-2026

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

Name _____ Phone No. _____

Address _____

City _____ State _____ Zip Code _____

PROJECT INFORMATION, SPECIFICATIONS, DRAWINGS

City of Boulder

Streetlight Standards



October 2026

Prepared for:

CITY OF BOULDER

Prepared by:

CLANTON & ASSOCIATES

LIGHTING DESIGN AND ENGINEERING

MULLER ENGINEERING COMPANY

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CHAPTER 1 GENERAL REQUIREMENTS

1.01 GENERAL

(A) Intent

These Boulder Street Lighting Standards describe general guidance for standards that will apply to street lighting infrastructure, equipment, procurement, construction, and installation. The standards listed here serve the purpose of ensuring the safety and quality of products, systems, and construction. City of Boulder's Energy Only Street Lighting (ESL) shall comply with these standards.

(B) Scope

The City will review, approve, and monitor the design and construction of all roadway and pedestrian lighting within the public right-of-way or public easements to ensure compliance with these Standards. The City has the sole authority for approving, accepting, or denying the design and construction of the lighting improvements.

(C) Abbreviations and Definitions

AFG – Above Finished Grade

AIC - Ampere Interrupting Capacity

ASCR – Aluminum Conductor Steel Reinforced

BFG – Below Finished Grade

CCT - Correlated Color Temperature

CRI - Color Rendering Index

EMT - Electrical Metallic Tubing

EPC – Electrical Polyvinyl Chloride

EPEC – Electrical Polyethylene Conduit

ESL – Energy only Street Lighting

GRC - Galvanized Rigid Steel Conduit

HDPE - High Density Polyethylene Conduit

IMC - Intermediate Metal Conduit

LCC - Lighting Control Center/Cabinet

LED - Light Emitting Diode

LER - Luminaire Efficacy Rating

LFMC – Liquid tight Flexible Metallic Conduit

LFNC – Liquid tight Flexible Non-metallic Conduit

Light Standard - Luminaire support structure with all mounting, mast arms, grounding and conduit/wiring access provisions.

Luminaire - Complete lighting fixture, including driver housing, lens, control device (i.e. photocell or node receptacle), and all mounting provisions.

Mast Arm - Luminaire mounting bracket attached to a pole or standard.

MPP - Meter Power Pedestal

Pole - Same definition as "Light Standard" above.

PVC - Polyvinyle Chloride

RNC – Rigid Non-metallic Conduit

SBCG – Soft drawn Bare Copper Ground

(D) Standards Organizations and Associations

(1) American Society for Testing and Materials (ASTM) – ASTM develops quality standards used globally. ASTM standards apply to materials such as: hot-dip galvanized steel (ASTM A153), spun cast prestressed concrete poles (ASTM C1089), high-strength alloy and stainless-steel bolting materials (ASTM A193), carbon steel bolts, studs, and threaded rods (ASTM A307), steel rebar (ASTM A615), hardened steel washers (ASTM F436).

(2) American Association of State Highway and Transportation Officials (AASHTO) – AASHTO is a non-profit organization whose publications cover roadside safety (AASHTO Roadside Design Guide) and material evaluation and testing (Manual for Assessing Safety Hardware). AASHTO standards apply to poles, mast arms, foundations, breakaway bases, soil conditions, and concrete.

(3) American National Standards Institute (ANSI) – The American National Standards Institute (ANSI) is a private, non-profit organization that administers and coordinates the U.S. voluntary standards and conformity assessment system. ANSI oversees the development of a broad range of standards, including construction equipment, energy distribution, and electronics.

(4) National Electrical Manufacturers Association (NEMA) – As an ANSI-accredited standards developer, NEMA brings together technical experts to ensure electrical and medical imaging technologies are safe, efficient, reliable, and interoperable. NEMA publishes standards for switchboards, panelboards, and enclosures for electrical equipment (1,000 volts maximum). NEMA enclosure types include Types 3, 3X, 3R, 3RX, 3S, 3SX, 4, 4X, 6, and 6P for outdoor locations.

(5) Institute of Electrical and Electronics Engineers (IEEE) – The IEEE develops and publishes standards covering communication networks, electronics, traffic lights, distributed energy resources (DER) like solar panels. The IEEE also publishes the National Electrical Safety Code (NESC).

(6) Underwriters Laboratories/Electrical Testing Laboratories (UL/ETL) – UL and ETL are among Nationally Recognized Testing Laboratories recognized by the Occupational Safety and Health Administration (OSHA). UL and ETL certify that certain products meet the requirements of both the construction and general industry OSHA electrical standards.

(7) Ingress Protection (IP) – IP ratings are developed by the International Electrotechnical Commission (IEC). Ingress protection ratings grade the resistance of an enclosure against the intrusion of dust or liquids to protect against deterioration or malfunction.

(8) National Electrical Code (NEC) – The NEC, published 3-year periodicals by the National Fire Protection Association (NFPA), addresses principles for the safe use of electricity and establishes standards for the installation of electrical systems. The NEC applies to circuits, conductors, raceways (such as conduits), and electrical equipment such as circuit breaker panels. NEC guidance mitigates hazards such as fire and shock that can arise from use of electricity. Engineers and electricians typically use the NEC. The administrative authority is usually a city, state, or local inspector.

(9) National Electrical Safety Code (NESC) – The NESC, published 5-year periodicals by the Institute of Electrical and Electronics Engineer (IEEE) sets the ground rules and guidelines for practical safeguarding of utility workers and the public during the installation, operation, and maintenance of electric supply, communication lines and associated equipment. The NESC is typically used by engineers, utility companies, and utility lineman. The administrative authority is typically the state public service or utility commission.

(10) NEC and NESC distinction: The governing use for the NESC will be used from supply lines and/or the utility transformer to the point of service. The point of service being the dividing point. The governing use for the NEC will be used from the point of service out to all other electrical, overcurrent protection, controls, and lighting devices. The majority of the work associated with Energy only Street Lighting (ESL) and Metered Street Lighting (MSL) will fall under the NEC. Typical exceptional places where the NESC will be applicable on the secondary side of the point of service are as follows:

(a) When City of Boulder acquires existing utility overhead service to lighting equipment on wooden utility poles.

(b) When City of Boulder acquires existing utility underground service to lighting equipment where triplex cable will not be replaced and therefore there is no raceway ground conductor provided in the existing service.

(c) When City of Boulder wants to use aluminum triplex direct bury cable for new installations of single lights at 120V. For circuits with multiple lights, conductors must be contained in conduit.

(11) Illuminating Engineering Society (IES) – The IES is the recognized technical and educational authority on illumination, whose mission is to improve the lighted environment by communicating lighting knowledge and translating that knowledge into actions that benefit the public. IES standards apply to luminaires and lighting applications, such as roadway, pedestrian, and tunnel lighting.

(12) Occupational Safety and Health Administration (OSHA) – OSHA exists to assure America's workers have safe and healthful working conditions free from unlawful retaliation. OSHA standards cover construction and repair activities, and apply to areas such as trenching, fall hazards, machinery, and construction equipment.

1.02 MATERIAL SPECIFICATIONS

(A) Conduit Material Specifications

(1) Below grade conduit shall be schedule 80 PVC or equivalent HDPE SDR 11:

(a) PVC shall comply with RNC: NEMA TC 2, Type EPC-80-PVC, UL 651.

(b) HDPE type EPEC-80 HDPE, complying with NEMA TC 7 and UL 651A.

(c) 2 inch (minimum) for all lighting branch circuits not crossing roadway.

(d) 3 inch (minimum) for all lighting branch circuits crossing roadway.

(e) All fittings shall match conduit and solvent cement attached approved for conduit.

(f) Minimum conduit burial:

(i) 24 inches below finished grade not crossing roadway.

(ii) 36 inches below finished grade crossing roadway.

(iii) 42 inches below finished grade crossing railroad.

(iv) 48 inches below finished grade crossing CDOT managed roadway.

(2) Above grade conduit shall be galvanized rigid steel conduit GRC:

(a) Hot dip galvanized.

(b) GRC shall comply with cULus-6, File Number: E37797, PROP65 - Compliant, ANSI C80.1, and approved for use by the NEC.

(c) Trade size conduit depending on conduit fill, minimum 1/2".

(d) Supports to be 2-hole conduit straps at a maximum of 5'-0" on centers. Straps material to be galvanized or stainless steel. Screws to be stainless steel.

(e) All fittings shall match conduit and NPT threaded fittings.

(B) Conductors Material Specifications:

(1) All conductors shall be rated for 600 volts.

(2) Aluminum conductors (for existing Xcel to convert to COB's ESL) and aluminum conductors (for new ESL and MSL installations).

(a) Aluminum shall be for existing Xcel & new streetlight direct bury cable.

(b) Aluminum shall be used for all new conductors in conduit for streetlights, unless approved by City of Boulder Lighting.

i. Critical Execution Rules for Aluminum

1. To ensure the installation is safe and doesn't fail at the pole connections, make sure the installation crew follows these code requirements:
2. Anti-Oxidation Compound is Mandatory (NEC 110.14): Aluminum oxidizes instantly when exposed to air, which creates electrical resistance and heat. Every wire termination inside the pole handholes must be coated in an approved paste (like Noalox or Penetrox) before tightening.
3. Verify Terminal Ratings: The inline breakaway fuse holders and terminal blocks inside the light pole bases must be explicitly stamped AL/CU (approved for both aluminum and copper). Standard copper-only lugs will cause a chemical reaction that destroys the connection over time.
4. Color Coding: Because THWN-2 aluminum is usually sold only in black for this gauge size, the installers must use color-coded electrical tape (white for neutral, green for ground) at every pole box to keep the phases clear.

(c) Copper shall be used for signal infrastructure

(d) Copper shall be used for pole wiring from the luminaire to the handhole.

(3) Conductors: Comply with NEMA WC 70/ICEA S-95-658.

(4) Conductor Insulation: Comply with NEMA WC 70/ICEA S-95-658 for Type THWN, THWN-2, XHHW-2.

(5) Direct bury triplex and quadplex cable shall be Underground Residential Distribution cable URD/URD2 aluminum or copper. Sizes and type to be indicated on the plans and details.

(6) Use solid for No. 10 AWG and smaller; use stranded for No. 8 AWG and larger. Exception: #6 soft drawn bare copper or #4 aluminum shall both be solid conductors for grounding to ground rod.

(C) Guidance for Grounding Electrode Conductors

(1) Table 250.66 (A) of the 2023 NFPA 70 - National Electrical Code states: "If the grounding electrode conductor or bonding jumper connected to a single or multiple rod, pipe, or plate electrode(s), or any combination thereof, as described in 250.52(A)(5) or (A)(7), does not extend on to other types of electrodes that require a larger size conductor, the grounding electrode conductor shall not be required to be greater than #6 AWG copper wire or #4 AWG aluminum wire" (NFPA 70, 2023). However, the Engineer can require a larger size grounding conductor installed when doubling as lightning protection for lowering resistance to ground.

(2) Where existing Xcel Energy infrastructure has used AWG #4 bare copper conductors for grounding electrode conductors, no change is needed. On previously Xcel-owned existing street lighting infrastructure, grounding electrode conductors do not need to change that

already meet or exceed NEC requirements. However, new installations can use AWG #6 solid bare copper grounding conductors and meet NEC requirements.

(D) Wire Connectors Specifications

(1) Splice box mounted, submersible, multi-port, insulated, connectors rated for direct burial.

(a) 4-ports minimum, (6-ports for ground connector in point of service pull box.)

(b) Dual rated for a wide range of aluminum or copper conductors (must be rated for both).

(c) Compact designed to enable installation in limited spaces.

(d) Field adjustable cable sleeves to enable installation on a wide range of conductor sizes.

(e) Standard aluminum set screws with 5⁄16" socket hex spread compressive forces over conductor strands, reduce strand damage, result in lower contact resistance, and provide high set-screw torque retention.

(f) Comply with ANSI C119.1 of the 2023 American National Standards Institute and ANSI C119.4 of the 2022 American National Standards Institute as well as Western Underground C.G. 2.5 of the 2004 version of Western Underground Committee Guidelines.

(2) Overhead open air ACSR cable connectors:

(a) H-tap compression connectors.

(b) Made of 1350 aluminum alloy

(c) Pre-filled with an oxide inhibitor held captive in the rib/connection area.

(d) Dual rated aluminum or copper conductor connectors. Aluminum-aluminum, aluminum-copper, or copper-copper.

(e) Tool operated fold-in tabs for positive tab interlock as tool closes.

(f) Comply with ANSI C119.4 of the 2022 American National Standards Institute requirements.

(g) Size of primary and secondary cables as indicated on plans and details.

(E) Splice/Pull Boxes Specifications:

(1) Flush In-Grade Boxes:

(a) Polymer concrete bottomless box size as indicated on plans & details.

(b) Heavy-Duty, TIER 22 non-deliberate traffic rated, 2-bolt covers.

(c) Engraved label in cover to read as indicated on plans & details.

a. For separation pull boxes, labelling shall include “BOULDER ESL” to meet separation requirements

(d) 12” deep, ¾” gravel or road base below box for drainage.

(e) Pull boxes shall be verified by a 3rd Party Nationally Recognized Independent Testing Laboratory as meeting all test provisions of American National Standards Institute/Society of Cable Telecommunications Engineers (ANSI/SCTE) 77, 2013 Specification for Underground Enclosure Integrity, Tier 22 rating. Pull boxes shall be Underwriters Laboratories (UL) listed. Certification documents shall be submitted with material submittals.

(2) Surface Boxes:

(a) Heavy-duty galvanized and painted (Stainless steel as an option) box size as indicated on plans & details.

(b) NEMA 3R/4 rated.

(c) Stainless steel screw for screwed covers or hinged with stainless steel clamped covers.

(d) Minimum 14ga. Steel.

(e) UL listed for electrical enclosures.

(F) Power Pedestals Specifications:

(1) Direct bury, galvanized painted steel pedestal.

(2) Maximum height: 50 inches

(3) Maximum width: 12 inches

(4) Non-metered, service entrance rated enclosure.

(5) Main Circuit Breaker size as indicated on plans and details.

(6) UL listed as service equipment.

(7) Provide engraved phenolic to read: “BOULDER ESL”

(G) Disconnect & Safety Switch Specifications:

(1) All switches shall be heavy-duty rated. Switch blades and jaws shall be fixable and plated copper. Switches shall have a pad lockable handle. Switches shall have defeatable door interlocks that prevent the door from opening when the handle is in the ON position (except for double-throw switches). The defeat mechanism shall be front accessible. Switches shall have deionizing arc chutes. Switch assembly and operating handle shall be an integral part of the enclosure base.

(2) Switches rated 30 A, 60 A, and 100A shall have reinforced, rejection type fuse clips. Switch blades shall be readily visible in the “ON” and “OFF” position. The switch operating mechanism shall be non-teasable, positive quick-make/quick-break type. Bail type mechanisms are not acceptable. Fusible switches shall be suitable for service

entrance equipment with factory installed neutrals. Switches shall have line terminal shields.

(3) All enclosures shall be NEMA 3R unless otherwise noted. Other types, where noted, shall be NEMA 4X watertight corrosion-resistant 316 stainless steel or NEMA 12 dust-tight and oil-tight special industry (dual NEMA 12/3R rating 30A, 60A, and 100A). All enclosures shall have factory installed neutral and ground terminal blocks. Nameplate shall be front cover mounted, containing a permanent record of switch type, ampere rating, and maximum voltage rating. 30 A, 60A, and 100 A, NEMA 4X or NEMA 12/3R enclosures shall be provided with draw-pull latches.

(4) Acceptance Testing Preparation:

(a) Test insulation resistance for each enclosed switch and circuit breaker, component, connecting supply, feeder, and control circuit.

(b) Test continuity of each circuit.

(5) Tests and Inspections:

(a) Perform each visual and mechanical inspection and electrical test stated in NEMA Acceptance Testing Specification. Certify compliance with test parameters.

(b) Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace them with new units and retest.

(H) Luminaire Specifications:

(1) Luminaire Photometric Data Testing Laboratory Qualifications: Provided by laboratories that are accredited under the National Volunteer Laboratory Accreditation Program for Energy Efficient Lighting Products or by an independent agency, with the experience and capability to conduct the testing indicated, that is an NRTL as defined by Standard 29 CFR 1910 of the 2026 Occupational Safety and Health Standards for General Industry.

(2) All luminaires shall be aluminum housing with LED engines.

(3) Drivers shall be integrated into housing with passive heat sink technology.

(4) Dead Load: Weight of luminaire and its horizontal and vertical supports, and supporting structure, applied as stated in the 4th Edition (2001) of the Standard Specifications for Structural Supports for Highway Signs, Luminaires and Traffic Signals, published by the American Association of State Highway and Transportation Officials (AASHTO LTS-4-M).

(5) Live Load: Single load of 500 lb. (2224 N), distributed as stated in AASHTO LTS-4-M.

(6) Ice Load: Load of 3 lb./sq. ft. (145 Pa) applied as stated in AASHTO LTS-4-M Ice Load Map.

(7) Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

(8) Comply with IEEE C2, National Electrical Safety Code.

(a) Horizontal clearances between lights and utility poles/overhead lines: Detail LD-12

(b) Vertical clearances between lights and overhead lines: Detail LD-13

(c) Neutral/ground bonding at poles: Detail FD-02

(d) Conductive parts grounding: Details FD-02, FD-05, FD-06, LD-10, and LD-11

(9) Comply with NFPA 70, National Electrical Code.

(e) Standard voltages and housing requirements: Details LD-05, LD-06, LD-07, LD-08, and LD-09

(f) Fusing and grounding: Details FD-02, FD-05, FD-06, LD-10, & LD-11.

(10 Control device: ANSI 7-pin NEMA receptacle.

(I) Light Standard Specifications:

(1) Metal light standards:

(a) Galvanized painted round tapered steel:

(i.) 12-inch bolt circle.

(ii.) Height of pole to be as indicated on the plans and details.

(iii.) Hand hole at 12 inches from bottom.

(iv.) Color: Black powder-coated

(b) Painted round tapered aluminum:

(i.) 12-inch bolt circle.

(ii.) Height of pole to be as indicated on the plans and details.

(iii.) Hand hole at 12 inches from bottom.

(iv.) Color: Black powder-coated

(2) Wooden utility poles for light standards:

(a) Direct bury wooden light standard

(b) Height of pole above finished grade to be as indicated on the plans and details.

(c) Wooden poles: use only utility grade, pressure-treated fir, class, specifications, and dimensions per standard ANSI 05.1 (Wood Poles: Specifications and Dimensions) of the 2022 American National Standards Institute.

(d) Wood poles to be Coastal Douglas Fir or approved equal with minimum class of pole to be Class 3. Wood poles to be manufactured per American National Standards Institute (ANSI) and AASHTO, latest revision. Poles to be pressure treated per American Wood Protection Associates (AWPA), 2022. The

preservative used shall be pentachlorophenol or approved equal. Poles to meet Plan Quality Control inspection.

(e) See **Section 4.01** Structural Notes and Requirements for specified embedment depth and soil compaction based on wind loading per City of Boulder standards.

CHAPTER 2 ENGINEERING DESIGN REQUIREMENTS

2.01 LUMINAIRE STANDARDS

(A) Introduction

(1) All luminaires shall:

(a) Have an LED light source with minimum 50,000-hour rated life (L70)

(b) All luminaires shall have a photocell receptacle ANSI C136.41 7-pin as indicated in ANSI C136.41-2025 – Roadway and Area Lighting Equipment–Dimming Control Between an External Locking Type Photocontrol and Ballast or Driver.

(c) Meet Dark Sky requirements.

(d) Support City-approved networked lighting controls

(d) Controls must support 0-10V dimming.

(e) DALI controls may be an option.

(e) Match City-standard finishes and optical distributions

(2) All luminaires shall be color: Black

(3) Unless otherwise approved, luminaires shall be from the City's Approved Products List (APL).

(B) Cobrahead Luminaires

(1) Street lighting shall be provided by cobrahead luminaires with the exception of areas where lighting is provided by post top luminaires. These luminaires are mounted on mast arms and are intended for typical street lighting installations that represent the majority of the City's lighting inventory. Standard details provide multiple roadway light distributions to address varying street widths and pole placements. Street lighting shall be specified as control-ready and equipped with 7-pin NEMA receptacles. Smart controls may be activated at the time of installation or in the future. Luminaire orientation, aiming, and control node placement are shown in the standard details (LD-05 and LD-06).

(a) Mounting: Mast arm, horizontal tenon

(b) Distribution types:

(i.) Type II – Narrow / residential streets, collectors, and minor arterials

(ii.) Type III – Principal arterials, freeways, and expressways

(iii.) Type IV – near parking lots, trails and cul de sacs

(c) Color Temperature: 2700K

(d) Optional shielding where required to limit glare and trespass.

(C) Pedestrian and Post-top Luminaires

(1) Pedestrian and post-top luminaires are used in pedestrian-oriented areas, including shared-use paths, downtown areas, some local streets, and other contexts where City-maintained pedestrian lighting is required. Standard mounting configurations include post-top mounting on vertical tenons. Pedestrian and post-top luminaires shall be control-ready, equipped with ANSI C136.41 7-pin receptacle and compatible with the City's smart lighting platform.

(a) Mounting:

(i.) Post-top (vertical tenon)

(b) Distribution types:

(i.) Type III – neighborhoods or open space/parks

(ii.) Type V – neighborhoods or open space/parks

(c) Color Temperature: 2700K

(d) Optional shielding where required to limit glare and trespass.

(D) Canopy Luminaires

(1) Canopy luminaires are used in a limited number of City-owned locations, such as parking structures, under overpasses or covered pedestrian areas. Canopy luminaires shall be selected from a limited set of City-approved models and shall have standardized mounting and electrical requirements. Standard details illustrate typical ceiling-mounted installations, including junction box locations and electrical connections. Canopy lighting must be approved by the City.

(E) Underpass Luminaires

(1) Underpass luminaires are used to illuminate multi-use paths where they pass under roadways.

(a) Mounting:

(i.) Wall-mounted

(b) Distribution types:

(i.) Wide

(c) Color Temperature: 2700K

2.02 POLES OR LIGHT STANDARDS

(A) Introduction

(1) Poles specified in these standards are intended to support the standard luminaires and mast arms. Pole selection shall be based on roadway type, context, structural requirements, and electrical service conditions, as further defined by City standards and details.

(B) Pole or Light Standard Types

(1) The following pole materials are approved for use within the City's standard street lighting system:

- (a) Black powder-coated galvanized steel (primary standard)
  - (i.) Pedestrian and post-top 14'-0" non-tapered steel (for post top luminaire)
  - (ii.) Pedestrian and post-top 20'-0" non-tapered steel (for post top luminaire)
  - (iii.) Streetlight 25'-0" non-tapered steel (for cobrahead luminaire)
  - (iv.) Streetlight 28'-6" tapered steel (for cobrahead luminaire)
    - a. Preferred for most applications
  - (v.) Streetlight 30'-0" tapered steel (for cobrahead luminaire)
  - (vi.) Streetlight 35'-0" tapered steel (for cobrahead luminaire)
- (b) Wood utility pole
  - (i.) Streetlight 25'-0" above grade mounting height (for cobrahead luminaire)
  - (ii.) Streetlight 30'-0" above grade mounting height (for cobrahead luminaire)
  - (iii.) Streetlight 35'-0" above grade mounting height (for cobrahead luminaire)

(2) Steel poles, attachments, and hardware shall be galvanized in accordance with Section 509.11 (Galvanized and Metallized Steel) of the 2025 (Version 1) CDOT Standard Specifications for Road and Bridge Construction.

(3) Power coating for galvanized steel and aluminum poles and attachments shall be in accordance with CDOT Standard Specification 715.04. The color shall be black unless otherwise specified and approved by the City. Surfaces to be powder coated shall be properly prepared and pretreated. For galvanized steel, outgassing shall be properly controlled to avoid potential failure of the duplex coating.

(4) Protection and preservation for wood utility poles shall be in accordance with Section 14.8 (Wood Structures) of the 2015 edition (First Edition with 2022 Interim Revisions) of the AASHTO LRFD Specifications for Structural Supports for Highway Signs, Luminaires, and Traffic Signals (AASHTO LTS).

(5) Pole details identify handhole locations and clearances for electrical and control components (LD-01 through LD-04 for handhole locations, LD-12 through LD-14 for clearances).

(6) Frangible breakaway bases shall be provided for light poles located in the clear zone, as defined in the AASHTO Roadway Design Guide. Exceptions to this standard include the following conditions where breakaway bases should not be provided:

(i.) Light poles that are protected behind a guardrail or curb

(ii.) Light poles located in close proximity (less than 1.5 x pole height) to sidewalks, pedestrian walkways, bike paths, or trails where a falling pole could fall on a pedestrian or cyclist.

(iii.) Light poles located in close proximity (less than 1.5 x pole height) to a building.

(7) Design shall be completed in accordance with the 2015 edition of the AASHTO LRFD Specifications for Structural Supports for Highway Signs, Luminaires, and Traffic Signals (AASHTO LTS), as modified below:

(a) Design wind speed (3 second gust) shall be in accordance with the current City of Boulder adopted code and amendments. Base wind speed for analysis shall be taken for the appropriate wind zone from Table 1609.3 (Basic Design Wind Speed) of the 2026 (Sept 16, 2026, Supplement 167) City of Boulder Municipal Code.

(i.) Low risk (300 year MRI / Risk Category I) shall be assigned to locations where failure could not cross a travelway, including pedestrian trails and sidewalks.

(ii.) Typical risk (700 year MRI / Risk Category II) shall be assigned to locations where failure could cross a travelway, including pedestrian trails and sidewalks, with an ADT less than 10,000 vehicles per day.

(iii.) High risk (1,700 year MRI / Risk Category III) shall be assigned to locations where failure could cross a lifeline travelway, or where ADT exceeds 10,000 vehicles per day.

(b) See Table 3.8-1 (Mean Recurrence Interval) of the 2015 edition (First Edition with 2022 Interim Revisions) of the LRFD Specifications for Structural Supports of Highway Signs, Luminaires, and Traffic Signals (AASHTO LTS) for additional information.

(8) Poles shall be rated for the attachments shown in the standard details, including luminaires, mast arms, and control equipment. Use of additional attachments not shown in the standards shall require City review and approval.

(9) Prior to procurement, the supplier shall submit pole shop drawings, product data, and structural design documentation for review. Submittals shall include, at a minimum, the pole type, material specifications, pole height, geometry, arm configuration, luminaire and attachment weights, EPA values, design wind speed, risk category, allowable EPA, handhole details, base plate and anchor rod details, breakaway base details where applicable, finish details, including a method statement for duplex coating, and foundation-level reactions.

(a) For direct-embedded wood poles, calculations shall be provided to demonstrate adequate embedment for stability. The calculations shall specify the required hole size and backfill materials requirements.

(b) Submittals shall clearly identify any assumptions, limitations, or installation restrictions.

(c) Standard poles may be accepted where the supplier can provide published design data demonstrating that the proposed configuration satisfies the City's required wind speed.

2.03 MAST ARMS

(A) Introduction

(1) Standard mast arms are provided for single-luminaire and double-luminaire roadway lighting installations. Mast arm lengths and configurations shall be limited to a set of predefined options appropriate for typical roadway conditions and pole placements. Mast arms shall be fabricated from galvanized steel and finished with a black powder-coated finish to match the associated poles unless otherwise approved.

(B) Mast Arm Types

(1) Standard details LD-01, LD-02, LD-04, LD-14, and LD-15 identify the allowable mast arm lengths and corresponding applications. Variations outside of these standard configurations are discouraged and require City review and approval. Mounting options for post-top luminaires and pedestrian luminaires are shown: tenon post-top mounting.

2.04 FOUNDATIONS AND BASES

(A) Introduction

(1) Standard details FD-01 through FD-06 illustrate typical pole installation methods, including concrete caisson and direct-buried foundations, as applicable. Installation methods may vary by pole material, height, and site conditions and shall follow manufacturer recommendations and City standards. Foundation details identify grounding provisions. Where required, breakaway bases shall be provided and detailed to meet applicable safety standards.

(B) Foundation Types

(1) Foundations with splice boxes (detail FD-02).

(2) Foundations without splice boxes (detail FD-01).

(3) Foundations details for light standard mounting

(i.) Located in hardscape (detail FD-03).

(ii.) Located in open landscape (details FD-01 and FD-02).

(iii.) Located in parking lots (detail FD-03).

(4) Components of typical foundations are illustrated and described, including anchor bolts, hand holes, in-line fuses, conductors, ground rods, and conduits.

(C) Breakaway Bases

(1) Cast aluminum breakaway transformer base with access handhole cover (detail LD-11).

(2) Painted powder-coat finished black to match pole and luminaire.

(3) Ground lug visible from the handhole opening.

(4) 12-inch bolt circle holes on top and bottom of transformer base.

2.05 SMART LIGHTING CONTROLS

(A) Introduction

(1) Smart lighting controls are recommended for City of Boulder lighting systems. Unless otherwise specified by the city, smart controls are required. All standard luminaires shall be specified as control-ready to support future deployment without replacement of luminaires.

(B) Smart Lighting Controls

(1) ANSI 7-pin control receptacles are required on all luminaires.

(a) When smart controls are not installed at the time of construction.

(i.) On metered lighting control systems, the luminaires shall be equipped with a shorting capacitor.

(ii.) On energy only street lighting (ESL) systems the luminaires shall be equipped with a long-life LED twist-lock photocell.

(b) The luminaire shall support 0-10V dimming and network metering capabilities.

(c) Smart lighting controls shall not require additional mesh network hardware to be installed

2.06 CABLES, CONDUCTORS, AND CONDUIT

(A) Introduction

(1) Sizing cables, conductors and conduits depends on several electrical engineering parameters. Cable and conductor sizes are determined by wire ampacity, voltage drop, overcurrent protection rating, derating factors to avoid overheating of wiring, and cost. Voltage drop in turn is determined by the resistivity of the conductor material: aluminum or copper. Copper conducts better than aluminum, and so it has a lower resistance and results in a lower voltage drop over the same distance. However, aluminum costs less than copper to purchase. Sizing of conduit depends on size and number of conductors within the conduit to be able to pull the conductors into the conduit.

(B) Cables, Conductors, and Conduit Sizing

(1) Conduit sizes are determined by the size and the number of conductors contained in the conduit. The NEC provides guidance about the allowable cross-sectional area of the conduit that can be filled by conductors. For most conduits, the allowable fill ratio is 40% of the interior area of the conduit.

(2) Circuit layouts are also crucial to sizing conductors. The number of streetlights – and the wattage, as a result – determine the current that conductors will carry and the voltage drop over the length of the wires. Increasing wire size reduces voltage drop over the same distance, but larger wires cost more per linear foot. Circuits can be laid out to avoid high voltage drop and large wire sizes.

(3) For more information about sizing conductors and conduits, refer to Section 2.06 E Electrical Engineering Parameters.

(4) One key difference in selecting conductors for overhead or underground applications is the difference in wire ampacity, the amount of current a conductor can carry. Ampacity depends on the material (copper or aluminum), the insulation type, and the size of the wire, and its installation type if underground (direct bury vs. in conduit). Conductors in free air generally have much higher ampacities than conductors in underground raceways, such as conduits. NEC Table 310.16 (NFPA 70, 2023) provides ampacities for conductors in raceway, cable, or earth (directly buried). NEC Table 310.17 (NFPA 70, 2023) provides ampacities of single-insulated conductors in free air. For example, AWG #6 copper wire with THWN insulation has an ampacity of 65 A in conduit and the same wire has an ampacity of 95 A in air. As a result, for the same loads, smaller wire sizes can be used in air than in buried conduits. Also, aluminum might be more viable for overhead applications because of the lighter weight than that of copper of the same size.

(5) When new lights are added, existing conductors must be evaluated for suitability to add lights to existing circuits. Adding a light to a circuit will change the overall wattage and the current that the circuit will carry. Voltage drop will change as well. The result is that existing wires may be undersized for the addition of new lights and may require replacement. When lights are removed from an existing circuit, loads decrease, and generally existing wires will be more than adequate for lights to remain. Where new lights will be installed close to existing overhead electric utility infrastructure, refer to the ODL CONST-J.80 (Streetlight Horizontal Clearance) and ODL CONST-J.90 (Streetlight Vertical Clearance) in the Electrical Wiring and Related Components for Streetlighting section of the 2025 Xcel Energy Electric Distribution Design and Construction Standards, Outdoor Lighting (ODL)

(6) The contractor shall install 1#12 AWG locate wire and 1,250 lb. nylon pull string with footage tape in all spare and empty conduits.

(7) Splices must be approved by City Engineering.

(8) Standard Cable Sizes

(a) #6 Triplex Underground Residential Distribution cable URD/URD2 is an Aluminum or Copper direct-burial distribution cable. The primary use of URD/URD2 cable is for service laterals to residential homes. Direct-bury cable is only allowed when feeding only one streetlight on a circuit.

i. This is preferred for most applications

(b) Xcel Energy predominantly uses #1/0 Triplex Aluminum URD/URD2 for the majority of their street lighting branch feeder circuits as they follow the NESC that does not require an equipment grounding conductor (EGC). In addition, aluminum weighs less and costs less than copper, making installation easier.

i. 1/0 Triplex may be approved by City of Boulder Lighting for circuits of 5 lights or a branch feed for multiple branches

(c) The NEC requires an Equipment Grounding Conductor (EGC) or a qualifying metal raceway per 250.118(A) (NFPA 70, 2023). Quadplex URD/URD2 aluminum cable could be used to provide an EGC, using the three phase wires as (2) Hots, (1) Neutral, and the 4th reduced size wire as the EGC. This approach would comply with the NEC. Quadplex cables usually start at #4 quadplex or 4-4-4-4 Tulsa aluminum. Larger #1/0 Quadplex cable typically lowers the fourth conductor to a #2, or 1/0,1/0,1/0,& 2 Notre Dame aluminum cable.

(A) Trenched Conduit

(1) Determining what conduit to use typically takes into account the requirements of the NEC as well as other factors. Cost and availability of conduits are typical considerations. Conduit location is another factor. PVC and HPDE are often used for trenched and bored conduit, respectively. HPDE is often used for bored conduit under existing roadways or surfaces such as streets and parking lots if the surface will not be fully reconstructed as part of the project.

(a) Galvanized Rigid Conduit (GRC) – use in above ground locations, such as mounted to structures, to avoid using PVC which will photodegrade in sunlight and become brittle. Use also for surface mounted conduit applications, such as for underdeck or underpass lighting. The NEC also permits rigid metal conduits to have a shallower burial depth of 6 inches compared to non-metallic conduits, such as PVC (NEC Table 300.7(A)) (NFPA 70, 2023).

(b) Polyvinyl Chloride Conduit (PVC) – use in trenched locations and in applications where conduit will be embedded in concrete, such as in the walls of some bike underpasses. PVC conduit can be buried at reduced depths if it is encased in concrete for protection from damage. This approach is taken when electrical conduits must cross over other buried utilities and sufficient clearance from the nearby utilities is not achievable.

(c) High Density Polyethylene (HDPE) – use for bored conduit under existing roadways that will remain to avoid cutting pavement to trench.

(2) Detectable warning tape is required by the NEC to be installed 12 inches above any direct-buried conduits installed at a depth of 18 inches or more below grade (300.7(D)(3) (NFPA 70, 2023). The warning ribbon reduces the risk of an accident — such as an electric shock or arc-flash incident — during excavation near underground service conductors that are not encased in concrete. The American Public Works Association (APWA) Uniform Color Code required electrical warning tape for cables and conduits to be red.

(A) Bored Conduit

(1) The work specified below documents the approved construction methods, procedures and materials for Directional Boring, also commonly called Horizontal Directional Drilling (HDD).

(a) HDD is a trenchless method for installing a product that serves as a conduit for liquids, gasses, or as a duct for pipe, cable, or wire line products. It is a multi-stage process consisting of site preparation and restoration, equipment setup, and drilling a pilot bore along a predetermined path and then pulling the product back through the drilled space. When necessary, enlargement of the pilot bore hole may be necessary to accommodate a product larger than the pilot bore hole size. This process is referred to as back reaming and is done at the same time as the product is being pulled back through the pilot bore hole.

(b) Accomplish alignment of the bore by proper orientation of the drill bit head as it is being pushed into the ground by a hydraulic jack. Determine orientation and tracking of the drill bit by an above ground radio detection device which picks up a radio signal generated from a transmitter located within the drill bit head. Then electronically translate the radio signal into depth and alignment. In order to minimize friction and prevent collapse of the bore hole, introduce a soil stabilizing agent (drilling fluid) into the annular bore space from the trailing end of the drill bit. The rotation of the bit in the soil wetted by the drilling fluid creates a slurry. The slurry acts to stabilize the surrounding soil and prevent collapse of the bore hole as well as provides lubrication.

(c) Select or design drilling fluids for the site-specific soil and ground water conditions. Confine free flowing (escaping) slurry or drilling fluids at the ground surface during pull back or drilling. Accomplish this by creating sump areas or vacuum operations to prevent damage or hazardous conditions in surrounding areas. Remove all residual slurry from the surface and restore the site to preconstruction conditions.

(B) Electrical Engineering Parameters

(1) Standard conductor sizes are listed in the following table. Wire size AWG #12 is shown for conductors installed in streetlight poles from the handhole to the luminaire at the top of the pole. Three standard wire sizes are shown for copper (Cu) and aluminum (Al) conductors. For spacing greater than 200’ between streetlights, or for more than 8 streetlights, calculate the voltage drop to the most distant streetlight and size conductors accordingly.

CONDUCTOR SIZING PER NUMBER OF LUMINAIRES UNDERGROUND FED TABLE					
# of LTS	**COPPER CONDUCTORS**	**ALUMINUM CONDUCTORS**	**VOLTAGE**	**LUMINAIRE & DISTANCE**	**DESCRIPTION OF SERVICE**
1	3#12 THWN-2, CABLE	N/A	120V-HOT,NEU,GND	N/A	In every light standard from fuse holder in handhole up to pole top lumianaire
1-2	3#10 THWN-2	3#8 THWN-2	120V-HOT,NEU,GND	<150W @MAX. 200' SPACING	From utility to point of service (POS) box and from POS to fuse holder in handhole
3-4	4#6 THWN-2	4#4 THWN-2	120/240V-HOT,HOT,NEU,GND	<150W @MAX. 200' SPACING	From utility to point of service (POS) box and from POS to fuse holder in handhole(alternate L1, L2 every other light)
5-8	4#2 THWN-2	4#1/0 THWN-2	120/240V-HOT,HOT,NEU,GND	<150W @MAX. 200' SPACING	From utility to point of service (POS) box and from POS to fuse holder in handhole(alternate L1, L2 every other light)
>8	Consult voltage drop calculator		120/240V-HOT,HOT,NEU,GND	<150W @MAX. 200' SPACING	From utility to point of service (POS) box and from POS to fuse holder in handhole(alternate L1, L2 every other light)

(2) Conductor ampacity: conductors must be sized to carry the current drawn by the electrical loads connected to a circuit. The current carrying capacity of a conductor is known as ampacity and has units of Amps (A). Conductor ampacities, based on the wire gauge and wire insulation type, are listed in Table 310.16 (Ampacities of Insulated Conductors in Raceway, Cable, or Earth (Directly Buried)) of the 2023 NFPA 70 - National Electrical Code. Wires must be sized to have an ampacity no less than 125% of the continuous load (NEC 210.19) (NFPA 70, 2023).

(a) Ampacity derating factors: wire ampacity must be derated when 3 or more current carrying conductors are contained in a raceway. Derating factors are listed in NEC table 310.15(C)(1) (NFPA 70, 2023):

(i.) 4-6 Conductors, 80% derating factor

(ii.) 7-9 Conductors, 70% derating factor

(iii.) 10-20 Conductors, 50% derating factor

(b) Overcurrent protection: the overcurrent protection for a circuit is typically provided by a circuit breaker or fuse, with a rating in Amps (A). Conductors must be sized to have an ampacity greater than or equal to the circuit breaker rating (NEC 240.4).

(c) Voltage drop: the NEC recommends that conductors are sized to limit the voltage drop to less than 5% for the combination of feeders and branch circuits, 2% for service laterals from power source to service entrance, and 3% to last device (NEC 215.4(A)(2)). Also, where circuit conductors are increased due to voltage drop, 250.122(D) requires an increase in circular mil area for the associated equipment grounding conductors. Limiting voltage drop to less than 5% is intended to ensure reliable operation of electrical equipment and avoid

inefficiencies and potential equipment damage due to undervoltage (NFPA 70, 2023).

(3) The NEC provides guidance on the sizing of conduit and on the physical layout of conduit.

(a) Conduit routing: The maximum amount of conduit bends permitted by the NEC before a pull box or junction box must be installed is 360 degrees in Article 352 (Rigid Polyvinyl Chloride Conduit (PVC), 352.24 (B)) of the 2023 NFPA 70 - National Electrical Code. Installations must include pull boxes or junction boxes where the sum of the bends in the conduit would otherwise exceed 360 degrees.

(b) Conduit sizing: the minimum size of conduit used as a raceway is determined by the percent of the interior area filled by conductors:

(4) The percent of cross section of conduit permitted to be filled with conductors and cables is provided in NEC Chapter 9, Table 1 (NFPA 70, 2023). The most common fill ratio is 40%, with more than 2 conductors or cables in the conduit.

(c) Conduit burial depth: NEC Table 300.7(A) lists minimum cover requirements for direct-buried cable, conduit, and other raceways. Non-metallic conduit, such as PVC, has a minimum burial depth of 18 inches, or 24 inches under streets, highways, roads, alleys, driveways, and parking lots (NFPA 70, 2023). Other requirements may apply to burial depth; CDOT requires conduit to be buried at a 48-inch burial depth.

(d) Common trenches: the NEC permits multiple electrical conduits to be installed in a single trench. Other requirements may apply, such as physical distances from other buried utilities.

(5) Standardizing typical sizes of wires and conduits can have benefits in terms of costs and maintenance. Bulk wire prices may save on cost and using a selection of typical wire sizes may help simplify inventory by reducing the stock of numerous specific wire sizes. Several considerations may be taken into account.

(a) Ampacity – lighting loads draw low currents in Amps. As a result, most wire sizes will be adequate in terms of ampacity. For example, AWG #10 wire with THWN-2 insulation has an ampacity of 40 A. Even when derated by 50% - for raceway containing 10-20 conductors (NEC 310.15(C)) (NFPA 70, 2023) – the ampacity is still 20 A and will be suitable for 20 amp lighting circuit.

(b) Voltage drop – increasing wire size will decrease voltage drop. AWG #6 wire will often reduce the voltage drop to under 3% for circuits feeding multiple lights up to hundreds of feet from the power source. Continuing to increase the wire size will increase cost, so consider modifying the layout of circuits if excessive voltage drop will be a concern.

2.07 PULL/SPLICE BOXES

(A) In-grade Pull/Splice Box Sizing

(1) All flush in-grade pull boxes shall be heavy duty, Tier 22 rated, bottomless, polymer concrete box with 2-bolt cover and a minimum of 12 inches deep of 3/4" diameter rock or crashed road base below box for drainage. There shall be (3) sizes of pull boxes used for the ESL lighting determined by the largest trade size conduit installed in the pull box as follows:

(a) 1/2"-2" conduit - pull box shall be type one - 11" x 18" x 12"dp. See detail #EL-02 & PS-04 for additional information.

(b) 3" conduit - pull box shall be type two - 13" x 24" x 12"dp. See detail #EL-02 & PS-04 for additional information.

(c) 4" conduit - pull box shall be type three - 17" x 30" x 12"dp. See detail #EL-02 & PS-04 for additional information.

(d) The cover shall be engraved with “BOULDER ESL” for the following:

(i.) All boxes with in-line fuses to provide separation from utility to the energy only street lighting system.

(ii.) All other boxes in the energy only street lighting system may be engraved differently, per City of Boulder Lighting discretion.

(e) The contractor shall be responsible for repairing damage caused by boring, trenching, cutting, drilling, etc. and performance of any related work in this contract area. The contractor shall furnish and install all barriers, pavement, landscape materials, etc. to restore the construction site outside of the pavement included in the roadway improvements to restore the construction site to pre-construction conditions.

(f) A minimum of 12 inches of 3/4-inch angular granite-gravel or road base shall be installed at the base of the pull box. The granite-gravel shall be free of dirt and debris and spread evenly to facilitate a level base for the pull box. The contractor shall compact to the same density of the in-situ soil prior to the installation of the granite-gravel to alleviate future settling.

(g) Tracer wire shall be attached to the trace test points on the underside of the pull box lid. Each tracer wire shall be attached to an individual trace point; no two wires shall be attached to the same point. The contractor shall coil an additional 6 feet of tracer wire inside the pull box to ensure that the trace wire will not disconnect from the test points when the lid is removed.

(h) Pull boxes shall be installed in areas that are easily accessible by maintenance personnel. The slope around the pull box shall not be steeper than 5:1.

(i) Concrete support rings may be approved on a case-by-case basis. Pull boxes installed with concrete aprons or pre-cast polymer concrete shall not be installed

above the grade of the apron. The concrete apron shall have a 1 percent slope away from the top of the pull box to allow for drainage. Pre-cast concrete aprons shall be installed per the manufacturer's recommendations. Unless otherwise shown on the plans, pull and splice boxes shall be installed so that the covers are level with the curb or sidewalk grade. Covers shall be level with the surrounding ground when no grade is established.

(j) Where practical, pull and splice boxes near curbs shall be placed adjacent to the back of the curb. Pull boxes adjacent to light standards shall be placed along the side of the foundations as shown on the plans.

2.08 ELECTRICAL SYSTEM PROTECTION

(A) Introduction

(1) Circuit overcurrent protection shall be provided for all circuits used for pedestrian and street lighting circuits per Article 240 (Overcurrent Protection) of the 2023 NFPA 70 - National Electrical Code for protection of conductor, device, and life safety of the public and maintenance personnel.

(B) Protection Types

(1) ESL point of service protection.

(a) Above grade

(i.) Overhead fed service to street lighting on wooden utility pole, NEMA 3R fusible disconnect switches and fuses on wooden utility pole at the start of the multiple streetlights overhead fed service. See detail PS-01 for mounting height and additional information.

(ii.) Direct bury non-metered service pedestal with service entrance rated main breaker for underground fed service to multiple streetlights. See details PS-02 and PS-05 for additional information.

(b) Below grade submersible fuse holders and fuses in flush in-grade pull boxes for underground fed service to multiple streetlights. See details PS-03 and PS-04 for additional information.

(2) Individual luminaire protection

(a) Overhead fed luminaire on wooden pole, NEMA 3R fusible disconnect switches and fuses on wooden utility pole with streetlight and mast arm on pole.

(b) Underground fed light standard, submersible rated, breakaway 2-pole fuse holder with fuses.

(i.) 120V circuit, (1) fuse for the hot conductor and (1) dummy fuse for the neutral conductor installed in the hand hole of the pole or transformer breakaway base.

(ii.) 240V circuit, (2) fuses for the (2) hot conductors installed in the hand hole of the pole or transformer breakaway base.

(B) In Handhole of Light Pole Protection Bill of Material

(1) Provide breakaway 2-pole, 250V rated, submersible, in-line fuse holder for 120V class CC fuses basis of design as follows:

(a) Fuse holder - Eaton/Bussman #HEX-AW-DRLC-A for phase and neutral conductors.

(b) Fuse - Eaton/Bussman FNQ-15 (1) required.

(c) Dummy fuse - Eaton/Bussman #NNB for neutral conductor (1) required.

(C) Flush In-grade Pull Box Point of Service Protection Bill of Material

(1) Provide non-breakaway, 60A, 250V rated, submersible, in-line fuse holder with set-screw lugs to fit 12-3 AWG CU & 12-2 AWG AL conductors for protection of underground point of service, basis of design:

(a) Fuse holder copper - Eaton/Bussman #HEJ-JJ for copper phase conductors' size 12-3 AWG. (2) required

(b) Fuse holder aluminum - Eaton/Bussman #HEJ-LL for aluminum phase conductors' size 12-2 AWG. (2) required

(c) Insulating boot - Eaton/Bussman #2A0660 for phase conductors. (2) required per fuse holder, (4) total.

(d) Fuse - Eaton/Bussman #SC-60. (2) required.

(D) Above Grade Pedestal Point of Service Protection Bill of Material

(1) Provide non-metered, NEMA 3R, direct bury, power pedestal, 100A (minimum), 120/240V-1PH-3W service entrance rated, with 60A-100A main breaker, full size neutral and ground bus, for protection of above ground point of service, basis of design:

(a) Pedestal - Milbank #U5949 or approved equal.

(b) Either 60A main breaker – 60A-2P, 250V or 100A main breaker – 100A-2P, 250V

(c) Label - label to read "BOULDER ESL", label shall be per ASTM D4956, Class 4

(d) Provide neutral to ground service entrance bonding.

(E) Overhead Fed Lighting Point of Service Protection Bill of Material

(1) Provide heavy duty, NEMA 3R, 60A-2P or 100A-2P, 120/240V-1PH-3W, service entrance rated, fusible safety switch with fuses, full size neutral and ground bus, for protection of overhead lighting point of service, basis of design:

(a) 60A safety switch - Square D #H222NRBCLR or approved equal.

i. Fuses- FRN-R-60 fuses. (2) required.

(b) 100A safety switch - Square D #H223NRBCLR or approved equal.

i. Fuses- FRN-R-100 fuses. (2) required.

(c) Provide neutral to ground service entrance bonding.

(d) Mounting - surface mount on wooden pole at +18'-0" above finished grade to the bottom of the switch.

(e) Label - shall be per ASTM D4956, Class 4 and read "BOULDER ESL

CHAPTER 3 ENERGY ONLY STREET LIGHTING

3.01 ENERGY ONLY STREET LIGHTING

(A) Introduction

(1) City of Boulder engaged with Xcel Energy to acquire their flat rate street lighting and agreed to an Energy Only Street Lighting (ESL) rate for all non-metered street lighting in the city. The acquisition was completed in 2024, and initial fusing (15A in the handhole to suffice for temporary fusing requirements) and LED conversions were completed through 2025 and 2026. The descriptions here describe the changes needed to accommodate these changes.

(B) Existing Electrical Infrastructure for Streetlights

(1) Point of service types for existing overhead fed streetlights (details PS-01, PS-03, PS-05)

(a) Single luminaire on non-distribution utility wooden pole.

(i.) Install NEMA 3R service entrance rated disconnect on pole, surface mounted on pole at 18'-0" above finished grade.

(ii.) Provide rigid conduit to weather-head at incoming service conductors.

(iii.) Provide rigid conduit to weather-head at secondary lighting system wiring.

(b) String of luminaires on non-distribution utility wooden pole.

(i.) Install NEMA 3R service entrance rated disconnect on first pole of string, surface mounted on pole at 18'-0" above finished grade. NOTE: Downstream installation of pull boxes (not at the point of service) shall not be required for existing string of light standards in this situation. However, pull boxes are required for new underground-fed light standard installations. See Section 3.01,(D),(2),(b) for this example of required pull box adjacent to new light standard installations.

(ii.) Provide neutral to ground service entrance bonding in this disconnect.

(iii.) Provide 3/4" x 8'-0" copper clad driven ground rod. Ground rod shall be buried a minimum 6 inches from and adjacent to the wooden pole where disconnect is installed.

(iv.) Provide #6 solid bare copper ground conductor in 1/2" GRC, 2-hole strapped to the wooden pole max. 5'-0" on center from the disconnect neutral bus to the ground rod at the base of the pole.

(v.) Exothermic weld or underground rated lug must be utilized to connect #6 Soft Drawn Bare Copper Ground to the ground rod.

(vi.) Provide rigid conduit to weather heads at incoming service conductors.

(c) Provide rigid conduit to weather-head at lighting system wiring. Single luminaire on distribution utility wooden pole.

(i.) Install new wooden pole direct buried 10’-0” from distribution pole. Refer to detail PS-01.

(ii.) Provide service lateral from distribution pole to new light standard with NEMA 3R disconnect on new light standard. Provide rigid conduit from disconnect to weather-head at service conductors and from disconnect to weather-head at lighting system wiring.

(iii.) Provide 3/4" x 8'-0" copper clad driven ground rod. Ground rod shall be buried a minimum 6 inches from and adjacent to the wooden pole where disconnect is installed.

(iv.) Provide #6 solid bare copper ground conductor in 1/2" GRC, 2-hole strapped to the wooden pole max. 5'-0" on center from the disconnect neutral bus to the ground rod at the base of the pole.

(v.) Exothermic weld or underground rated lug must be utilized to connect #6 Soft Drawn Bare Copper Ground to the ground rod.

(2) Point of service types for existing underground service to the lighting system (details PS-02, PS-04)

(a) Single light standard on underground feeder from transformer or Xcel service pedestal to light standard.

(i.) Provide flush in-grade point of separation box with submersible fuse holders, fuses, and intercept existing below grade conduit and conductors from transformer/pedestal to existing light standard. Contractor shall install point of service separation pull boxes as close as practically possible to the Xcel source.

(ii.) Route conductors in existing conduit from the new point of separation box to the existing light standard.

(iii.) Maintain Xcel standard ground rod, fuse holders, and fuses at pole handhole.

(b) String of light standards on underground feeder from transformer or Xcel service pedestal to underground lighting system.

(i) Provide flush in-grade point of separation box with submersible fuse holders, fuses, and intercept existing below grade conduit and conductors from transformer/pedestal to the first existing light standard in the string. Contractor shall install point of service separation pull boxes as close as practically possible to the Xcel source.

(ii) Route conductors in existing conduit from the new point of separation box to the first existing light standard in the string.

(iii) Maintain the Xcel standard underground feeders at the remaining light standards on the string.

(iv) Maintain Xcel standard ground rod, fuse holders, and fuses at each light standard.

New Electrical Infrastructure for Streetlights

(1) Point of Service to new lighting infrastructure:

(a) Provide new underground feeder from existing/new pad mounted transformer or ground mounted service pedestal with 5'-0" coiled conductors at transformer and routed out to the first light standard.

(b) Provide new flush in-grade point of separation box with (2) 4-pole submersible phase connectors, (1) 6-pole ground/neutral submersible connector, (2) fuse holders, and fuses. Contractor shall install point of service separation pull boxes as close as practically possible to the Xcel source.

(c) Provide new (2) phase conductors (or (1) phase conductor in situations with two or less lights), (1) neutral conductor and (1) raceway ground conductor feeder from the point of separation box to the string of light standard. For lighting strings with three or more lights, phase conductors shall alternate L1 and L2 to every other light standard in the string with a pull box adjacent to each light standard.

(d) Provide (2) 4-pole submersible phase connectors, (1) 4-pole submersible neutral connector, and (1) 6-pole submersible ground connector in each pull box adjacent to the light standard. Provide (1) 3/4" x 10'-0" copper clad driven ground rod in each of these pull boxes with 1#6 green copper conductor from ground rod and the submersible ground connector. Provide 1#6 green copper conductor from the submersible ground connector to the ground lug in the back of the light standard handhole. Route this ground conductor, phase conductor and neutral conductor through the light standard feeder from the pull box to the light standard. 2#12 copper conductor (one white and one black) with the 1#6 green ground conductor to each light standard.

(e) Provide (1) 2-pole, submersible, breakaway fuse holder with (1) FRN-15-amp fuse for hot conductor and (1) dummy fuse for the neutral conductor. Connect the one #6 green ground conductor to the lug in the light standard hand hole.

(C) Bill of Materials for Overhead and Underground Fed Streetlight Point of Delivery

(1) Electrical separation of lighting from existing overhead utility infrastructure to overhead fed lighting bill of material:

(a) Heavy-Duty, 60Amp, 250VS, 2-pole, NEMA-3R, fusible disconnect switch with separate neutral & ground bars bonded as service entrance equipment. Phenolic label to read: "BOULDER ESL" This disconnect shall be installed on the first wooden pole light standard in a string of overhead fed light standards. Disconnects shall only be installed on city owned light standards.

(b) FRN-R-60 type fuses sized to the load side conductor size if different than #4 CU.

(c) 2-inch Galvanized Rigid steel Conduit (GRC) length as required to make incoming and outgoing raceways.

(d) 2-inch aluminum weather head service entrance clamp. (2) required, (1) incoming line and (1) outgoing load.

(e) 3/4" x 8'-0" copper clad driven ground rod. Ground rod shall be buried a minimum 6 inches from and adjacent to the wooden pole where disconnect is installed.

(f) #6 solid bare copper conductor stapled to wooden pole 3'-0" on center from the disconnect to the ground rod.

(g) ACSR #2 Duplex “Sparrow” for incoming line and outgoing load to other wooden pole mounted streetlights. Limit span length to 200’ ideally, or 250’ max with special requirements.

(h) Provide (2) H-tap compression crimp connectors to terminate the (1) hot and (1) neutral/ground conductors from the disconnect to the overhead ACSR cable for both line and load wiring, (6) connectors needed per disconnect switch. 2-8 AWG run and tap, if run cable is larger than #2 AWG contractor shall upsize H-tap to match run cable size.

(2) Electrical separation of lighting from existing overhead utility infrastructure to underground fed lighting bill of material:

(a) Provide a Tier 22 rated, 11" x 18" x 12"dp. flush to grade, bottomless, polymer concrete splice/pull box as the point of service with a minimum of 12 inches deep of 3/4" diameter rock or crashed road base below box for drainage. Cover shall be engraved: "BOULDER ESL".

(i.) Provide 60A, 250V, submersible fuse holders with 60A fuses, insulating boots and fuses, (2) of each required.

(ii.) Provide (3) 4-terminal and (1) 6-terminal, submersible, insulated lug connectors.

(iii.) Provide 3/4" x 10'-0"Lg. copper clad driven ground rod. Exothermic weld or underground rated lug connect conductors to ground rod.

(iv.) Provide the conductors as listed in the "CONDUCTOR SIZING PER NUMBER OF LUMINAIRES UNDERGROUND FED TABLE”. Refer to **Section 2.06(B)**. These conductors shall be in 2" PVC secondary conduit to each streetlight.

(v.) Provide conductor sizes as listed in the "CONDUCTOR SIZING PER NUMBER OF LUMINAIRES UNDERGROUND FED TABLE" in a 2" PVC conduit to the overhead utility pole and coil enough conductors at the base of the pole to reach the transformer terminals or serving secondary at the top on the pole. Conductors routed up the pole and attachments to the transformer shall be installed by Xcel Energy's Forces. City streetlighting will provide complete bill of materials for connection, including coiled conductors, U-guards, and any applicable materials for Xcel to install on their infrastructure

(vi.) Contractor shall install point of service separation pull boxes as close as practically possible to the Xcel source.

(b) At each underground fed streetlight provide a Tier 22 rated, 11" x 18" x 12"dp. flush to grade, bottomless, polymer concrete splice/pull box with a minimum of 12 inches deep of 3/4" diameter rock or crashed road base below box for drainage, mount adjacent to streetlight foundation.

(i.) Provide (3) 4-terminal, (1) 6-terminal submersible, insulated lug connectors, (2) hots, (1) neutral shall be the 4-terminal type and (1) 6-terminal type for the ground. Splice/pull box can be deleted if only one streetlight is served by this point of service, route 2-inch conduit directly into light standard foundation.

(ii.) Provide 3/4" x 10'-0" copper clad driven ground rod in each splice/pull box or 6 inches from light standard foundation if only one streetlight is served by this point of service.

(iii.) Provide 2-inch PVC conduit from splice/pull box to the light standard at each streetlight with 2#12THWN CU & 1#6 CU Ground. Route #6 solid bare copper ground conductor from the ground lug connector to the ground rod and from the ground rod to the ground lug at the back of the handhole of the streetlight pole. If this is only one streetlight served by this point of service, then provide 1#6 solid bare copper ground conductor from the light standard to the ground rod in 1/2" conduit.

(iv.) Provide breakaway submersible fuse holder for each hot conductor or breakaway submersible connector if one of the conductors is a neutral conductor in the handhole of the streetlight pole.

(3) Electrical separation of lighting from existing underground utility infrastructure to underground fed lighting bill of material:

(a) Provide a Tier 22 rated, 11" x 18" x 12"dp. flush to grade, bottomless, polymer concrete splice/pull box as the point of service with a minimum of 12-inches deep

of 3/4" diameter rock or crashed road base below box for drainage. Cover shall be engraved: "BOULDER ESL" Upstream cable to power source in this scenario is Xcel Energy triplex cable.

(i.) Provide 60A, 250V, submersible fuse holders with 60A fuses, insulating boots and fuses, (2) of each required.

(ii.) Provide (3) 4-terminal and (1) 6-terminal, submersible, insulated lug connectors.

(iii.) Provide 3/4" x 10'-0"Lg. copper clad driven ground rod. Exothermic weld or underground rated lug connect conductors to ground rod.

(iv.) Provide #4 CU triplex cable or refer to "CONDUCTOR SIZING PER NUMBER OF LUMINAIRES UNDERGROUND FED TABLE" for cable specifications to provide in 2" PVC conduit from this pull box to the Utility transformer or service pedestal. Coil 5'-0" of cable at utility transformer or service pedestal. Xcel Energy (or Xcel Energy Approved Contractors) shall connect cable to utility transformer or service pedestal

(v.) Provide the conductors as listed in the "CONDUCTOR SIZING PER NUMBER OF LUMINAIRES UNDERGROUND FED TABLE". Refer to **Section 2.06(B)**. These conductors shall be in 2" PVC secondary conduit to each streetlight.

(b) At each underground fed streetlight provide a Tier 22 rated, 11" x 18" x 12"dp. flush to grade, bottomless, polymer concrete splice/pull box adjacent to streetlight foundation with a minimum of 12-inches deep of 3/4" diameter rock or crashed road base below box for drainage.

(i.) Provide (3) 4-terminal, (1) 6-terminal submersible, insulated lug connectors, (2) hots, (1) neutral shall be the 4-terminal type and (1) 6-terminal type for the ground.

(ii.) Provide 3/4" x 10'-0" copper clad driven ground rod in each splice/pull box.

(iii.) Provide 2-inch PVC conduit from splice/pull box to the light standard at each streetlight with 2#12THWN CU & 1#6GND. Route #6 solid bare copper ground conductor from the ground lug connector to the ground rod and from the ground rod to the ground lug at the back of the handhole of the streetlight pole.

(iv.) For modifications to the existing system, additive pull box installations at each pole shall not be required aside from the separation/delivery point.

(c) Provide breakaway submersible 2-pole fuse holder with fuse for each hot conductor and dummy fuse if one of them is a neutral conductor in the handhole of the streetlight pole.

CHAPTER 4 STRUCTURAL AND CIVIL REQUIREMENTS

4.01 STRUCTURAL NOTES AND REQUIREMENTS

(A) General Structural Notes:

(1) Structural design and details for these foundation standards were taken from CDOT Safety Standard S-613-1 (dated 05/01/2020). Pursuant to Note 7, Sheet 3 of 6, foundation embedment shown herein have been increased by 12” to account for the special wind region.

(2) CDOT Standard S-613-1 assumed soil properties:

(a) Soil Density = 110 lbs./CF

(b) Soil Cohesion = 750 lbs./Sq Ft for medium stiff cohesive soil

(c) Soil Angle = 30° for medium dense cohesionless soil

(d) Resistance Factor = 0.40 for flexure

(3) Details shown herein for non-breakaway installations shall not be used within the limits of the roadway clear zone. Clear zone is defined as the traversable roadside area, beginning at the edge of traveled way, provided for the recovery of errant vehicles. The required clear zone width shall be established by the project roadway designer in accordance with the current edition of the AASHTO Roadside Design Guide. For installations within the roadway clear zone, refer to CDOT Standard S-613-1 for breakaway base requirements and minimum offset requirements.

(4) When using standard foundations, the designer shall verify that the Effective Projected Area (EPA) of the specified equipment does not exceed the assumed EPA for the selected foundation detail.

(B) Design Data:

(1) Concrete for luminaire foundations shall be air-entrained CDOT Class BZ per Section 601 (Structural Concrete) of the 2025 (Version 1) CDOT Standard Specifications for Road and Bridge Construction.

(2) Reinforcing steel shall be Grade 60 per Section 602 of the CDOT Standard Specifications for Road and Bridge Construction.

(C) Non-Standard Installations:

(1) Light standards with multiple luminaires, or light standards including banners or other attachments, shall be designed by a structural engineer licensed in the State of Colorado.

(2) Light standards in locations with poor soil conditions shall be designed by a structural engineer licensed in the State of Colorado. Examples of these locations include:

(a) Soils with a high organic content or saturated silt or clay.

(b) Sites that won't support the weight of the drilling rig.

(c) Non-homogeneous foundation soils.

(d) Firm bedrock.

(3) Where site-specific foundations are required, details and calculations shall be submitted to the Project Manager for review prior to construction.

(4) The following shall apply to site-specific foundation designs:

(a) Design shall be completed in accordance with the 2015 (First Edition with 2022 Interim Revisions) of the AASHTO LRFD Specifications for Structural Supports for Highway Signs, Luminaires, and Traffic Signals (AASHTO LTS).

(b) Design wind speed (3 second gust) shall be in accordance with the current City of Boulder adopted code and amendments. Base wind speed for analysis shall be taken for the appropriate wind zone from Table 1609.3. Retrieved and reprinted here for convenience (retrieval date: August 3, 2026).

Basic Design Wind Speed (V) in MPH according to Wind Zone and Building Risk Category			
Wind Zone	**Risk Category I**	**Risk Category II**	**Risk Category III and IV**
East of Broadway	140	150	160
West of Broadway	155	165	175

(c) The Effective Projected Area (EPA) shall be taken as the effective projected area normal to the wind of the complete pole assembly. The designer shall obtain EPA values from the pole and luminaire manufacturer's published data (or sealed calculations provided by the manufacturer).

4.02 LIGHTING LOCATION PARAMETERS

(A) Introduction

(1) Final luminaire placement shall be determined based on site conditions, roadway or pedestrian context, code requirements, and utility clearances. Lighting placement should be informed by current Illuminating Engineering Society (IES) recommendations, applicable utility requirements, and accessibility considerations.

(B) Luminaire Orientation and Placement

(1) Luminaires shall be located and oriented to provide illumination within the Right of Way, in such a way to minimize glare, light trespass, and uplight. Mast arms shall be aligned perpendicular to the direction of the roadway being illuminated. Luminaires should be positioned to avoid conflict with trees, signage, and other site features. All luminaires shall be installed level and plumb such that the light source is horizontal.

(2) Luminaire locations and pole placements shall maintain required clearances from overhead and underground utilities in accordance with Xcel Energy requirements (details LD-12 and LD-13) and other applicable utility standards. Coordination with utility providers is required where lighting is located near power lines, transformers, vaults, or service connections. Pole foundations, handholes, and underground conduits shall be located to avoid conflicts with existing or proposed utilities and to maintain access for maintenance. Luminaires shall be located outside of sight triangles.

City specific clearance requirements are also listed in the DCS. The 2025 DCS version has the following sections noting clearances. Section 4.06 Separation of Utilities has the horizontal and vertical separation requirements. This includes utility separation of water, sewer, storm, other electric/gas/telecom assets, and trees. DCS Section 2.12 Street Lighting has information regarding clearances for back of curb. DCS Section 2.10 Emergency Access Lanes has information regarding emergency vehicle clearance requirements.

(3) Lighting placement within pedestrian areas shall comply with Americans with Disabilities Act (ADA) requirements. Poles and foundations shall be located outside of pedestrian circulation zones and maintain required clear widths of 5’-0” minimum along sidewalks, curb ramps, and shared-use paths. Lighting equipment shall not reduce accessible routes or create obstructions for pedestrians.

(C) Luminaire Spacing Guidance

(1) Luminaire spacing guidance is provided as an initial planning tool and rough gauge for evaluating appropriate streetlight spacing based on typical multiples of luminaire mounting height. These spacing ranges are not a substitute for project-specific lighting design. All new street lighting installations shall be designed and verified to meet the applicable Illuminating Engineering Society (IES) RP-8 lighting criteria, including required light levels, uniformity, glare control, and other relevant performance requirements.

(2) Streetlight and pedestrian light spacing on arterial streets with continuous lighting should be based on mounting height, street width, lumen output, and light distribution. Generally:

(a) Streetlights are typically spaced at approximately 5 to 7 times the mounting height, depending on roadway width, pole setback, and light distribution type (Type II, III, or IV).

(b) Pedestrian and post-top luminaires are typically spaced at approximately 5 to 6 times the mounting height.

(3) Continuous lighting is not required/warranted on local and collector streets in residential areas with speed limits under 30 mph.

CHAPTER 5 UTILITY COORDINATION PROCEDURES AND PROTOCOLS

1. Utility Data Collection and Record Research

Objective: Establish baseline utility information and initiate utility owner communication.
Project Timeline: At onset of project through preliminary design.

- **1.1 Obtain Record Maps and Existing Utility Information**
  - Submit an Excavator Single Address Request via 811 Colorado (*Work Type: Engineering)*. On larger projects, dividing the project area into smaller segments and submitting multiple 811 requests may help improve the efficiency and clarity of record responses.
  - Receive facility and mapping data from all utility providers within the requested limits, including underground and overhead electric, gas, fiber, and small-cell infrastructure.
    - **Note:** 811 does not contain private utility information, such as a property's utility connection off a main service line. If applicable, this should be coordinated with the affected landowner.
  - Utilize the City of Boulder GIS database to obtain all available utility and infrastructure records.
  - Identify what SUE Quality Level is required for this effort. Refer to the 2022 edition of ASCE 38-22 (Standard Guideline for Investigating and Documenting Existing Utilities, published by the American Society of Civil Engineers) for guidance on quality levels and criteria for selecting the appropriate level. Quality Level C generally requires survey, while B and A generally require Subsurface Utility Engineering (SUE) services.

> A field walkthrough of the project area can help identify unique circumstances early on, e.g. overhead utility poles with multiple parties on them.

- **1.2 Identify and Document Utility Contacts, Contextual Data Collection**
  - Contact all utility providers noted on the 811 response to introduce the project and establish an engineering point of contact for coordination. Many utility owners include direction on their coordination process in their 811 response, or have a uniform process:
    - For Xcel, establish an engineering point of contact through the Builder's Call Line by submitting an "Apply for Service" request.
    - For Lumen/Centurylink, establish an engineering point of contact through their online relocation request portal.
    - Identify a City of Boulder point of contact through the 811 process with Traffic Operations, Lighting, Water, Stormwater, and Wastewater.
  - Document utility contact information, including names, phone numbers, email addresses, and any other pertinent details that arise from introductory conversations or site walks.
  - Identify existing Right of Way (ROW), utility easements and agreements related to utilities in the project limits.

- Assess existing infrastructure to identify which utility owners have attachments on each pole or structure. This may include electric, telecom, fiber, small-cell carriers, and any City-owned equipment. Document joint-use or multi-party attachments for each pole or structure, noting ownership, attachment type, and established coordination procedures.
  - Specifically, work with City of Boulder Lighting to confirm applicable terms in active agreements, which may include:
    - Street Light System Purchase and Sale Agreement (2024)
    - Franchise Agreement Between City of Boulder and Xcel
    - Small Cell Facilities in Public Right-Of-Way Terms and Conditions
    - CenturyLink Pole Attachment Agreements with City of Boulder
    - Transition Agreement between City of Boulder and Xcel
    - AT&T Small Cell Attachment Agreement

Deliverables:

- Existing Utility Design File - Compile record maps and GIS base layers into CAD or other referenceable format for project design plans.
- Records and Locations of Existing Utility Easements and Agreements
- Utility Owners and Contact List
- Correspondence Tracker – To document anecdotal information received from utility contact to inform design.

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2. Create Utility Conflict Matrix and Assess Project Conditions

Objective: Identify conflicts, determine responsibilities, and document constraints.
Project Timeline: Following data collection and initiation of preliminary design.

- **2.1 Create and Populate Utility Conflict Matrix**
  - Mark all known facilities within project limits and highlight critical avoidances. Document potential relocations comparing the existing utilities to the proposed preliminary design and organize them in a conflict matrix.
  - When reviewing conflicts, it is important to also consider overhead utilities and the proximity of proposed infrastructure and equipment during construction.
- **2.2 Determine Project Constraints and Responsibilities**
  - Determine if the project or the utility owner is responsible for the cost of relocation on each utility in conflict.
  - Determine roadway jurisdiction and any federal funding triggers.
  - Confirm needs for easement replacements or encroachment agreements.
    - Use existing agreements and coordination to determine which utility owners must be coordinated with, and whether attachment agreements or joint-use permits affect the project schedule.

- **2.3 Utility Service Component – For Projects anticipating new or revised utility service needs:**
    - Identify the point of separation (POS) (point of delivery (POD)), as well as the installation type and desired tariff rate (Energy Only Streetlighting (ESL), Metered Intersection (MI), Metered Street Lighting (MSL)) for any existing infrastructure present. Coordinate with City of Boulder Lighting on the options available to the project with the existing service infrastructure.
        - Document whether the POS/POD is located at a pole, splice box, cabinet, transformer, handhole, etc.
    - Review utility record maps and City of Boulder GIS database to identify conflicts with any underground utilities that may conflict with the proposed service route.
    - Capture the Contractor's responsibilities in coordinating with Xcel and City of Boulder for new or modified service requests. See section 6.3 for best practices.

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3. Coordination

Objective: Establish recurring communication with utility owners and City of Boulder Lighting to define clear responsibilities, timelines, and actions for relocations and conflict mitigation that support the project schedules. Communication methods (phone, email, or meetings) should be selected based on what best supports the project's needs and level of complexity.

Project Timeline: Initiate following Data Collection and maintain through project design.

- **3.1 Conduct Initial Coordination**
    - Review project scope, schedule, existing company-owned facilities and any existing agreements (franchise, access permit, utility permit, etc.) with utility owners.
        - Present the assessment of pole attachments to each utility owner and confirm attachment ownership, joint-use status, and any pending permit obligations between utilities.
        - Verify POS and POD with each utility to establish service boundaries and the corresponding responsibility for work and coordination.
    - Identify potential undergrounding opportunities under the 1% Underground Program (Xcel-specific).
- **3.2 Follow Up Coordination**
    - Review potential conflict locations and confirm whether conflicts require avoidance, adjustment, or relocation. Consult accordingly with the design team to refine design based on utility provider feedback.
    - Update project schedules, including the utility relocation timeline, design milestones, and construction windows.
        - For each potential conflict, verify:
            - The party responsible for relocation costs, including reimbursement eligibility.
            - The party responsible for obtaining easement replacements, if applicable.
            - The party responsible for performing the construction.
            - Any required notice periods that utilities must make to customers ahead of service cut-overs.
            - Any moratorium periods for utility owners where no cutovers are permitted (example: holiday season)
            - Anticipated design, approval, mobilization, and construction timelines.

- For conflicts involving poles or shared structures, confirm which utilities have attachments and identify any required sequencing (e.g., which utility must detach or relocate first based on owner/operator requirements).
- Document dependencies between utility parties for attachment removals or relocations.

- **3.3 Utility Service Component (New or Revised Service)**
  - Coordinate with City of Boulder Lighting on potential power source options. Depending on the scope of the project, it may be possible to adjust within City of Boulder's infrastructure, or an additional service from Xcel may be required. For example, it is not allowed to add service for additional lighting onto an MI installation.
  - Confirm service requirements and billing type (ESL, MSL, MINMTR). ESL is generally used unless there are future plans to add additional electrical components, such as irrigation or holiday lighting, in which case NMTR or MSL may be most applicable
  - Establish maintenance and power service responsibilities for any proposed lighting configuration, confirming which party owns, maintains, and energizes each component of the system.

Initiate coordination with City of Boulder Lighting early in the process to allow time to determine the appropriate service need for your project.

Deliverables:

- Meeting minutes, including decisions and action items.
- Action log with responsible entity and deadlines.
- Relocation schedules
- Preliminary Utility Conflict Matrix

4. Preliminary Design Deliverable

Objective: Establish initial design framework and utility conflicts.
Project Timeline: Preliminary Design Milestone, often 30%.

- **4.1 Refine the Utility Conflict Matrix**
  - Identify preliminary resolution strategies for anticipated conflicts.
  - Confirm vertical and horizontal separation requirements are achieved per Section 4.06 (Separation of Utilities) of the 2024 edition of the City of Boulder Design and Construction Standards.
- **4.2 Develop Preliminary Utility Plan Sheets or Information**
  - Depict all known utilities with appropriate quality-level callouts per the 2022 version of ASCE 38-22 (Standard Guideline for Investigating and Documenting Existing Utilities, published by the American Society of Civil Engineers).

    Utility conflict information may be included on other discipline plan sheets if standalone utility relocation plans are not warranted.

  - Show existing easements, and any required new easements or permit areas.
  - Show proposed infrastructure locations and identify any necessary utility adjustments or relocations along with the party responsible.
- **4.3 Prepare Preliminary Cost Estimates**
  - Prepare preliminary cost estimates that account for expected utility relocation and adjustment costs, ROW acquisition or permitting costs, and document project risks that may influence schedule or budget.
- **4.4 Review Preliminary Design Package with the Utility Owners**
  - Distribute the utility conflict plan information to utility owners.
  - Review initial conflicts and proposed relocations with utility owners to discuss risks and confirm relocation is constructable and can meet the project schedule.
  - For utilities requiring easement replacements to be acquired by the project, review proposed easement linework.
    - ROW needs are typically defined early in the design process. If the project is acquiring easements for utility providers, close coordination is needed to commit to the easement replacement limits prior to final design.

Deliverables:

- Preliminary Utility Conflict Plans
- Preliminary Utility Conflict Matrix
- Preliminary Cost Estimate
- SUE Plans and Report (if applicable)
- Quality Level A Testhole Investigation Needs
- ROW Easement Acquisition Linework for Utility Relocations

5. Final Design Deliverable and Clearances

Objective: Finalize design documents, verify utility locations through pot holes performed by SUE, resolve all remaining conflicts, and secure all utility clearances required for permitting, advertisement, and construction.

Project Timeline: Following Preliminary Design through Final Design milestone.

- **5.1 Coordinate and Complete Utility Investigations (Pot holes)**
  - Identify conflict locations where additional SUE investigation (QL-A) is required to support final design.
  - Direct the SUE provider to perform pot holing to verify horizontal and vertical utility locations as required, confirm clearances, and resolve discrepancies.

    SUE Law includes specific requirements for when QL-A certified testholes are required.

    - Colorado SUE law dictates the situations when QL-A pot holes are required on a project. Due to the certification required for QL-A, they can be expensive. For projects with several pot holes anticipated, consider using a utility investigation company for non-QLA pot holes where possible to save budget.
  - Coordinate pot hole work with utility owners to avoid damage and encourage successful identification.
  - Incorporate pot hole results into utility plans, the Utility Conflict Matrix, and design decisions.
- **5.2 Finalize the Utility Conflict Matrix, Utility Plan Sheets and Cost Estimate**
  - Incorporate pot hole findings, preliminary design review comments, and all final design updates.
  - Update the final cost estimate to reflect confirmed relocations and adjustments, ROW acquisition and permitting needs.
- **5.3 Prepare Utility Specifications**
  - Develop and refine project specifications related to utility work, including trenching, boring, protection of existing utilities, clearance requirements, and in-construction coordination.
  - Include utility-owner standards and agency requirements as applicable.
    - Coordinate with utility owners to include any conflict-specific notes or requirements.
  - Ensure consistency between specifications, utility plan sheets, the Utility Conflict Matrix, and the Utility Clearances discussed below. The specifications are the place to transfer project commitments and requirements during design to the contractor doing the project work.
- **5.4 Obtain Utility Clearances**

  This step should occur after construction funding is confirmed and the project schedule is defined, so utility owners receive accurate timelines and commitments.

  - Draft and finalize utility letter agreements for identified relocations and adjustments, clearly defining roles, responsibilities and timelines.
  - Submit utility letter agreements to the appropriate utility owners for signature and request signed versions for the final package.

Review utility clearance letters against the utility specification to ensure they are consistent.

    - For CDOT oversight projects, prepare and submit Utility Clearance Requests to the Region's Utility Manager.
    - Obtain formal utility clearances (e.g., no-conflict confirmation, relocation commitments, or completed relocation documentation) from each affected utility owner.
- **5.5 Utility Service Component (New or Revised Service)**
    - Finalize utility documentation and asset records.
        - If the project involves a new or relocated service, prepare materials for GIS asset management, including new service points, identification, and facility attributes.

- **5.6 Review Final Design Package with Utility Owners**
  - Review with utility owners to confirm final details and relocation plans.
  - Verify utility owner design, approval, material procurement, and construction timelines.

Deliverables:

- Final Utility Conflict Plans
- Final Utility Specifications
- Final Utility Conflict Matrix
- Final Cost Estimate
- SUE Plans, Report, and Pot hole Logs
- CDOT Utility Clearance (if applicable)
- Final Concurrence on Utility Clearances with Owners

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6. Construction Implementation and Documentation

Objective: Oversee utility relocations, construction activities, and energization/cut-over efforts, ensuring all work is coordinated, documented, and restored to required standards.
Project Timeline: During construction and post-construction through closeout.

- **6.1 Relocation Work**
  - Issue the Notice to Proceed (NTP) only after confirming that all required utility relocations are complete or that a relocation schedule has been formally accepted.

    > Stake key points like ROW limits and relocation alignment to walk through at pre-con meetings with utility contractors can help ensure relocations are done once.

  - Confirm Xcel Energy's de-energization, switching, and safety procedures for any work impacting electric facilities.
  - Hold a pre-construction meeting with the contractor and any required representatives to review final utility specifications, remaining utility considerations, traffic control requirements, and site access needs.
- **6.2 Best Practices During Construction**
  - Ensure a smooth handoff of design assumptions, utility information, and coordination commitments from pre-construction to construction teams.
  - Conduct recurring (typically weekly) coordination meetings between the City of Boulder, contractor, utility owners and field staff.
  - Monitor any adjustments required due to unforeseen field conditions.
  - Document all field modifications, utility adjustments, unforeseen conflicts, and changes to agreed-upon relocation strategies.
  - Ensure all disturbed areas are restored in accordance with "as good or better condition".
- **6.3 Best Practices When Coordinating New or Modified Service Requests**
  - For metered installations:

- Ensure all final meter locations are clear, and that each new or modified meter has an assigned address from City of Boulder Planning and Development.
- All applicable City of Boulder meter inspections need to be complete prior to scheduling the Xcel representative work through the Xcel Builder's Call Line. Follow City of Boulder permitting guidelines and plan to be onsite during the inspection to resolve any issues.

- Similarly, plan to have materials and electrical contractor onsite when Xcel performs their work to adapt to changing circumstances and complete the work.
- Work with the billing team to categorize and ensure bills are tied to the exact address and infrastructure is entered into GIS.
    - All infrastructure, cable, conduit, pullboxes, poles, and connections to the utility system shall be tracked and entered

Make sure billing is transferred from Contractor to City of Boulder once new service lines are installed.

- For non-metered installations:
    - Complete all work leading up to the transformer that the service will be provided from prior to the Xcel representative coming onsite. Do not open or modify the transformer as that is Xcel's property. The Xcel representative will open the transformer and observe the connection work, then perform any required terminations.
    - Coordinate with City of Boulder lighting to add the new service line and asset to its GIS System as well as its non-metered inventory with Xcel.
        - All infrastructure, cable, conduit, pullboxes, poles, and connections to the utility system shall be tracked and entered
- If any connections require work on an Xcel distribution pole, an Xcel-approved electrical contractor must perform the work. This is typically conducted by a Builder's Call Line request to Xcel for them to conduct the work and bill the City.
- For all service requests, record the location of the point of delivery/separation for as-built documentation.

- **6.4 Post-Construction Documentation**
    - Obtain as-builts from utility owner for relocated utilities and/or Contractor for work performed as part of the project. As-builts should include attributable GIS information and geo-referenced photos of exposed subsurface infrastructure to the extent practicable.
        - For projects with new or revised COB utility service needs, as-built documentation should include such details as Xcel-serving equipment, POS and POD locations, and all downstream asset information including pullbox, pole, cable, and conduit information. Coordinate with City of Boulder Lighting and GIS teams to update assets and GIS records.
    - Record any agreements or easements with the appropriate party (often County Assessor).

APPENDIX A - NATIONAL ELECTRIC CODE (NEC) REFERENCES

NEC Article 210.19 Conductors – Minimum Ampacity and Size

(A) General.

Branch-circuit conductors shall have an ampacity not less than the larger of the following and comply with 110.14(C) for equipment terminations:

(2) Where a branch circuit supplies continuous loads or any combination of continuous and noncontinuous loads, the minimum branch-circuit conductor size shall have an ampacity not less than the noncontinuous load plus 125 percent of the continuous load in accordance with 310.14.

Exception to (1): If the assembly, including the branch-circuit OCPDs, is listed for operation at 100 percent of its rating, the ampacity of the branch-circuit conductors shall be permitted to be not less than the sum of the continuous load plus the noncontinuous load in accordance with 110.14(C).

(3) The minimum branch-circuit conductor size shall have an ampacity not less than the maximum load to be served after the application of any adjustment or correction factors in accordance with 310.15.

Exception to (1) and (2): Where a portion of a branch circuit is connected at both its supply and load ends to separately installed pressure connections as covered in 110.14(C)(2), an allowable ampacity in accordance with 310.15 not less than the sum of the continuous load plus the noncontinuous load shall be permitted. No portion of a branch circuit installed under this exception shall extend into an enclosure containing either the branch-circuit supply or the branch-circuit load terminations.

NEC Article 215.4 (A)(2) Feeders – Minimum Rating and Size - Ampacity Adjustment or Correction Factors

The minimum feeder conductor size shall have an ampacity not less than the maximum load to be served after the application of any adjustment or correction factors in accordance with 310.14.

NEC Article 250.66(A) Guidance for Grounding Electrode Conductors

If the grounding electrode conductor or bonding jumper connected to a single or multiple rod, pipe, or plate electrode(s), or any combination thereof, as described in 250.52(A)(5) or 250.52(A)(7), does not extend on to other types of electrodes that require a larger size conductor, the grounding electrode conductor shall not be required to be larger than 6 AWG copper wire or 4 AWG aluminum or copper-clad aluminum wire.

NEC Article 250.118(A) Types of Equipment Ground Conductors - Permitted

Each equipment grounding conductor run with or enclosing circuit conductors shall be one or more, or a combination, of the following as part of an effective ground-fault current path:

(1) A copper, aluminum, or copper-clad aluminum conductor that is solid or stranded; insulated, covered, or bare; and in the form of a wire or a busbar of any shape
(2) Rigid metal conduit
(3) Intermediate metal conduit
(4) Electrical metallic tubing
(5) Listed flexible metal conduit meeting all of the following conditions:
    a. The conduit is terminated in listed fittings.
    b. The circuit conductors contained in the conduit are protected by OCPDs rated at 20 amperes or less.
    c. The size of the conduit does not exceed metric designator 35 (trade size 1¼).
    d. The combined length of flexible metal conduit, flexible metallic tubing, and liquid-tight flexible metal conduit in the same effective ground-fault current path does not exceed 1.8 m (6 ft).
    e. If flexibility is necessary to minimize the transmission of vibration from equipment or to provide flexibility for equipment that requires movement after installation, a wire-type equipment grounding conductor or a bonding jumper in accordance with 250.102(E)(2) is installed.
    f. If flexible metal conduit is constructed of stainless steel, a wire-type equipment grounding conductor or bonding jumper in accordance with 250.102(E)(2) is installed.
(6) Listed liquid-tight flexible metal conduit meeting all of the following conditions:
    a. The conduit is terminated in listed fittings.
    b. For metric designators 12 through 16 (trade sizes ⅜ through ½), the circuit conductors contained in the conduit are protected by OCPDs rated at 20 amperes or less.
    c. For metric designators 21 through 35 (trade sizes ¾ through 1¼), the circuit conductors contained in the conduit are protected by OCPDs rated not more than 60 amperes and there is no flexible metal conduit, flexible metallic tubing, or liquid-tight flexible metal conduit in metric designators 12 through 16 (trade sizes ⅜ through ½) in the effective ground-fault current path.
    d. The combined length of flexible metal conduit, flexible metallic tubing, and liquid-tight flexible metal conduit in the same effective ground-fault current path does not exceed 1.8 m (6 ft).
    e. If flexibility is necessary to minimize the transmission of vibration from equipment or to provide flexibility for equipment that requires movement after installation, a wire-type equipment grounding conductor or a bonding jumper in accordance with 250.102(E)(2) is installed.
    f. If liquid-tight flexible metal conduit contains a stainless steel core, a wire-type equipment grounding conductor or a bonding jumper in accordance with 250.102(E)(2) is installed.
(7) Flexible metallic tubing, if the tubing is terminated in listed fittings and meets the following conditions:
    a. The circuit conductors contained in the tubing are protected by OCPDs rated at 20 amperes or less.
    b. The combined length of flexible metal conduit, flexible metallic tubing, and liquid-tight flexible metal conduit in the same effective ground-fault current path does not exceed 1.8 m (6 ft).

c. The tubing contains circuit conductors not exceeding 1000 volts, nominal.

(8) Armor of Type AC cable as provided in 320.108 containing circuit conductors not exceeding 600 volts, nominal

(9) Copper sheath of mineral-insulated, metal-sheathed cable Type MI containing circuit conductors not exceeding 600 volts, nominal

(10) Type MC cable that provides an effective ground-fault current path in accordance with one or more of the following:

a. It contains an insulated or uninsulated equipment grounding conductor complying with 250.118(A)(1).

b. The combined metallic sheath and uninsulated equipment grounding/bonding conductor of interlocked metal-tape-type MC cable is listed and identified as an equipment grounding conductor.

c. The metallic sheath or the combined metallic sheath and equipment grounding conductors of the smooth or corrugated tube-type MC cable is listed and identified as an equipment grounding conductor.

(11) Cable trays as permitted in 392.10 and 392.6

(12) Cable bus framework as permitted in 370.60(1)

(13) Other listed electrically continuous metal raceways and listed auxiliary gutters.

(14) Surface metal raceways listed for grounding and containing circuit conductors not exceeding 600 volts, nominal.

Informational Note: See Article 100 for a definition of effective ground-fault current path.

NEC Article 250.122(D) Size of Equipment Ground Conductors (EGCs) – Increased in Size

If ungrounded conductors are increased in size for any reason other than as required in 310.15(B) or 310.15(C), wire-type EGCs, if installed, shall be increased in size proportionately to the increase in circular mil area of the ungrounded conductors.

Exception: EGCs shall be permitted to be sized by a qualified person to provide an effective ground-fault current path in accordance with 250.4(A)(5) or 250.4(B)(4).

Table 300.7(A) Wiring Methods and Materials – Underground Installations – Minimum Cover Requirements

Direct-buried cable, conduit, or other raceways shall be installed to meet the minimum cover requirements of Table 300.7(A).

Table 300.7(A) Minimum Cover Requirements, 0 to 1000 Volts ac, 1500 Volts dc, Nominal, Burial in Millimeters (Inches)

Location of Wiring Method or Circuit	Type of Wiring Method or Circuit									
	Column 1 Direct Burial Cables or Conductors		Column 2 Rigid Metal Conduit or Intermediate Medal Conduit		Column 3 Electrical Metallic Tubing, Nonmetallic Raceways Listed for Direct Burial Without Concrete Encasement, or Other Approved Raceways		Column 4 Residential Branch Circuits Rated 120 Volts or Less with GFCI Protection and Maximum Overcurrent Protection of 20 Amperes		Column 5 Circuits for Control of Irrigation and Landscape Lighting Limited to Not More Than 30 Volts and Installed with Type UF or in Other Identified Cable or Raceway	
	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.
All locations not specified below	600	24	150	6	450	18	300	12	150	6
In trench below 50 mm (2 in.) thick concrete or equivalent	450	18	150	6	300	12	150	6	150	6
Under a building	0 (in raceway or Type MC or Type MI cable identified for direct burial)	0	0	0	0	0	0 (in raceway or Type MC or Type MI cable identified for direct burial)	0	0 (in raceway or Type MC or Type MI cable identified for direct burial)	0
Under minimum of 102 mm (4 in.) thick concrete exterior slab with no vehicular traffic and the slab extending not less than 152 mm (6 in.) beyond the underground installation	450	18	100	4	100	4	150 (direct burial) 100 (in raceway)	6 4	150 (direct burial) 100 (in raceway)	6 4

[1] A lesser depth shall be permitted where specified in the installation instructions of a listed low-voltage lighting system.

[2] A depth of 150 mm (6 in.) shall be permitted for pool, spa, and fountain lighting, installed in a nonmetallic raceway, limited to not more than 30 volts where part of a listed low-voltage lighting system.

Notes:

1. Cover shall be defined as the shortest distance in mm (in.) measured between a point on the top surface of any direct-buried conductor, cable, conduit, or other raceway and the top surface of finished grade, concrete, or similar cover.
2. Raceways approved for burial only where concrete encased shall require a concrete envelope not less than 50 mm (2 in.) thick.
3. Lesser depths shall be permitted where cables and conductors rise for terminations or splices or where access is otherwise required.
4. Where one of the wiring method types listed in Columns 1 through 3 is used for one of the circuit types in Columns 4 and 5, the shallowest depth of burial shall be permitted.

Table 300.7(A) Minimum Cover Requirements, 0 to 1000 Volts ac, 1500 Volts dc, Nominal, Burial in Millimeters (Inches)

Location of Wiring Method or Circuit	Type of Wiring Method or Circuit									
	Column 1 Direct Burial Cables or Conductors		Column 2 Rigid Metal Conduit or Intermediate Medal Conduit		Column 3 Electrical Metallic Tubing, Nonmetallic Raceways Listed for Direct Burial Without Concrete Encasement, or Other Approved Raceways		Column 4 Residential Branch Circuits Rated 120 Volts or Less with GFCI Protection and Maximum Overcurrent Protection of 20 Amperes		Column 5 Circuits for Control of Irrigation and Landscape Lighting Limited to Not More Than 30 Volts and Installed with Type UF or in Other Identified Cable or Raceway	
	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.
Under streets, highways, roads, alleys, driveways, and parking lots	600	24	600	24	600	24	600	24	600	24
One- and two-family dwelling driveways and outdoor parking areas, and used only for dwelling-related purposes	450	18	450	18	450	18	300	12	450	18
In or under airport runways, including adjacent areas where trespassing is prohibited	450	18	450	18	450	18	450	18	450	18

[1] A lesser depth shall be permitted where specified in the installation instructions of a listed low-voltage lighting system.

[2] A depth of 150 mm (6 in.) shall be permitted for pool, spa, and fountain lighting, installed in a nonmetallic raceway, limited to not more than 30 volts where part of a listed low-voltage lighting system.

Notes:

1. Cover shall be defined as the shortest distance in mm (in.) measured between a point on the top surface of any direct-buried conductor, cable, conduit, or other raceway and the top surface of finished grade, concrete, or similar cover.
2. Raceways approved for burial only where concrete encased shall require a concrete envelope not less than 50 mm (2 in.) thick.
3. Lesser depths shall be permitted where cables and conductors rise for terminations or splices or where access is otherwise required.
4. Where one of the wiring method types listed in Columns 1 through 3 is used for one of the circuit types in Columns 4 and 5, the shallowest depth of burial shall be permitted.

NEC Article 300.7(D)(3) Wiring Methods and Materials – Underground Installations – Protection from Damage

Conductors and cables shall be protected from damage in accordance with 300.7(D)(1) through 300.7(D)(4).

(1) Emerging from Grade.
Direct-buried conductors and cables emerging from grade and specified in Columns 1 and 4 of Table 300.7(A) shall be protected by enclosures or raceways extending from the minimum cover distance below grade required by 300.7(A) to a point at least 2.5 m (8 ft) above finished grade. In no case shall the protection be required to exceed 450 mm (18 in.) below finished grade.

(2) Conductors Entering Buildings.
Conductors entering a building shall be protected to the point of entrance.

(3) Service Conductors and Raceways.
Direct buried service conductors and raceways containing service conductors that are not encased in concrete and that are buried 450 mm (18 in.) or more below grade shall have their location identified by a warning ribbon that is placed in the trench at least 300 mm (12 in.) above the underground installation.

(4) Enclosure or Raceway Damage.
Where the enclosure or raceway is subject to physical damage, the conductors shall be installed in electrical metallic tubing, rigid metal conduit, intermediate metal conduit, RTRC-XW, Schedule 80 PVC conduit, or equivalent.

NEC Table 310.15(C)(1) Adjustment Factors for More Than Three Current-Carrying Conductors

The ampacity of each conductor shall be reduced as shown in Table 310.15(C)(1) where the number of current-carrying conductors in a raceway or cable exceeds three, installed without maintaining spacing for a continuous length longer than 600 mm (24 in.).

The ampacity of each conductor shall be reduced as shown in Table 310.15(C)(1) where single conductors or multiconductor cables not installed in raceways are installed without maintaining spacing for a continuous length longer than 600 mm (24 in.).

Each current-carrying conductor of a paralleled set of conductors shall be counted as a current-carrying conductor.

Where conductors of different systems, as provided in 300.5, are installed in a common raceway or cable, the adjustment factors shown in Table 310.15(C)(1) shall apply only to the number of power and lighting conductors.

Table 310.15(C)(1) Adjustment Factors for More Than Three Current-Carrying Conductors

Number of Conductors*	Percent of Values in Table 310.16, 310.18, and 310.20 as Adjusted for Ambient Temperature if Necessary
4-6	80
7-9	70
10-20	50
21-30	45
31-40	40
41 and above	35

*Number of conductors is the total number of conductors in the raceway or cable, including spare conductors. The count shall be adjusted in accordance with 310.15(E) and 310.15(F). The count shall not include conductors that are connected to electrical components that cannot be simultaneously energized.

NEC Table 310.16 Ampacities of Insulated Conductors with Not More Than Three Current-Carrying Conductors in Raceway, Cable, or Earth (Directly Buried)

The ampacities shall be as specified in Table 310.16 where all of the following conditions apply:

(1) Conductors are rated 0 volts through 2000 volts.
(2) Conductors are rated 60°C (140°F), 75°C (167°F), or 90°C (194°F).
(3) Wiring is installed in a 30°C (86°F) ambient temperature.
(4) There are not more than three current-carrying conductors.

Table 310.16 Ampacities of Insulated Conductors with Not More Than Three Current-Carrying Conductors in Raceway, Cable, or Earth (Directly Buried)

Size AWG or kcmil	Temperature Rating of Conductor (See Table 310.4(1))						Size AWG or kcmil
	60°C (140°F)	75°C (167°F)	90°C (194°F)	60°C (140°F)	75°C (167°F)	90°C (194°F)	
	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE, ZW	Types TBS, SA, SIS, FEP, FEPB, MI, PFA, RHH, RHW-2, THHN, THHW, THW-2, THWN-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN, Z, ZW-2	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE	Types TBS, SA, SIS, THHN, THHW, THW-2, THWN-2, RHH, RHW-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN	
	Copper			Aluminum or Copper-clad Aluminum			
16	10	15	20	-	-	-	-
14	15	20	25	10	15	20	14
12	20	25	30	15	20	25	12
10	30	35	40	25	30	35	10
8	40	50	55	35	40	45	8
6	55	65	75	40	50	55	6
4	70	85	95	55	65	75	4
3	85	100	115	65	75	85	3
2	95	115	130	75	90	100	2
1	110	130	145	86	100	115	1
1/0	125	150	170	100	120	135	1/0
2/0	145	175	195	115	135	150	2/0
3/0	165	200	225	130	155	175	3/0
4/0	195	230	260	150	180	205	4/0

Notes:

1. Section 310.15(B) shall be referenced for ampacity correction factors where the ambient temperature is other than 30°C (86°F).
2. Section 310.15(C) shall be referenced for more than three current-carrying conductors.
3. Section 310.16 shall be referenced for conditions of use.
4. Ampacity for 18 AWG copper Type MI and Type MTW shall be 14 amperes at 90°C (194°F).
5. Ampacity for 16 AWG copper Type MI and Type MTW shall be 18 amperes at 90°C (194°F).

[1] Section 240.4(D) shall be referenced for conductor overcurrent protection limitations, except as modified elsewhere in the code.

[2] Ampacity shall be used for adjustment or correction only.

[3] Ampacity shall be applicable only to copper-clad aluminum conductors.

Table 310.16 Ampacities of Insulated Conductors with Not More Than Three Current-Carrying Conductors in Raceway, Cable, or Earth (Directly Buried)

Size AWG or kcmil	Temperature Rating of Conductor (See Table 310.4(1))						Size AWG or kcmil
	60°C (140°F)	75°C (167°F)	90°C (194°F)	60°C (140°F)	75°C (167°F)	90°C (194°F)	
	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE, ZW	Types TBS, SA, SIS, FEP, FEPB, MI, PFA, RHH, RHW-2, THHN, THHW, THW-2, THWN-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN, Z, ZW-2	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE	Types TBS, SA, SIS, THHN, THHW, THW-2, THWN-2, RHH, RHW-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN	
	Copper			Aluminum or Copper-clad Aluminum			
250	215	255	290	170	205	230	250
300	240	285	320	195	230	260	300
350	260	310	350	210	250	280	350
400	280	335	380	225	270	305	400
500	320	380	430	260	310	350	500
600	350	420	475	285	340	385	600
700	385	460	520	315	375	425	700
750	400	475	535	320	385	435	750
800	410	490	555	330	395	445	800
900	435	520	585	355	425	480	900
1,000	455	545	615	375	445	500	1,000
1,250	495	590	665	405	485	545	1,250
1,500	525	625	705	435	520	585	1,500
1,750	545	650	735	455	545	615	1,750
2,000	555	665	750	470	560	630	2,000

Notes:

1. Section 310.15(B) shall be referenced for ampacity correction factors where the ambient temperature is other than 30°C (86°F).
2. Section 310.15(C) shall be referenced for more than three current-carrying conductors.
3. Section 310.16 shall be referenced for conditions of use.
4. Ampacity for 18 AWG copper Type MI and Type MTW shall be 14 amperes at 90°C (194°F).
5. Ampacity for 16 AWG copper Type MI and Type MTW shall be 18 amperes at 90°C (194°F).

[1] Section 240.4(D) shall be referenced for conductor overcurrent protection limitations, except as modified elsewhere in the code.

[2] Ampacity shall be used for adjustment or correction only.

[3] Ampacity shall be applicable only to copper-clad aluminum conductors.

NEC Table 310.17 Ampacities of Single-Insulated Conductors in Free Air

The ampacities shall be as specified in Table 310.17 where all of the following conditions apply:

(1) Conductors are rated 0 volts through 2000 volts.
(2) Conductors are rated 60°C (140°F), 75°C (167°F), or 90°C (194°F).
(3) Wiring is installed in a 30°C (86°F) ambient temperature.

Table 310.17 Ampacities of Single-Insulated Conductors in Free Air

Size AWG or kcmil	Temperature Rating of Conductor (See Table 310.4(1))						Size AWG or kcmil
	60°C (140°F)	75°C (167°F)	90°C (194°F)	60°C (140°F)	75°C (167°F)	90°C (194°F)	
	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, ZW	Types TBS, SA, SIS, FEP, FEPB, MI, PFA, RHH, RHW-2, THHN, THHW, THW-2, THWN-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN, Z, ZW-2	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN	Types TBS, SA, SIS, THHN, THHW, THW-2, THWN-2, RHH, RHW-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN	
	Copper			Aluminum or Copper-clad Aluminum			
18	-	-	18	-	-	-	-
16	-	-	24	-	-	-	-
14*	25	30	35	-	-	-	-
12*	30	35	40	25	30	35	12*
10*	40	50	55	35	40	45	10*
8	60	70	80	45	55	60	8
6	80	95	105	60	75	85	6
4	105	125	140	80	100	115	4
3	120	145	165	95	115	130	3
2	140	170	190	110	135	150	2
1	165	195	220	130	155	175	1
1/0	195	230	260	150	180	205	1/0
2/0	225	265	300	175	210	235	2/0
3/0	260	310	350	200	240	270	3/0
4/0	300	360	405	235	280	315	4/0

Notes:
1. Section 310.15(B) shall be referenced for ampacity correction factors where the ambient temperature is other than 30°C (86°F).
2. Section 310.17 shall be referenced for conditions of use.
*Section 240.4(D) shall be referenced for conductor overcurrent protection limitations, except as modified elsewhere in the code.

Table 310.17 Ampacities of Single-Insulated Conductors in Free Air

Size AWG or kcmil	Temperature Rating of Conductor (See Table 310.4(1))						Size AWG or kcmil
	60°C (140°F)	75°C (167°F)	90°C (194°F)	60°C (140°F)	75°C (167°F)	90°C (194°F)	
	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE, ZW	Types TBS, SA, SIS, FEP, FEPB, MI, PFA, RHH, RHW-2, THHN, THHW, THW-2, THWN-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN, Z, ZW-2	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE	Types TBS, SA, SIS, THHN, THHW, THW-2, THWN-2, RHH, RHW-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN	
	Copper			Aluminum or Copper-clad Aluminum			
250	340	405	455	265	315	355	250
300	375	445	500	290	350	395	300
350	420	505	570	330	395	445	350
400	455	545	615	355	425	480	400
500	515	620	700	405	485	545	500
600	575	690	780	455	545	615	600
700	630	755	850	500	595	670	700
750	655	785	885	515	620	700	750
800	680	815	920	535	645	725	800
900	730	870	980	580	700	790	900
1,000	780	935	1,055	625	750	845	1,000
1,250	890	1,065	1,200	710	855	965	1,250
1,500	980	1,175	1,325	795	950	1,070	1,500
1,750	1,070	1,280	1,445	875	1,050	1,185	1,750
2,000	1,155	1,385	1,560	960	1,150	1,295	2,000

Notes:

1. Section 310.15(B) shall be referenced for ampacity correction factors where the ambient temperature is other than 30°C (86°F).
2. Section 310.17 shall be referenced for conditions of use.

*Section 240.4(D) shall be referenced for conductor overcurrent protection limitations, except as modified elsewhere in the code.

NEC Article 352.24 (B) Rigid Polyvinyl Chloride Conduit (PVC) – Installation – Bends – Number in One Run

The total degrees of bends in a conduit run shall not exceed 360 degrees between pull points.

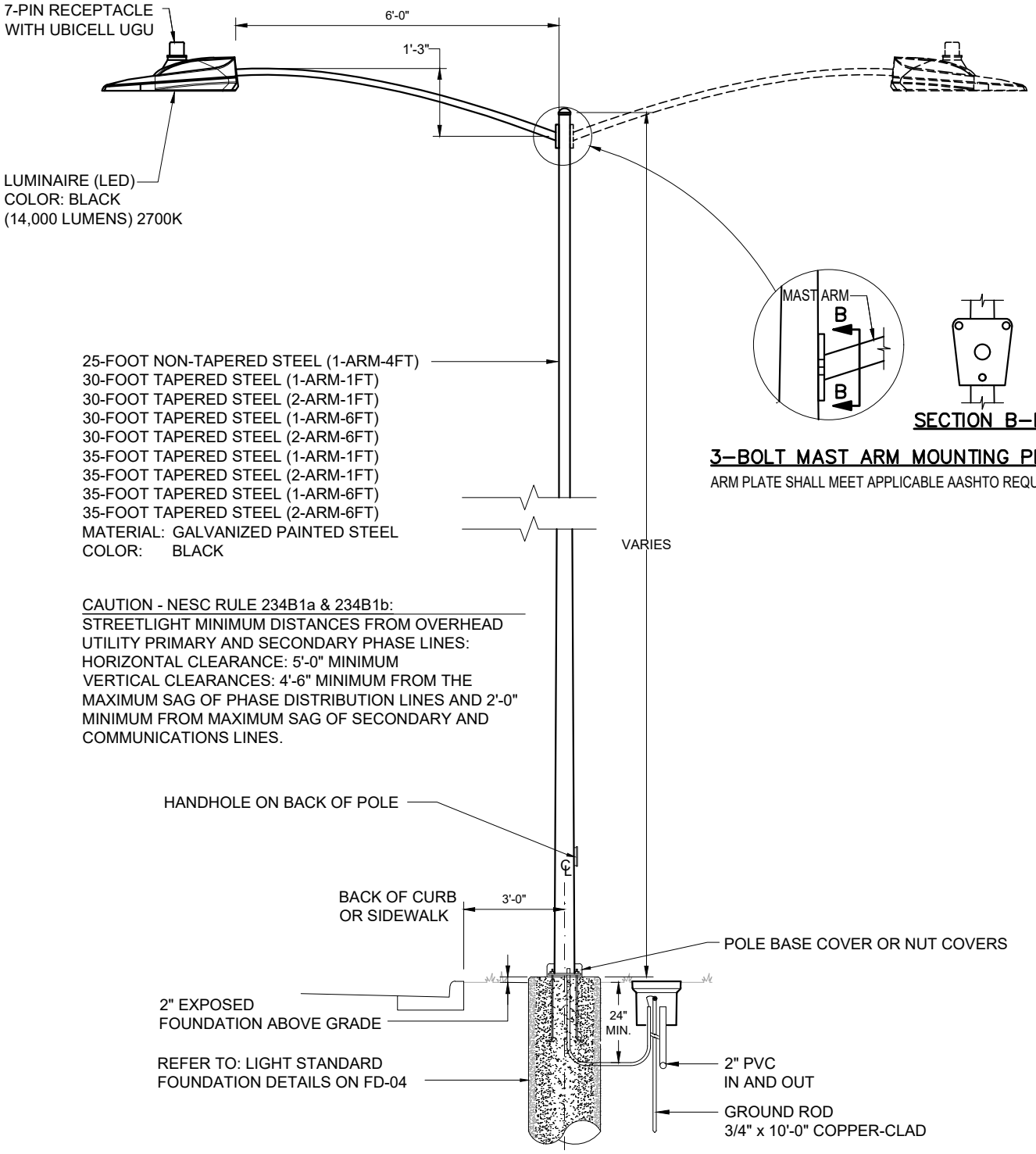
Chapter 9, Table 1, Conduit Fill

Table 1 Percent of Cross Section of Conduit and Tubing for Conductors and Cables	
Number of Conductors and/or Cables	Cross-Sectional Area (%)
1	53
2	31
Over 2	40

Note (2) - Table 1 applies only to complete conduit or tubing systems and is not intended to apply to sections of conduit or tubing used to protect exposed wiring and cable from physical damage.

All references in these standards cite the following version of the NFPA:

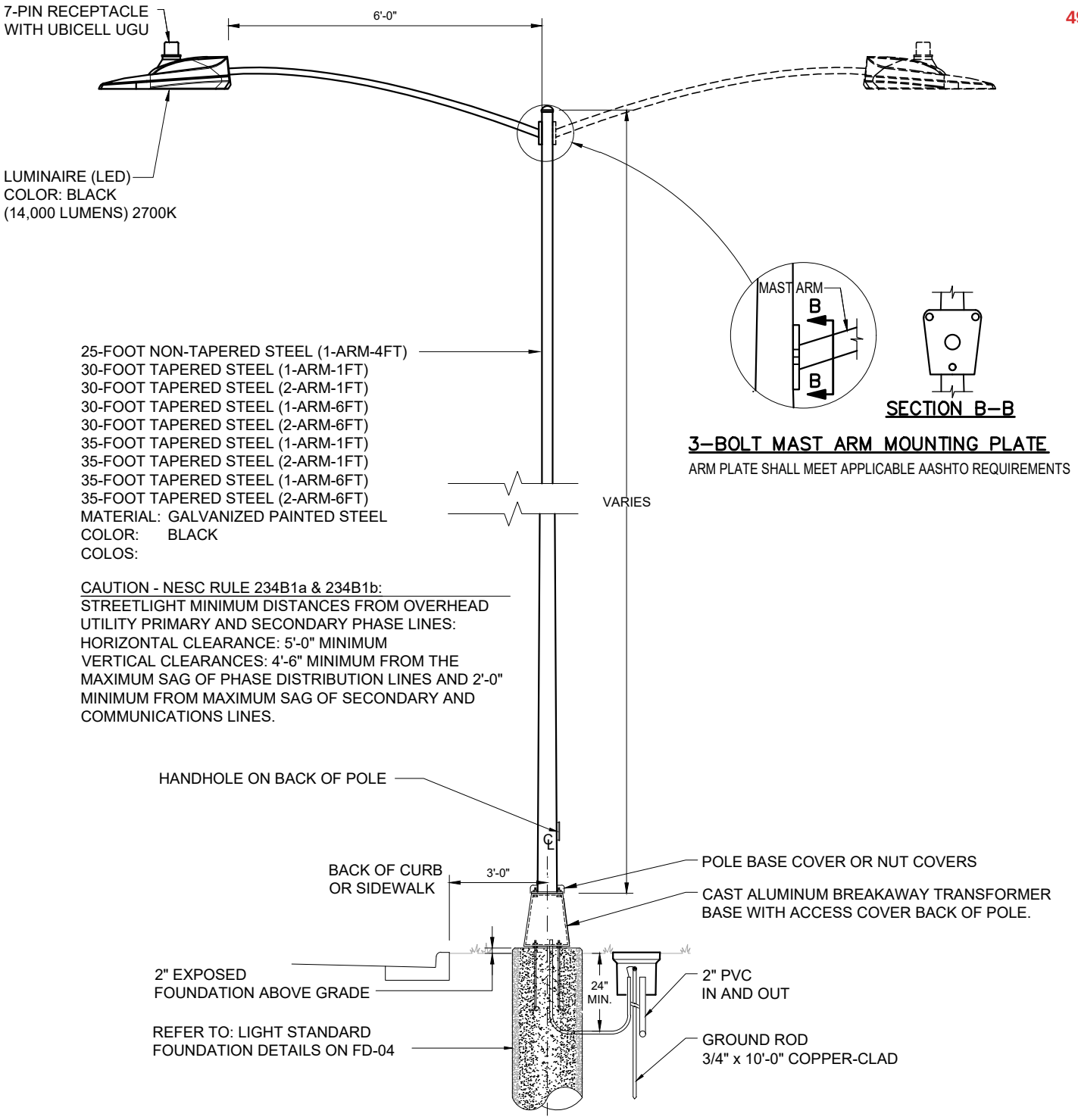
NFPA 70®, National Electrical Code® (Quincy: National Fire Protection Association, 2023)



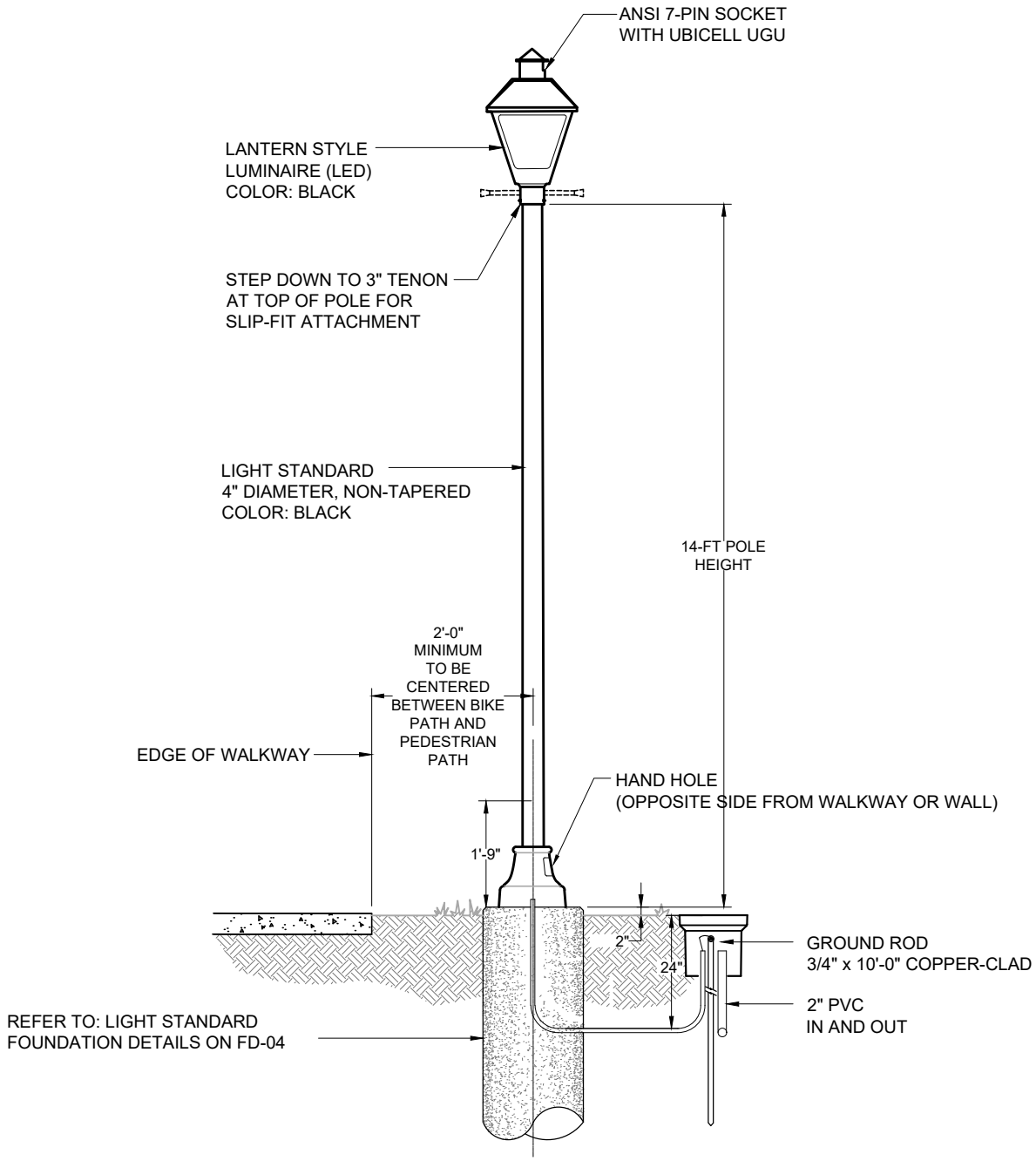
3–BOLT MAST ARM MOUNTING PLATE

ARM PLATE SHALL MEET APPLICABLE AASHTO REQUIREMENTS

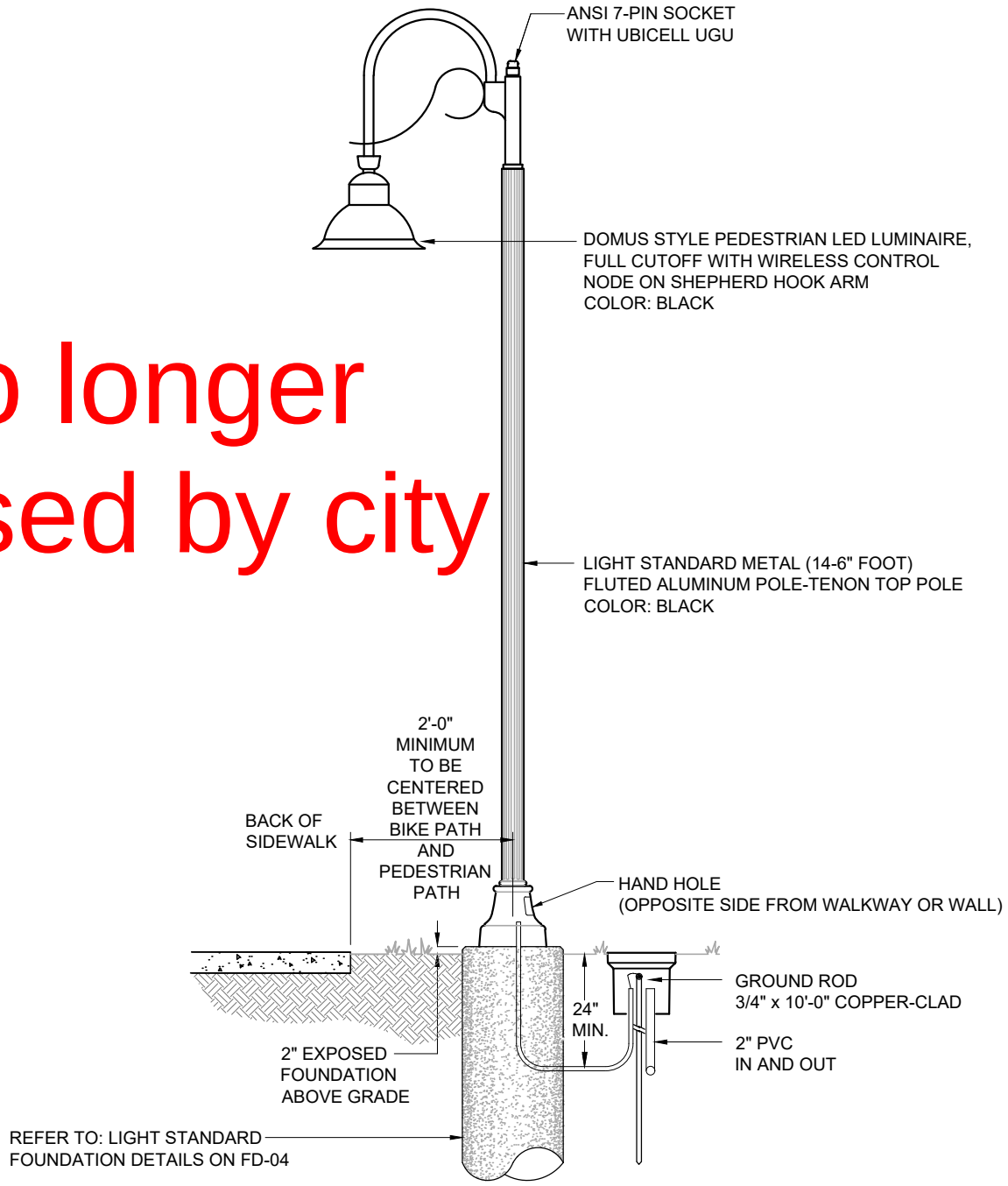
CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	ROADWAY LIGHT STANDARD – WITHOUT BREAKAWAY BASE	STD. DWG. NO. LD-01



CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	ROADWAY LIGHT STANDARD – WITH BREAKAWAY BASE	STD. DWG. NO. LD-02

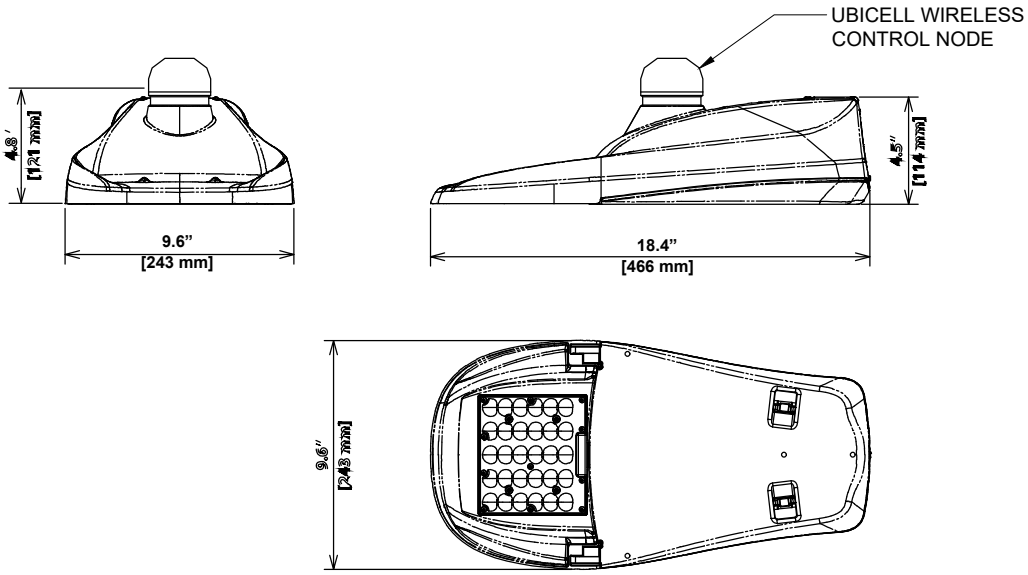


CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	Typical Post Top SL Standard Details	STD. DWG. NO. LD-03



no longer
used by city

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	PEDESTRIAN LIGHT WITH SHEPHERD'S HOOK ARM	STD. DWG. NO. LD-04



Distribution types:
Type II – Narrow / residential streets, collectors, and minor arterials
Type III – Principal arterials, freeways, and expressways
Type IV – near parking lots, trails and cul de sacs

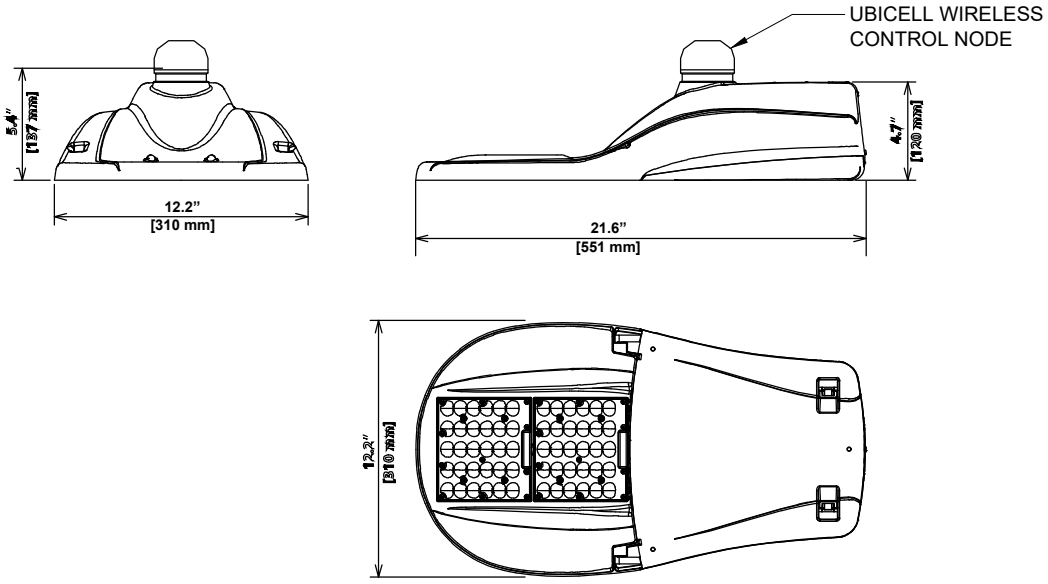
Color Temperature: 2700K

Wattages: based on roadway and location

LUMINAIRE DETAIL NOTES

1. LUMINAIRE INCLUDES 7-PIN PHOTOCELL (OR CONTROL NODE) RECEPTACLE.
2. REFER TO SECTION 2.02 FOR POLE INFORMATION AND SECTION 4.04 FOR FOUNDATION INFORMATION.
3. THE STANDARD LED LUMINAIRE DRIVER WITH THIS LUMINAIRE SUPPORTS VOLTAGES 120 - 277V. TYPICAL VOLTAGES FOR LED LIGHTING INCLUDE 120 V, 208V, 240V, OR 277V.
4. ACCESSORIES INCLUDE HSS (HOUSE SIDE SHIELD), CSS (CUL-DE-SAC SHIELD), AND FSS (FRONT SIDE SHIELD)
5. HORIZONTAL 2-BOLT SLIP FIT MOUNT TO 1-1/4" TO 2" (1-5/8" TO 2-3/8" O.D.) DIAMETER MAST ARM.
6. LUMINAIRE WEIGHT: 7.7 LBS, EPA: 0.39 SQFT
7. Luminaire specs to be approved by city before install

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	COBRAHEAD – SMALL LUMINAIRE DETAIL	STD. DWG. NO. LD-05



Distribution types:
Type II – Narrow / residential streets, collectors, and minor arterials
Type III – Principal arterials, freeways, and expressways
Type IV – near parking lots, trails and cul de sacs

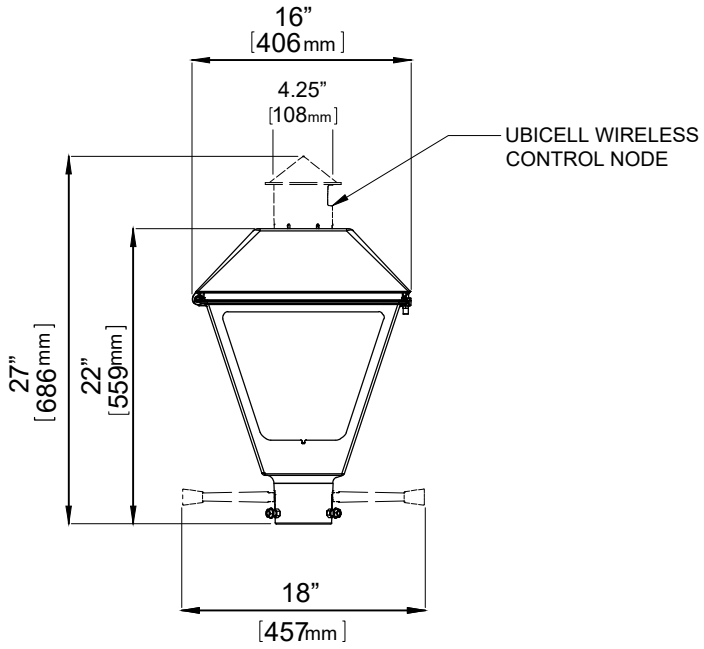
Color Temperature: 2700K

Wattages: based on roadway and location.

LUMINAIRE DETAIL NOTES

1. LUMINAIRE INCLUDES 7-PIN PHOTOCELL (OR CONTROL NODE) RECEPTACLE.
2. REFER TO SECTION 2.02 FOR POLE INFORMATION AND SECTION 4.04 FOR FOUNDATION INFORMATION.
3. THE STANDARD LED LUMINAIRE DRIVER WITH THIS LUMINAIRE SUPPORTS VOLTAGES 120 - 277V. TYPICAL VOLTAGES FOR LED LIGHTING INCLUDE 120 V, 208V, 240V, OR 277V.
4. ACCESSORIES INCLUDE HSS (HOUSE SIDE SHIELD), CSS (CUL-DE-SAC SHIELD), AND FSS (FRONT SIDE SHIELD)
5. HORIZONTAL 2-BOLT SLIP FIT MOUNT TO 1-1/4" TO 2" (1-5/8" TO 2-3/8" O.D.) DIAMETER MAST ARM.
6. LUMINAIRE WEIGHT: 11 LBS, EPA: 0.44 SQFT
7. Luminaire specs to be approved by city before install

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	COBRAHEAD – MEDIUM LUMINAIRE DETAIL	STD. DWG. NO. LD-06



Distribution types:
Type III – neighborhoods or open space/parks
Type V – neighborhoods or open space/parks

Color Temperature: 2700K

LUMINAIRE DETAIL NOTES

1. LUMINAIRE INCLUDES 7-PIN PHOTOCELL (OR CONTROL NODE) RECEPTACLE.
2. REFER TO SECTION 2.02 FOR POLE INFORMATION AND SECTION 4.04 FOR FOUNDATION INFORMATION.
3. THE STANDARD LED LUMINAIRE DRIVER WITH THIS LUMINAIRE SUPPORTS VOLTAGES 120 - 277V. TYPICAL VOLTAGES FOR LED LIGHTING INCLUDE 120 V, 208V, 240V, OR 277V.
4. ACCESSORIES INCLUDE HSS (HOUSE SIDE SHIELD), CSS (CUL-DE-SAC SHIELD), AND FSS (FRONT SIDE SHIELD)
5. SLIP-FIT ATTACHMENT TO POLE TOP TENON FROM 2-3/8" TO 3" DIAMETERS.
6. LUMINAIRE WEIGHT: 19.62 LBS, EPA: 1.58 SQFT
7. Luminaire specs to be approved by city before install

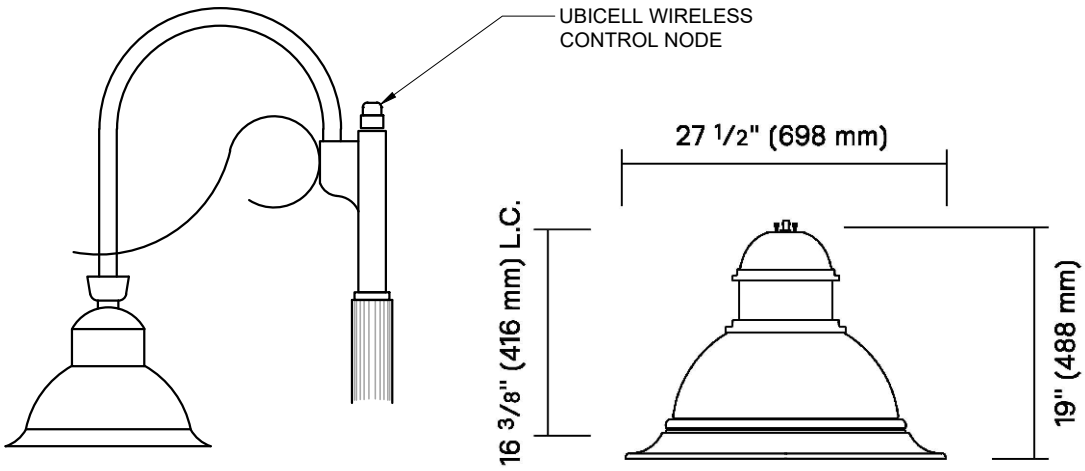
CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	POST TOP LUMINAIRE DETAIL	STD. DWG. NO. LD-07



Luminaire ID	Description	Lumen Output	Max Watts	CCT	Finish	IES Distribution	BUG Rating
W2-27-W	Wall Mount	6,000 - 6,500	60	2700 K	Black	Wide	NOT LISTED

LUMINAIRE DETAIL NOTES

1. NO PHOTOSENSOR WHEN USED IN A PEDESTRIAN UNDERPASS FOR 24/7 OPERATION. BUTTON-TYPE PHOTOCELL WHEN LOCATED OUTSIDE OF UNDERPASS.
2. THE STANDARD LED LUMINAIRE DRIVER WITH THIS LUMINAIRE SUPPORTS VOLTAGES 120 - 277V. TYPICAL VOLTAGES FOR LED LIGHTING INCLUDE 120 V, 208V, 240V, OR 277V.
3. MOUNT TO WALL RECESSED J-BOX (PREFERRED). KNOCKOUTS ON TOP AND BOTTOM ALLOW FOR SURFACE CONDUIT ENTRY.
4. LUMINAIRE WEIGHT: 9.9LBS



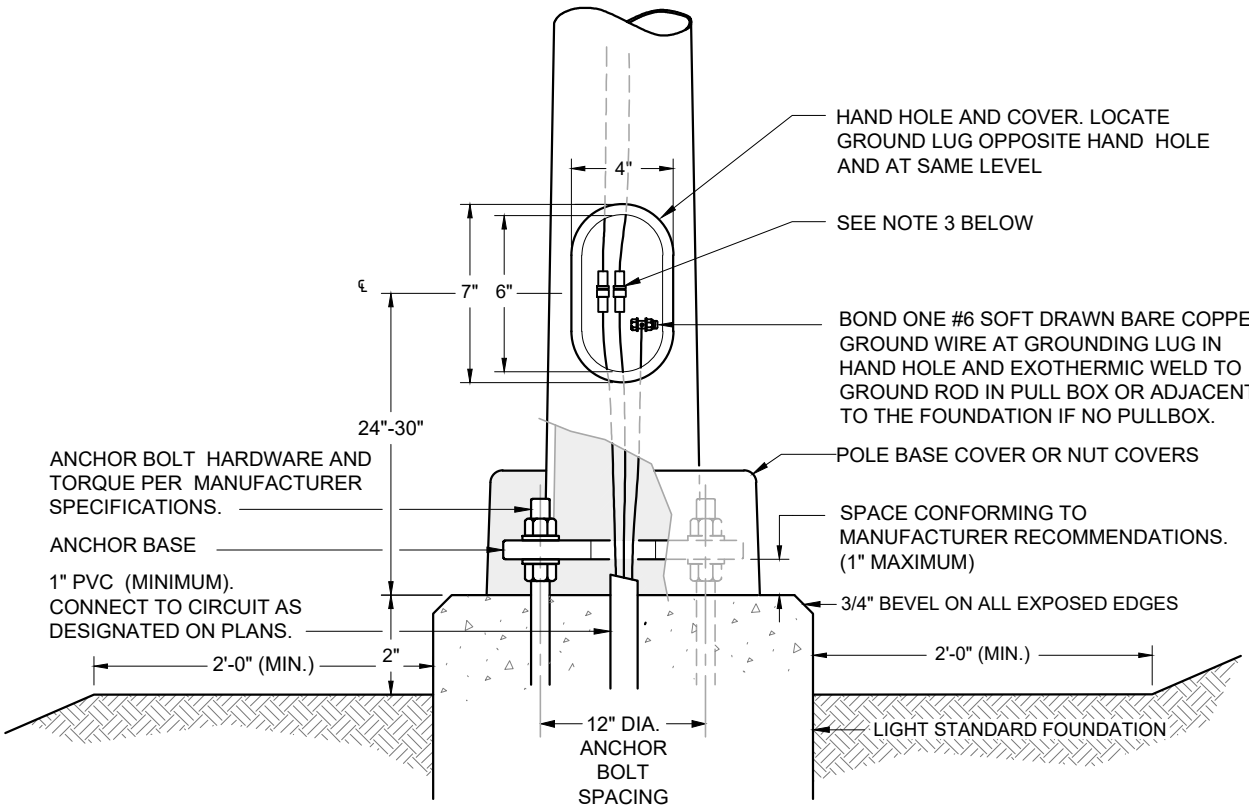
Luminaire ID	Description	Lumen Output	Max Watts	CCT	Finish	IES Distribution	BUG Rating
D1-27-T2	Domus Style Pendant (Small)	3,000-4,000	40	2700 K	Black	Type 2	B1-U0-G1
D1-27-T3	Domus Style Pendant (Large)	3,000-4,000	0	2700 K	Black	Type 3	B1-U0-G1

LUMINAIRE DETAIL NOTES

1. 7-PIN PHOTOCELL (OR CONTROL NODE) RECEPTACLE IS MOUNTED ON TOP OF DECORATIVE POLE ARMATURE.
2. REFER TO DETAIL LD-04 FOR POLE AND FOUNDATION DETAIL.
3. THE STANDARD LED LUMINAIRE DRIVER WITH THIS LUMINAIRE SUPPORTS VOLTAGES 120 - 277V. TYPICAL VOLTAGES FOR LED LIGHTING INCLUDE 120V, 208V, 240V, OR 277V.
4. ACCESSORIES INCLUDE HSS (HOUSE SIDE SHIELD)
5. VERTICAL SLIP FIT MOUNTING SIZE TO BE COORDINATED WITH THE MANUFACTURER.
6. LUMINAIRE & ARMATURE WEIGHT: 43 LBS, EPA: 1.26 SQFT

no longer
used by city

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	PEDESTRIAN LIGHT WITH SHEPHERD'S HOOK ARM LUMINAIRE DETAIL	STD. DWG. NO. LD-09



NOTES

1. LIGHT STANDARD FOUNDATIONS SHALL BE PRECAST CONCRETE.

2. ALL CONDUCTORS SHALL BE SIZED IN CONFORMANCE WITH N.E.C. REQUIREMENTS 3/C COPPER #12 AWG CABLE MINIMUM.

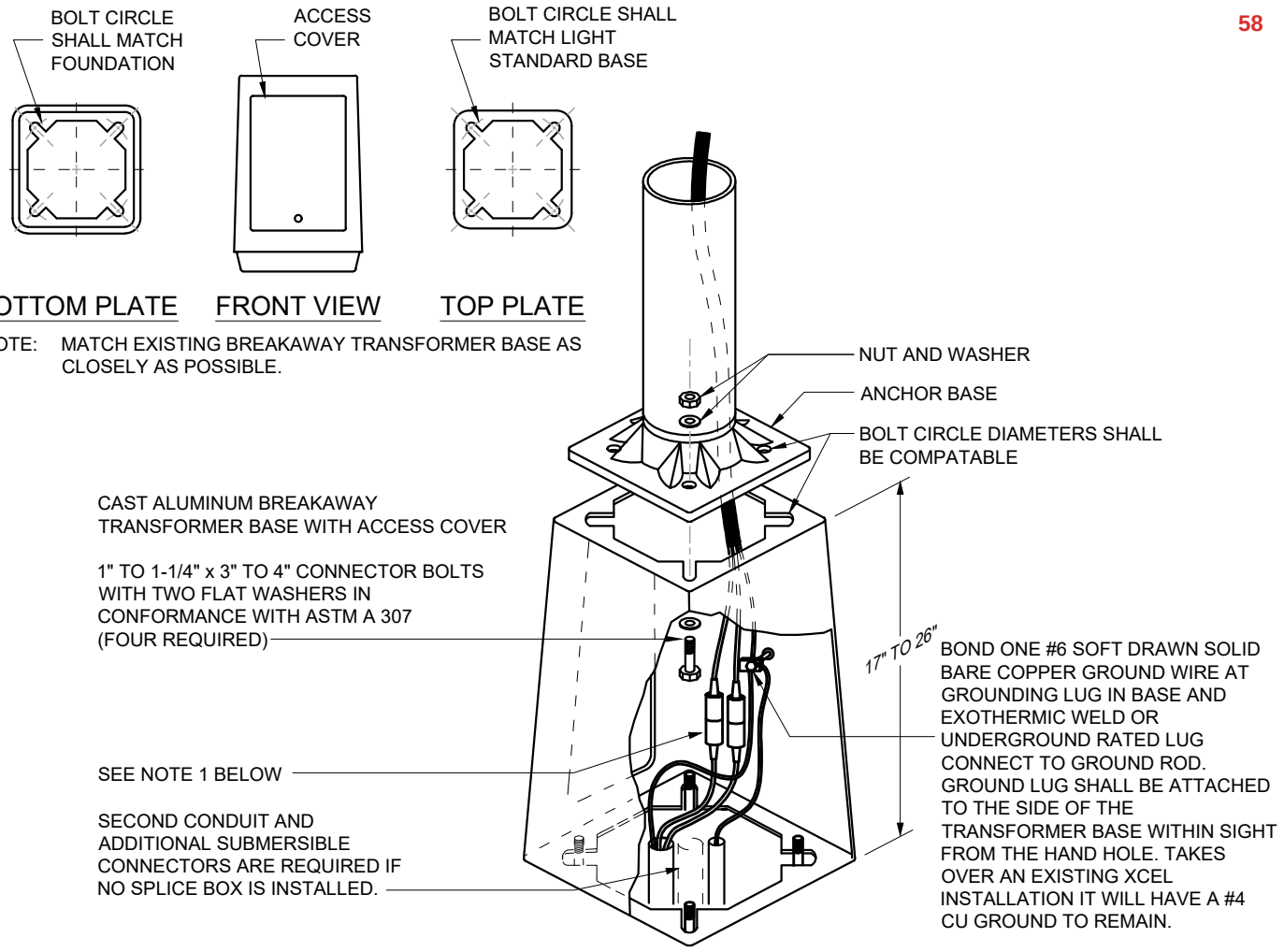
3. BREAKAWAY 2-POLE IN-LINE FUSE HOLDER FOR 240V CLASS CC FUSES BASIS OF DESIGN ARE AS FOLLOWS:
   FUSE HOLDER - EATON/BUSSMAN #HEX-AW-DRLC-A
   FUSE - EATON/BUSSMAN FNQ-15 (2) REQUIRED.

   BREAKAWAY 2-POLE IN-LINE FUSE HOLDER FOR 120V CLASS CC FUSES BASIS OF DESIGN ARE AS FOLLOWS:
   FUSE HOLDER - EATON/BUSSMAN #HEX-AW-DRLC-A FOR PHASE AND NEUTRAL CONDUCTORS.
   FUSE - EATON/BUSSMAN FNQ-15 (1) REQUIRED.
   DUMMY FUSE - EATON/BUSSMAN #NNB FOR NEUTRAL CONDUCTOR (1) REQUIRED.

4. THIS DETAIL ASSUMES THAT THERE IS A PULL BOX ADJACENT TO THE FOUNDATION. IF THIS IS THE LAST LIGHT ON THIS CIRCUIT AND NO PULL BOX REQUIRED, THEN THERE WILL NEED TO BE AN ADDITIONAL CONDUIT IN THE FOUNDATION FOR THE GROUND CONDUCTOR TO THE GROUND ROD ADJACENT TO THE FOUNDATION (NOT SHOWN).

5. FOR USE ONLY OUTSIDE CLEAR ZONE OR NEAR CROSSWALKS IN THE CLEAR ZONE OR IN PROTECTED INSTALLATIONS (I.E. INSTALLED BEHIND GUARDRAILS).

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	TYPICAL NON–BREAKAWAY POLE BASE DETAIL	STD. DWG. NO. LD-10



NOTE: MATCH EXISTING BREAKAWAY TRANSFORMER BASE AS CLOSELY AS POSSIBLE.

NOTES

1. BREAKAWAY 2-POLE IN-LINE FUSE HOLDER FOR 240V CLASS CC FUSES BASIS OF DESIGN ARE AS FOLLOWS:
   1. FUSE HOLDER
   2. FUSE - FNQ-15 (2) REQUIRED.

   BREAKAWAY 2-POLE IN-LINE FUSE HOLDER FOR 120V CLASS CC FUSES BASIS OF DESIGN ARE AS FOLLOWS:
   1. FUSE HOLDER - FOR PHASE AND NEUTRAL CONDUCTORS.
   2. FUSE - FNQ-15 (1) REQUIRED.
   3. DUMMY FUSE - FOR NEUTRAL CONDUCTOR (1) REQUIRED.

2. 120V NESC WIRING SCENARIO:
   PROVIDE (2) 4-TERMINAL, (1) 6-TERMINAL SUBMERSIBLE, INSULATED LUG CONNECTOR, (2) HOTS SHALL BE 4-TERMINAL TYPE AND (1) 6-TERMINAL TYPE FOR NEUTRAL/GROUND.

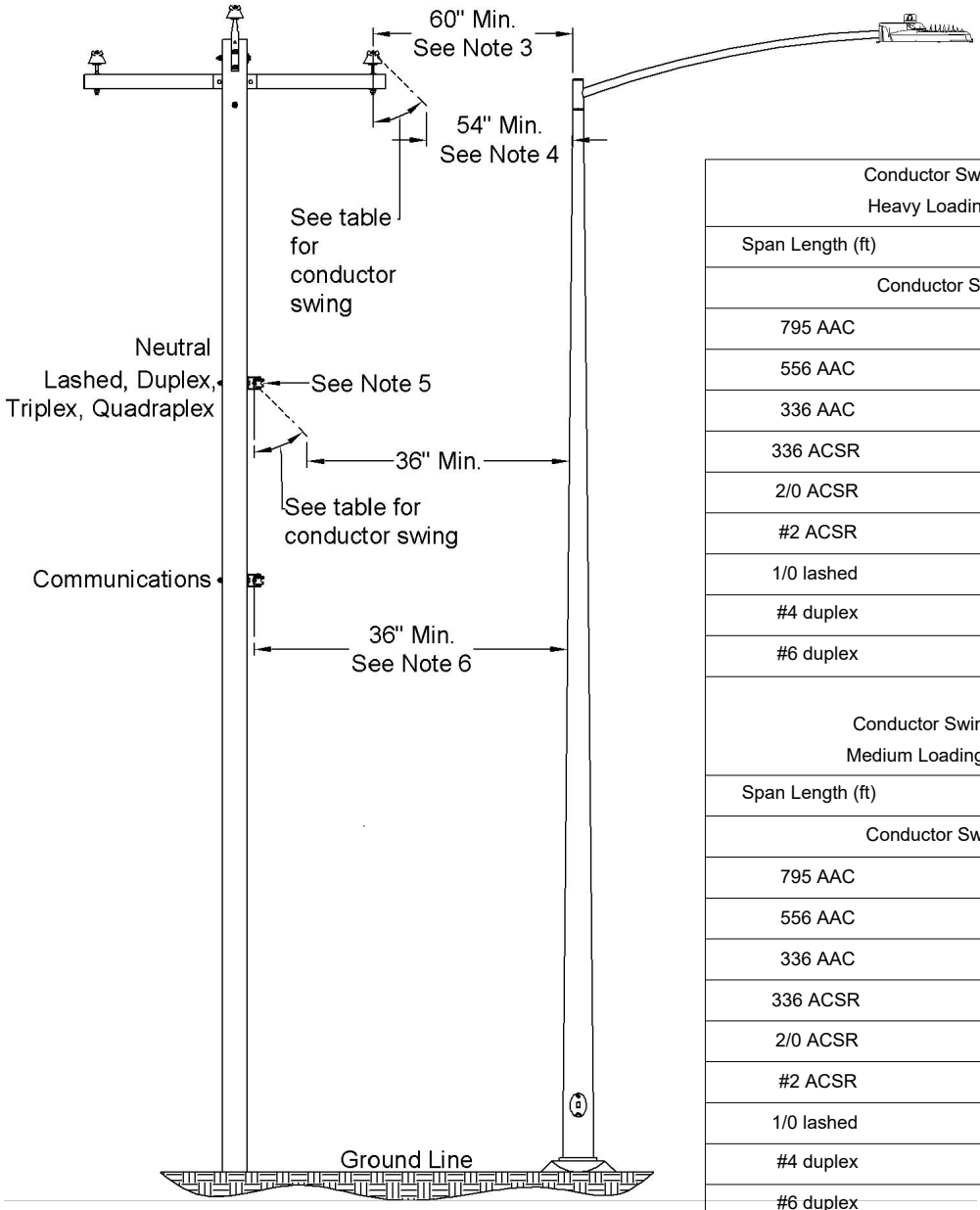
   120V NEC WIRING SCENARIO:
   PROVIDE (3) 4-TERMINAL, (1) 6-TERMINAL SUBMERSIBLE, INSULATED LUG CONNECTOR, (2) HOTS, (1) NEUTRAL SHALL BE 4-TERMINAL TYPE AND (1) 6-TERMINAL TYPE FOR GROUND.

   240V NEC WIRING SCENARIO:
   PROVIDE (3) 4-TERMINAL, SUBMERSIBLE, INSULATED LUG CONNECTOR, (2) HOTS, (1) GROUND.

   ONLY FOR USE IN CLEAR ZONE AND NOT NEAR CROSSWALKS IN THE CLEAR ZONE.

3.

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	TYPICAL BREAKAWAY POLE BASE DETAIL	STD. DWG. NO. LD-11



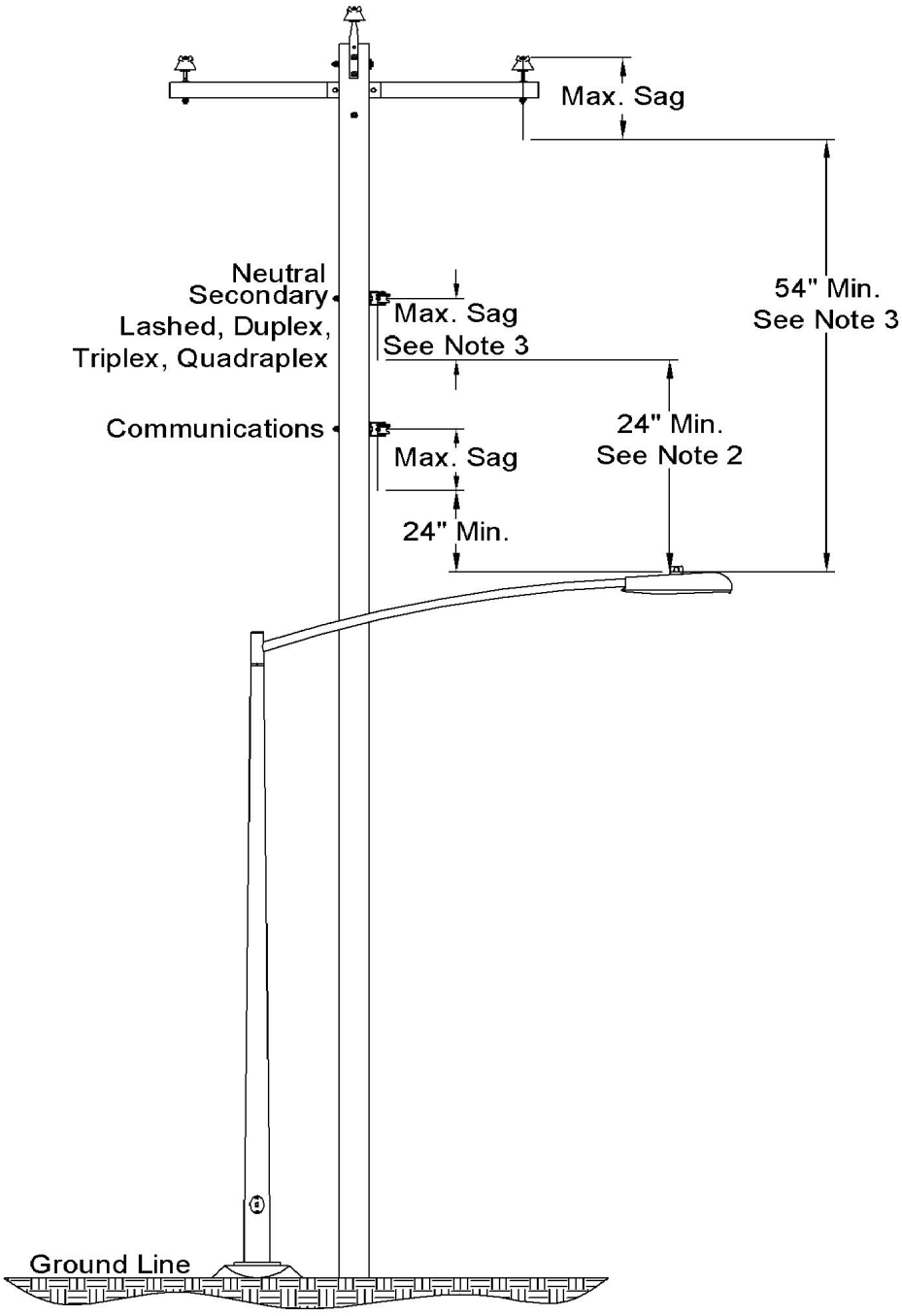
Conductor Swing Table Heavy Loading District					
Span Length (ft)	100	150	200	250	300
Conductor Swing (in)					
795 AAC	7	12	19	26	-
556 AAC	9	15	22	29	-
336 AAC	10	17	25	34	-
336 ACSR	7	12	18	26	33
2/0 ACSR	6	10	17	25	33
#2 ACSR	7	13	21	38	63
1/0 lashed	14	25	39	65	-
#4 duplex	13	31	63	-	-
#6 duplex	18	51	100	-	-

Conductor Swing Table Medium Loading District					
Span Length (ft)	100	150	200	250	300
Conductor Swing (in)					
795 AAC	6	10	16	22	-
556 AAC	7	12	18	26	-
336 AAC	8	14	21	30	-
336 ACSR	5	9	16	22	-
2/0 ACSR	4	9	14	21	29
#2 ACSR	5	11	18	27	36
1/0 lashed	12	23	36	51	-
#4 duplex	11	22	34	-	-
#6 duplex	13	26	49	-	-

Notes:

1. A separation of 7-1/2 feet is sufficient for most spans when conductor is displaced by wind. Less clearance is needed when not at midspan or spans and sags are reduced.
2. See page K-31 for sag change along length of span.
3. NESC Rule 234B1a - Clearance between phase conductor, open wire secondary and street light must be 5 feet when at rest.
4. NESC Rule 234B1b - Clearance between street light and primary conductor with 6lb/ft² wind at 60 F° final sag must be 4-1/2 feet minimum clearance when displaced by wind
5. NESC Rule 234B1b - Clearance between street light and open wire secondary with 6lb/ft² wind at 60 F° final sag must be 3-1/2 feet minimum clearance when displaced by wind.
6. Clearance between communications, neutrals, lashed, duplex, triplex, and quadraplex and streetlights without wind is 3 feet
7. Refer to detail sheet K-21, Xcel Energy Outdoor Lighting (ODL) Manual (2023).

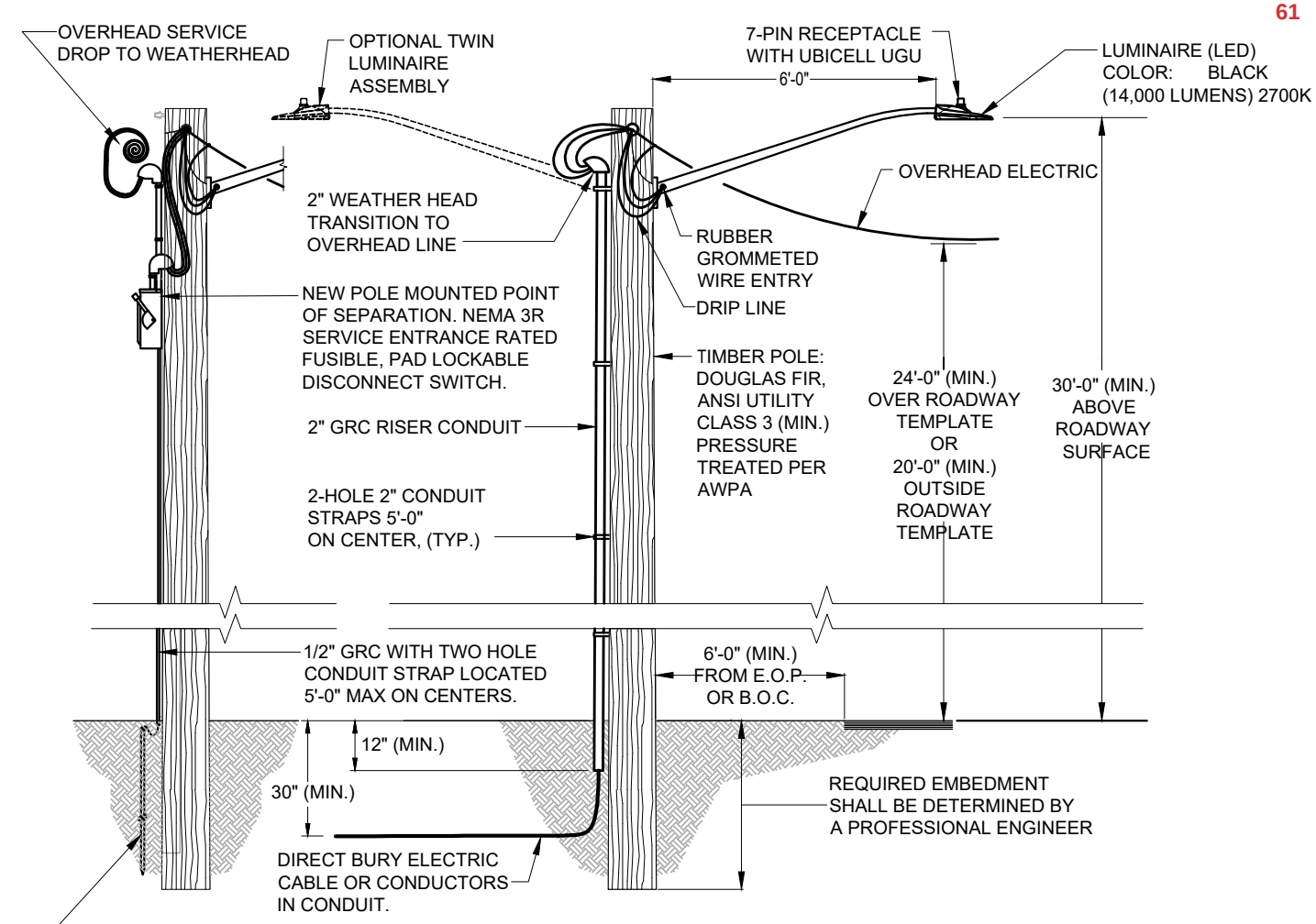
CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	HORIZONTAL CLEARANCE OF STREETLIGHT TO OVERHEAD ELECTRICAL CONDUCTORS	STD. DWG. NO. LD-12



Notes:

1. Clearances are to be applied at maximum sag conditions pages K-28, K-28.50 and K-29.
2. NESC permits neutrals (230C3) and communication to have clearance of 2 feet.
3. Open wire secondary use primary clearance of 54" Per NESC Rule 234B2.
4. Per OSHA 1910-269, workers that come within 10 feet of energized conductor or equipment need to be Supply Space (Electric Supply Zone) qualified.
7. Refer to detail sheet K-22, Xcel Energy Outdoor Lighting (ODL) Manual (2023).

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	VERTICAL CLEARANCE OF STREETLIGHT TO OVERHEAD ELECTRICAL CONDUCTORS	STD. DWG. NO. LD-13



OVERHEAD FED

3/4"x 8'-0" COPPER-CLAD DRIVEN GROUND ROD WITH EXOTHERMIC WELD OR UNDERGROUND LUG CONNECT CONDUCTOR TO ROD. 8'-0" GROUND ROD USED WHEN DIRECT BURIED BELOW GRADE.

UNDERGROUND FED

CAUTION - NESC RULE 234B1a & 234B1b:
STREETLIGHT MINIMUM DISTANCES FROM OVERHEAD UTILITY PRIMARY AND SECONDARY PHASE LINES: HORIZONTAL CLEARANCE: 5'-0" MINIMUM VERTICAL CLEARANCES: 4'-6" MINIMUM FROM THE MAXIMUM SAG OF PHASE DISTRIBUTION LINES AND 2'-0" MINIMUM FROM MAXIMUM SAG OF SECONDARY AND COMMUNICATIONS LINES.

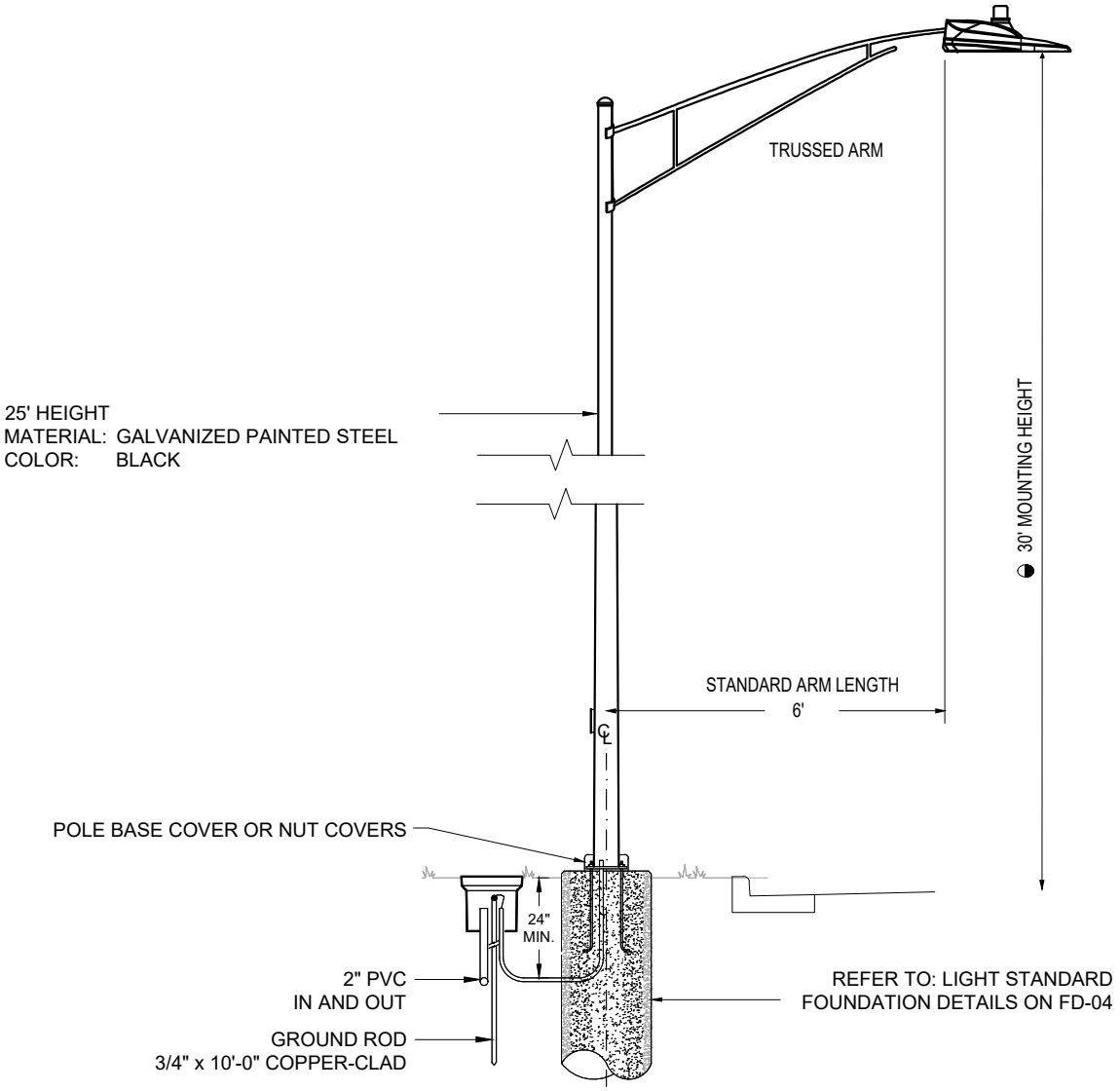
LUMINAIRE MOUNTING HEIGHTS

25-FOOT WOOD (1-ARM-6FT)	35-FOOT WOOD (1-ARM-1FT)
25-FOOT WOOD (2-ARM-6FT)	35-FOOT WOOD (2-ARM-1FT)
30-FOOT WOOD (1-ARM-6FT)	MAST ARM COLOR: BLACK
30-FOOT WOOD (2-ARM-6FT)	

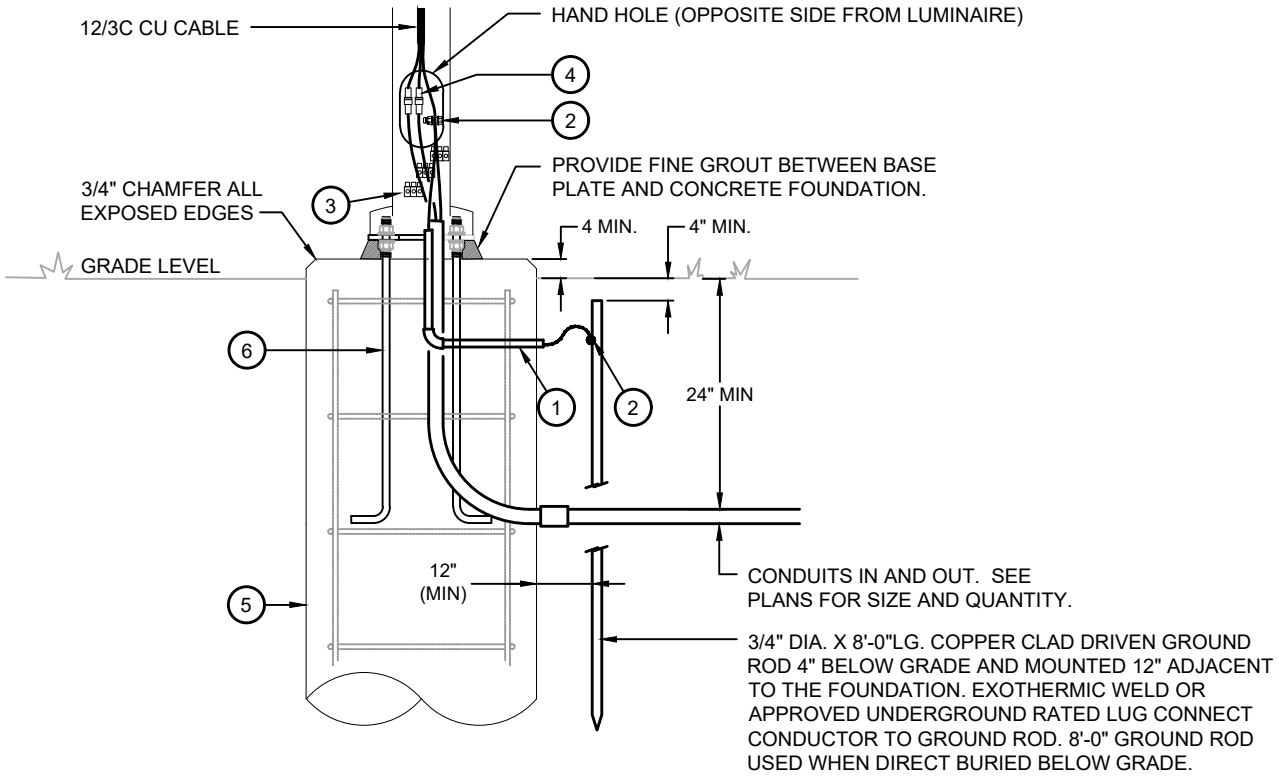
NOTES:

1. REFER TO DETAIL PS-01 FOR DISCONNECT MOUNTING HEIGHT.
2. THIS DETAIL APPLIES FOR THE FIRST STREETLIGHT FOLLOWING THE POINT OF SEPARATION FED WITH OVERHEAD ELECTRIC LINES.
3. Disconnect is only applicable for first light in OH string.

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	TYPICAL WOODEN UTILITY POLE LIGHTING DETAILS	STD. DWG. NO. LD-14



CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	LIGHT STANDARD WITH TRUSSED MAST ARM DETAIL	STD. DWG. NO. LD-15



GENERAL NOTES:

1. THIS DETAIL IS PROVIDED TO SHOW ELECTRICAL COMPONENTS. REFER TO FD-04 FOR STRUCTURAL REQUIREMENTS.
2. THIS DETAIL SHOWS A POLE BASE IN SOFTSCAPE. REFER TO DETAIL FD-03 FOR POLE BASE IN HARDSCAPE DETAIL.
3. REFER TO DETAIL LD-07 FOR FRANGIBLE (BREAKAWAY) BASE REQUIRED IN CLEAR ZONE INSTALLATIONS.
4. THIS ALTERNATIVE DETAIL IS ONLY FOR A FEED TO A SINGLE STREET LIGHT.

FOUNDATION DETAIL NOTES

(1) 1/2" PVC SLEEVE INSTALLED IN FOUNDATION FOR GROUND CONDUCTOR.

(2) EXOTHERMIC WELD OR UNDERGROUND RATED LUG (1#6 SOFT DRAWN BARE CU) TO GROUND ROD ADJACENT TO FOUNDATION AND THE OTHER END CONNECTED TO GROUNDING LUG IN POLE BASE HAND HOLE. ALSO CONNECT GREEN SYSTEM GROUND CONDUCTOR TO GROUNDING LUG.

(3) PROVIDE (3) #4-14 AWG, 3-TERMINAL, INSULATED LUG CONNECTOR, (2) HOTS, (1) NEUTRAL/GROUND INSTALLED IN POLE HAND HOLE, BASIS OF DESIGN: POLARIS #IPL4-3 4-14 AWG OR APPROVED EQUAL. MAXIMUM WIRE SIZE SHALL BE #4 AWG COPPER OR ALUMINUM. LEAVE ENOUGH SLACK IN WIRING TO REMOVE THESE CONNECTORS 8-12 INCHES FROM THE HANDHOLE.

(4) AT EACH POLE BASE HAND HOLE, PROVIDE A BREAKAWAY, WATERPROOF, IN-LINE FUSE HOLDER BY EATON OR APPROVED EQUAL WITH FNQ-R 15A FUSE FOR EACH HOT AND A BREAKAWAY, WATERPROOF CONNECTOR ON NEUTRAL (IF REQUIRED).

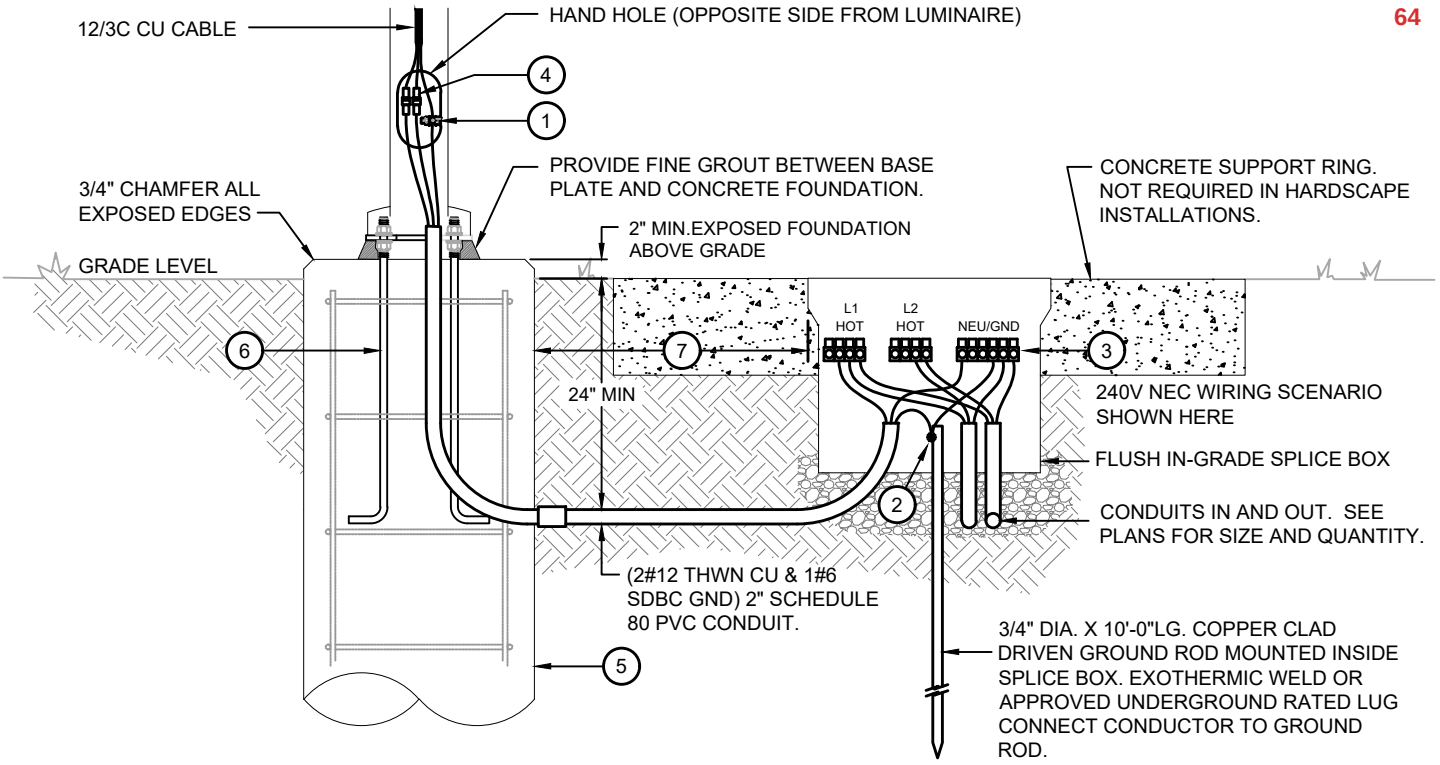
BREAKAWAY 2-POLE IN-LINE FUSE HOLDER FOR 240V CLASS CC FUSES BASIS OF DESIGN ARE AS FOLLOWS:
1. FUSE HOLDER - EATON/BUSSMAN #HEX-AW-DRLC-A
2. FUSE - EATON/BUSSMAN FNQ-15 (2) REQUIRED.

BREAKAWAY 2-POLE IN-LINE FUSE HOLDER FOR 120V CLASS CC FUSES BASIS OF DESIGN ARE AS FOLLOWS:
1. FUSE HOLDER - EATON/BUSSMAN #HEX-AW-DRLC-A FOR PHASE AND NEUTRAL CONDUCTORS.
2. FUSE - EATON/BUSSMAN FNQ-15 (1) REQUIRED.
3. DUMMY FUSE - EATON/BUSSMAN #NNB FOR NEUTRAL CONDUCTOR (1) REQUIRED.

(5) LIGHT STANDARD FOUNDATION EMBEDMENT DEPTH, DIMENSIONS AND REINFORCEMENT SHALL BE PER FD-04 OR AS DESIGNED BY A STRUCTURAL ENGINEER LICENSED IN THE STATE OF COLORADO.

(6) ANCHOR BOLT CIRCLE SHALL BE 12 INCH DIAMETER AND CENTERED ON FOUNDATION.

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	FOUNDATION ELECTRICAL DETAIL - WITHOUT SPLICE BOX	STD. DWG. NO. FD-01



GENERAL NOTES:

1. THIS DETAIL IS PROVIDED TO SHOW ELECTRICAL COMPONENTS. REFER TO FD-04 FOR STRUCTURAL REQUIREMENTS.
2. THIS DETAIL SHOWS A POLE BASE IN SOFTSCAPE. REFER TO DETAIL FD-03 FOR POLE BASE IN HARDSCAPE DETAIL.
3. REFER TO DETAIL LD-07 FOR FRANGIBLE (BREAKAWAY) BASE REQUIRED IN CLEAR ZONE INSTALLATIONS.
4. THIS DETAIL WITH PULL BOX WILL BE STANDARD FOR FEEDING A CIRCUIT WITH MULTIPLE STREET LIGHTS.

FOUNDATION DETAIL NOTES

(1) BOND RACEWAY GROUNDING CONDUCTOR TO GROUNDING LUG IN THE POLE BASE HAND HOLE.

(2) EXOTHERMIC WELD OR UNDERGROUND RATED LUG (1#6 SOFT DRAWN BARE CU) TO GROUND ROD ADJACENT TO FOUNDATION AND THE OTHER END CONNECTED TO GROUNDING LUG IN POLE BASE HAND HOLE.

(3) 120V NESC WIRING SCENARIO:
PROVIDE (2) 4-TERMINAL, (1) 6-TERMINAL SUBMERSIBLE, INSULATED LUG CONNECTOR, (2) HOTS SHALL BE 4-TERMINAL TYPE AND (1) 6-TERMINAL TYPE FOR NEUTRAL/GROUND. BASIS OF DESIGN: ABB ELASTIMOLD #USB43S and #USB63S RESPECTIVELY.
120V NEC WIRING SCENARIO:

(4) BREAKAWAY 2-POLE IN-LINE FUSE HOLDER FOR 120V CLASS CC FUSES BASIS OF DESIGN ARE AS FOLLOWS:
1. FUSE HOLDER - EATON/BUSSMAN #HEX-AW-DRLC-A FOR PHASE AND NEUTRAL CONDUCTORS.
2. FUSE - EATON/BUSSMAN FNQ-15 (1) REQUIRED.
3. DUMMY FUSE - EATON/BUSSMAN #NNB FOR NEUTRAL CONDUCTOR (1) REQUIRED.

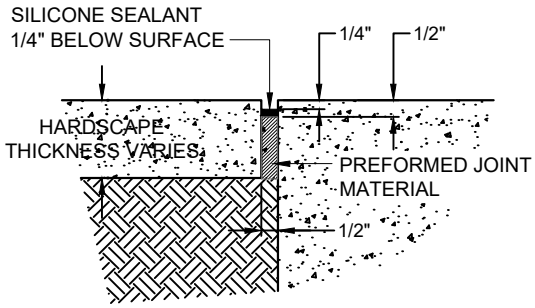
(5) LIGHT STANDARD FOUNDATION EMBEDMENT DEPTH, DIMENSIONS AND REINFORCEMENT SHALL BE PER FD-04 OR AS DESIGNED BY A STRUCTURAL ENGINEER LICENSED IN THE STATE OF COLORADO.

(6) ANCHOR BOLT CIRCLE SHALL BE 12 INCH DIAMETER AND CENTERED ON FOUNDATION.

(7) CLOSEST EDGE OF PULL BOX SHALL BE 1" TO 3" FROM FOUNDATION EXCEPT WHEN FIELD CONDITIONS DICTATE OTHERWISE.

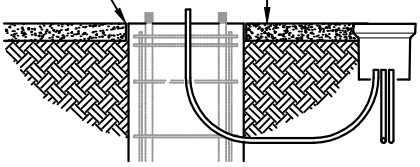
TYPICAL 120V NESC WIRING ESL CIRCUITRY

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	FOUNDATION ELECTRICAL DETAIL WITH SPLICE BOX	STD. DWG. NO. FD-02



HARDSCAPE EXPANSION JOINT DETAIL

1/2" THICK EXPANSION JOINT SHALL BE INSTALLED WHERE FOUNDATION ABUTS CONCRETE OR OTHER FIXED STRUCTURE. EXPANSION JOINT MATERIAL SHALL EXTEND THE FULL DEPTH OF CONTACT SURFACE.

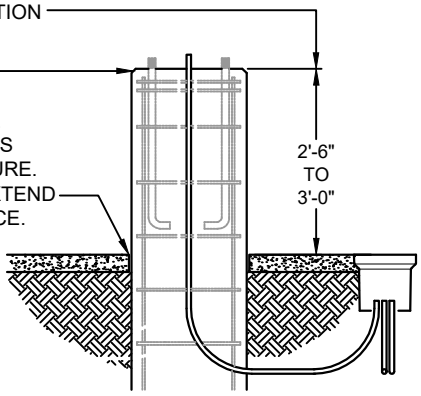


LIGHT STANDARD FOUNDATION IN HARDSCAPE

FOUNDATION HEIGHT ABOVE GRADE IS IN ADDITION TO THE REQUIRED FOUNDATION EMBEDMENT DEPTH.

3/4" CHAMFER ALL EXPOSED EDGES

1/2" THICK EXPANSION JOINT SHALL BE INSTALLED WHERE FOUNDATION ABUTS CONCRETE OR OTHER FIXED STRUCTURE. EXPANSION JOINT MATERIAL SHALL EXTEND THE FULL DEPTH OF CONTACT SURFACE.

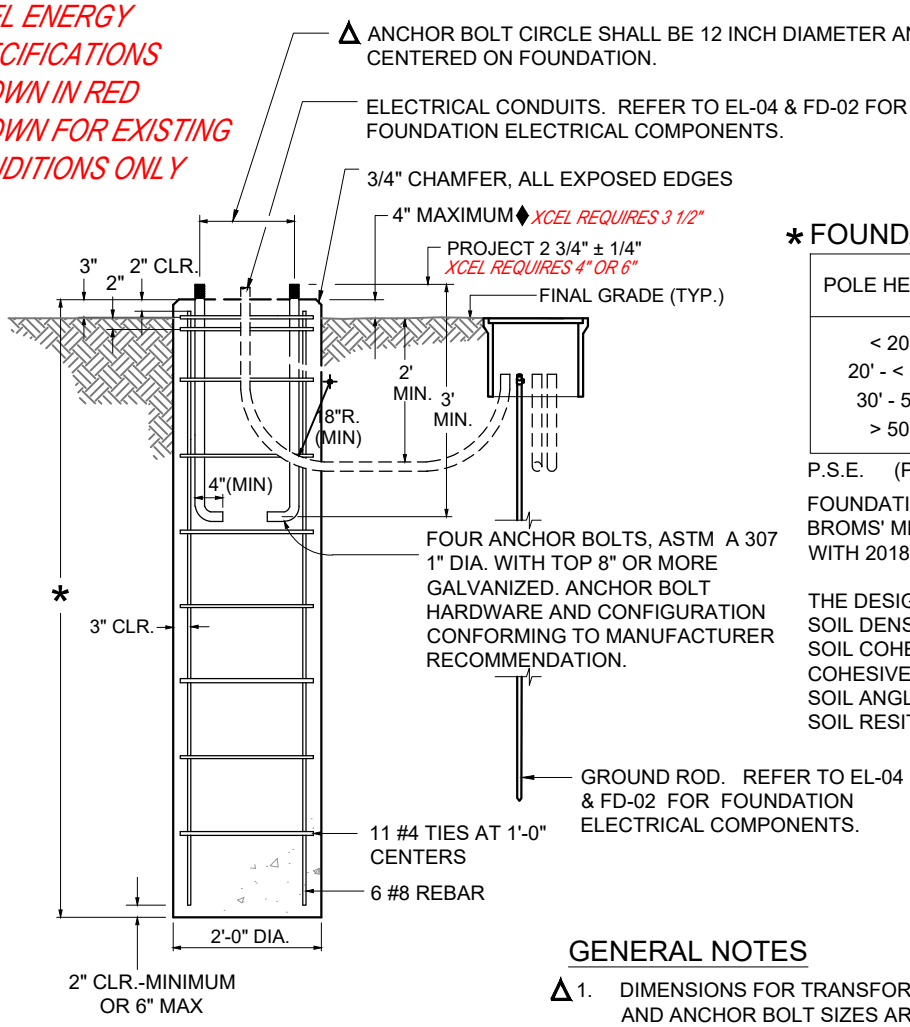


LIGHT STANDARD FOUNDATION IN PARKING LOT

WHERE LIGHT STANDARD FOUNDATIONS OCCUR IN OR AROUND PARKING AREAS AND ARE LOCATED LESS THAN 2'-0" BEHIND CURB, OR WHERE UNPROTECTED BY CURBS, THE FOUNDATION SHOULD BE EXTENDED A MINIMUM OF 2'-6" VERTICALLY, IN ADDITION TO FOUNDATION DEPTH LISTED IN THE FOUNDATION SCHEDULE ON SHEET FD-04, TO PROTECT THE LIGHT STANDARD FROM DAMAGE AND/OR KNOCK-DOWN DUE TO VEHICLE CONTACT.

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	LIGHT STANDARD FOUNDATION DETAIL IN HARDSCAPE	STD. DWG. NO. FD-03

XCEL ENERGY SPECIFICATIONS SHOWN IN RED SHOWN FOR EXISTING CONDITIONS ONLY



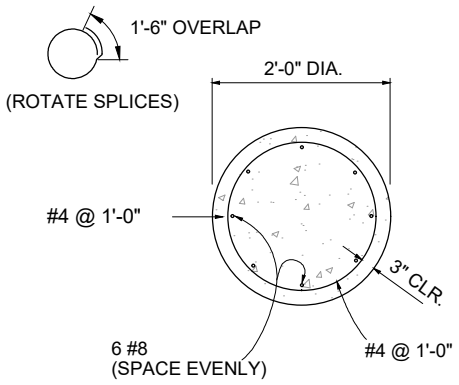
* FOUNDATION SCHEDULE

POLE HEIGHT	FOUNDATION DEPTH	FOUNDATION DIAMETER
< 20'	9'-0"	24"
20' - < 30'	10'-0"	24"
30' - 50'	13'-0"	24"
> 50'	P.S.E.	P.S.E.

P.S.E. (PER STRUCTURAL ENGINEER)

FOUNDATION DESIGN DATA:
BROMS' METHOD USING AASHTO LRFD LTS 1ST, 2015 WITH 2018 INTERIMS.

THE DESIGN ASSUMES FOLLOWING SOIL PARAMETERS:
SOIL DENSITY = 110 LB/CF
SOIL COHESION = 750 LB/SQFT FOR MEDIUM STIFF COHESIVE SOIL
SOIL ANGLE = 30° FOR MEDIUM DENSE COHESIONLESS
SOIL RESITANCE FACTOR = 0.4 FOR FLEXURE.



TYPICAL FOUNDATION SECTION

GENERAL NOTES

Δ 1. DIMENSIONS FOR TRANSFORMER BASE, ANCHOR BASE AND ANCHOR BOLT SIZES ARE VARIABLE FOR THE HEIGHT OF THE LIGHT STANDARD AND THE MAST ARM CONFIGURATION. ALL COMPONENTS SHALL FIT AND SHALL ACCOMMODATE THE REQUIREMENTS OF THE LIGHT STANDARD SUPPLIED.

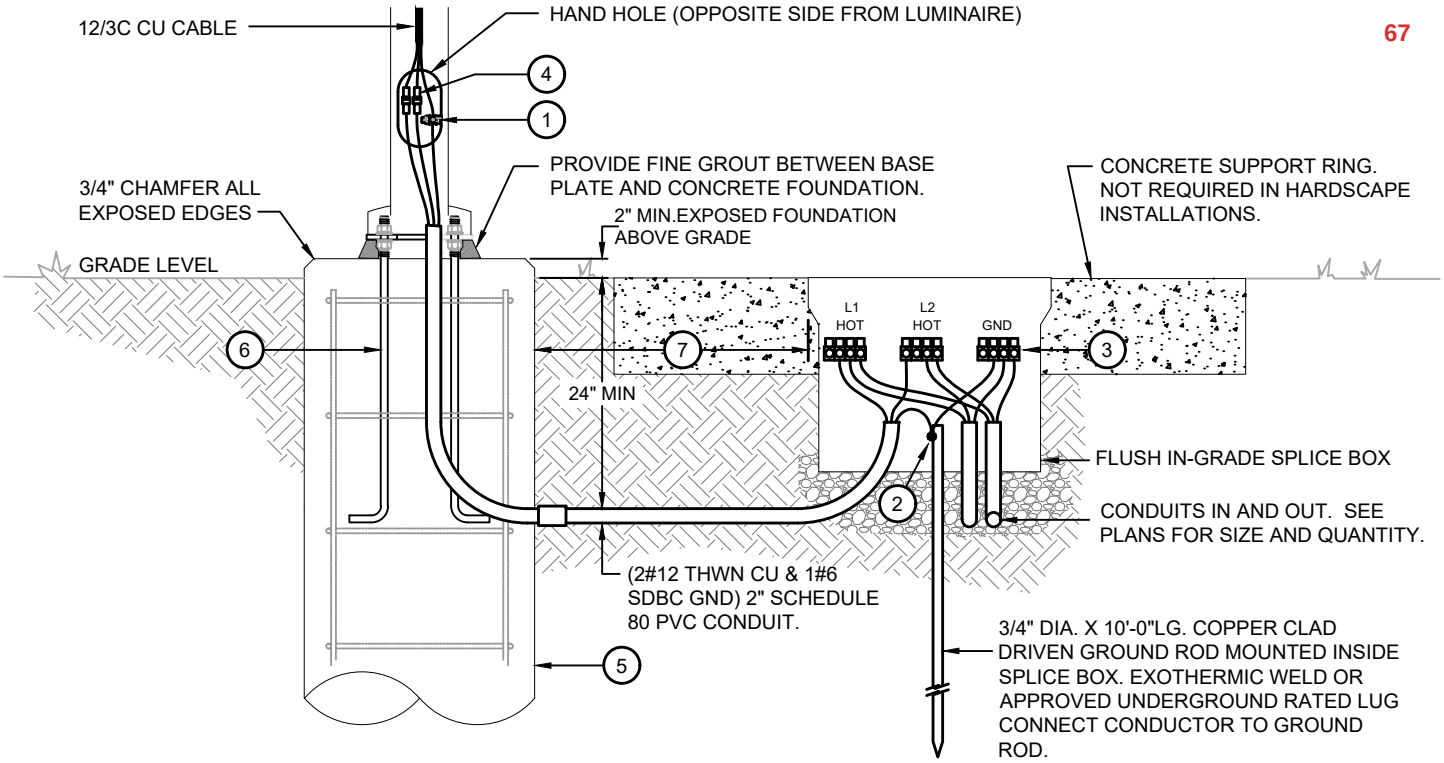
2. CONCRETE SHALL BE AIR-ENTRAINED CLASS BZ PER CDOT STANDARD 601. REINFORCEMENT SHALL BE GRADE 60 PER CDOT STANDARD 602.'.

♦ 3. WHERE FOUNDATION IS LOCATED IN THE SIDEWALK, THE TOP OF THE FOUNDATION SHALL BE FLUSH WITH THE TOP OF THE SIDEWALK CONFORMING TO ADA REQUIREMENTS.

* 4. FOUNDATION DIMENSIONS PER FOUNDATION SCHEDULE ABOVE. LIGHT STANDARDS HIGHER THAN 50'-0" OR WITH MULTIPLE LUMINAIRES OR BANNERS, OR VARYING SOIL OR WIND CONDITIONS, SHALL BE DESIGNED BY A STRUCTURAL ENGINEER LICENSED IN THE STATE OF COLORADO AND SHOWN ON THE PLANS.

XCEL REQUIRES CCD TO BE RESPONSIBLE FOR THE REMOVAL AND REPLACEMENT OF ANY FOUNDATIONS IF DAMAGED AND IF NOT SPECIFIED BY THE XCEL ENERGY OUTDOOR LIGHTING MANUAL GUIDE LINES. (48" Dp. x 24" Dia. PRECAST FOUNDATION. XCEL CAT ID# 54541)

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	LIGHT STANDARD FOUNDATION STRUCTURAL DETAIL	STD. DWG. NO. FD-04



GENERAL NOTES:

1. THIS DETAIL IS PROVIDED TO SHOW ELECTRICAL COMPONENTS. REFER TO FD-04 FOR STRUCTURAL REQUIREMENTS.
2. THIS DETAIL SHOWS A POLE BASE IN SOFTSCAPE. REFER TO DETAIL FD-03 FOR POLE BASE IN HARDSCAPE DETAIL.
3. REFER TO DETAIL LD-07 FOR FRANGIBLE (BREAKAWAY) BASE REQUIRED IN CLEAR ZONE INSTALLATIONS.
4. THIS DETAIL WITH PULL BOX WILL BE STANDARD FOR FEEDING A CIRCUIT WITH MULTIPLE STREET LIGHTS.

FOUNDATION DETAIL NOTES

(1) BOND RACEWAY GROUNDING CONDUCTOR TO GROUNDING LUG IN THE POLE BASE HAND HOLE.

(2) EXOTHERMIC WELD OR UNDERGROUND RATED LUG (1#6 SOFT DRAWN BARE CU) TO GROUND ROD ADJACENT TO FOUNDATION AND THE OTHER END CONNECTED TO GROUNDING LUG IN POLE BASE HAND HOLE.

(3) 240V NEC WIRING SCENARIO:
PROVIDE (3) 4-TERMINAL, SUBMERSIBLE, INSULATED LUG CONNECTOR, (2) HOTS, (1) GROUND. BASIS OF DESIGN: ABB ELASTIMOLD #USB43S.

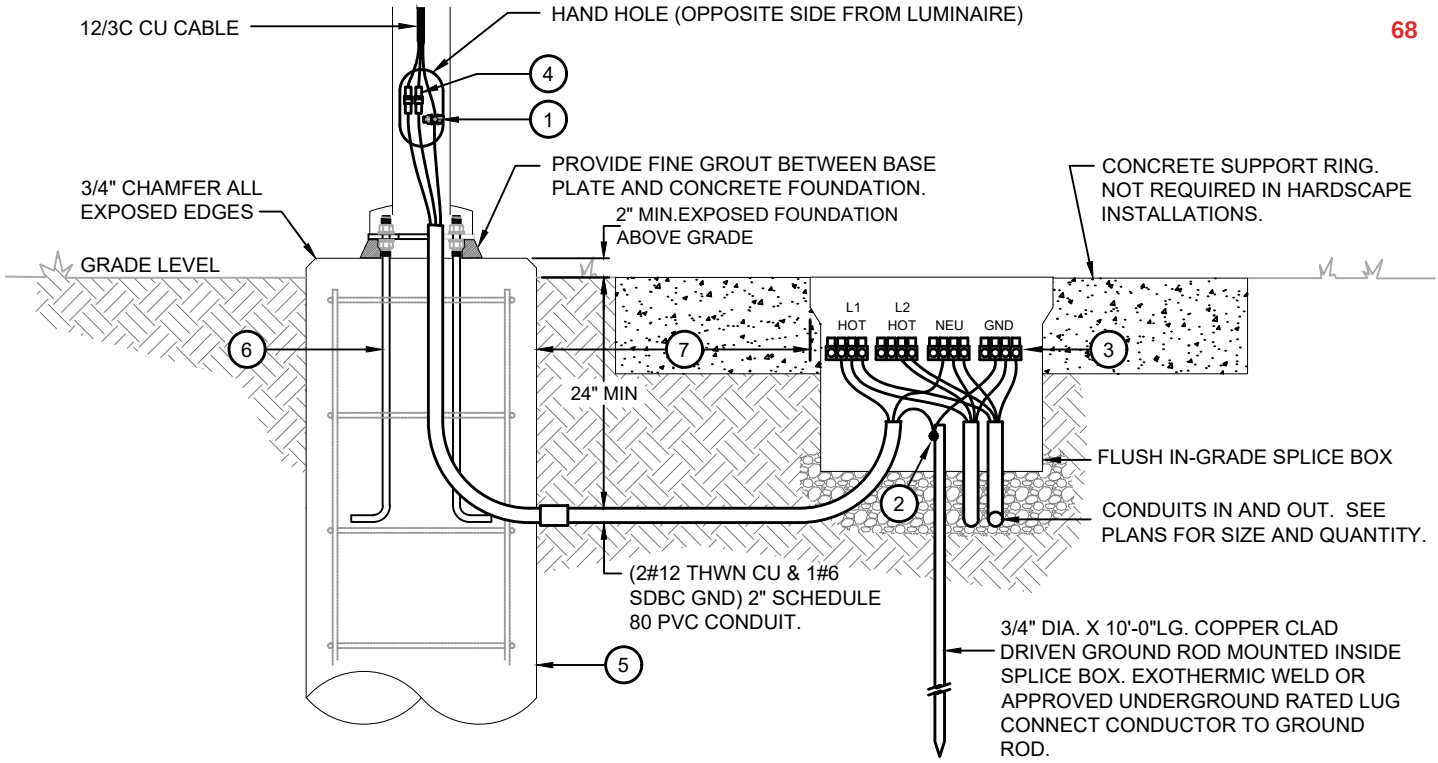
(4) AT EACH POLE BASE HAND HOLE, PROVIDE A BREAKAWAY, WATERPROOF, IN-LINE FUSE HOLDER BY EATON OR APPROVED EQUAL WITH FNQ-R 15A FUSE FOR EACH HOT AND A BREAKAWAY, WATERPROOF CONNECTOR ON NEUTRAL (IF REQUIRED).
BREAKAWAY 2-POLE IN-LINE FUSE HOLDER FOR 240V CLASS CC FUSES BASIS OF DESIGN ARE AS FOLLOWS:
1. FUSE HOLDER - EATON/BUSSMAN #HEX-AW-DRLC-A
2. FUSE - EATON/BUSSMAN FNQ-15 (2) REQUIRED.

(5) LIGHT STANDARD FOUNDATION EMBEDMENT DEPTH, DIMENSIONS AND REINFORCEMENT SHALL BE PER FD-04 OR AS DESIGNED BY A STRUCTURAL ENGINEER LICENSED IN THE STATE OF COLORADO.

(6) ANCHOR BOLT CIRCLE SHALL BE 12 INCH DIAMETER AND CENTERED ON FOUNDATION.

(7) CLOSEST EDGE OF PULL BOX SHALL BE 1" TO 3" FROM FOUNDATION EXCEPT WHEN FIELD CONDITIONS DICTATE OTHERWISE.

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	FOUNDATION ELECTRICAL DETAIL FOR 240V NEC WIRING	STD. DWG. NO. FD-05



GENERAL NOTES:

1. THIS DETAIL IS PROVIDED TO SHOW ELECTRICAL COMPONENTS. REFER TO FD-04 FOR STRUCTURAL REQUIREMENTS.
2. THIS DETAIL SHOWS A POLE BASE IN SOFTSCAPE. REFER TO DETAIL FD-03 FOR POLE BASE IN HARDSCAPE DETAIL.
3. REFER TO DETAIL LD-07 FOR FRANGIBLE (BREAKAWAY) BASE REQUIRED IN CLEAR ZONE INSTALLATIONS.
4. THIS DETAIL WITH PULL BOX WILL BE STANDARD FOR FEEDING A CIRCUIT WITH MULTIPLE STREET LIGHTS.

FOUNDATION DETAIL NOTES

(1) BOND RACEWAY GROUNDING CONDUCTOR TO GROUNDING LUG IN THE POLE BASE HAND HOLE.

(2) EXOTHERMIC WELD OR UNDERGROUND RATED LUG (1#6 SOFT DRAWN BARE CU) TO GROUND ROD ADJACENT TO FOUNDATION AND THE OTHER END CONNECTED TO GROUNDING LUG IN POLE BASE HAND HOLE.

(3) 120V NEC WIRING SCENARIO:
PROVIDE (4) 4-TERMINAL, SUBMERSIBLE, INSULATED LUG CONNECTOR, (2) HOTS, (1) NEUTRAL SHALL BE 4-TERMINAL TYPE AND (1) 4-TERMINAL TYPE FOR GROUND. BASIS OF DESIGN: ABB ELASTIMOLD #USB43S.

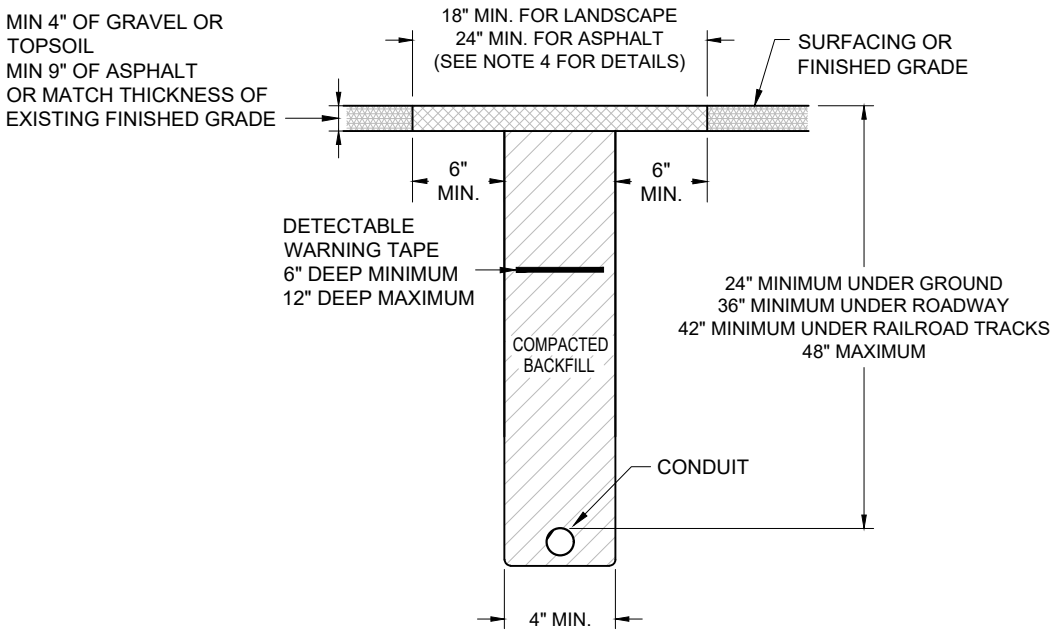
(4) BREAKAWAY 2-POLE IN-LINE FUSE HOLDER FOR 120V CLASS CC FUSES BASIS OF DESIGN ARE AS FOLLOWS:
1. FUSE HOLDER - EATON/BUSSMAN #HEX-AW-DRLC-A FOR PHASE AND NEUTRAL CONDUCTORS.
2. FUSE - EATON/BUSSMAN FNQ-15 (1) REQUIRED.
3. DUMMY FUSE - EATON/BUSSMAN #NNB FOR NEUTRAL CONDUCTOR (1) REQUIRED.

(5) LIGHT STANDARD FOUNDATION EMBEDMENT DEPTH, DIMENSIONS AND REINFORCEMENT SHALL BE PER FD-04 OR AS DESIGNED BY A STRUCTURAL ENGINEER LICENSED IN THE STATE OF COLORADO.

(6) ANCHOR BOLT CIRCLE SHALL BE 12 INCH DIAMETER AND CENTERED ON FOUNDATION.

(7) CLOSEST EDGE OF PULL BOX SHALL BE 1" TO 3" FROM FOUNDATION EXCEPT WHEN FIELD CONDITIONS DICTATE OTHERWISE.

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	FOUNDATION ELECTRICAL DETAIL FOR 120V NEC WIRING	STD. DWG. NO. FD-06



DETAIL NOTES

1. CONTRACTOR SHALL COORDINATE BORING, DRIVING, OR TRENCHING WITH OTHER UNDERGROUND UTILITIES. CONTRACTOR SHALL USE COMMON TRENCHES AT ALL ROAD CROSSINGS WHEREVER POSSIBLE. USE BORING TO INSTALL CONDUIT UNDER EXISTING ROADWAYS AND SURFACES SUCH AS PARKING LOTS WHEN THE SURFACE WILL NOT BE FULLY RECONSTRUCTED.

2. WHENEVER POSSIBLE, CONDUIT OR CABLE SHALL BE INSTALLED BY BORING, DRIVING, OR ANY OTHER ACCEPTABLE MEANS UNDER CONCRETE UNITS. OPEN CUTTING SHALL BE USED ONLY UNDER SPECIAL CIRCUMSTANCES AND ONLY WITH APPROVAL OF TRANSPORTATION AND MOBILITY.

3. MINIMUM WIDTH AND TYPE OF RESTORATION TO BE DETERMINED BY TRANSPORTATION AND MOBILITY INSPECTOR, BASED ON CONTRACTOR'S PRE-ACTIVITY PHOTOS, TO MATCH PRE-EXISTING CONDITIONS.

4. SOD REPLACEMENT SHALL BE A MINIMUM OF 18" IN WIDTH. ASPHALT REPLACEMENT SHALL BE A MINIMUM OF 24" IN WIDTH. CONCRETE REPLACEMENT SHALL BE PER THE DEPARTMENT OF TRANSPORTATION AND MOBILITY TRANSPORTATION STANDARDs.

5. ANY HARDSCAPE (CONCRETE OR PAVERS) SHALL BE REPLACED IN FULL PANELS OR PAVERS OF THE SAME TYPE, COLOR, AND SIZE AS BEFORE.

6. 1-#12 AWG LOCATE WIRE AND A NYLON OR POLYESTER PULL TAPE WITH 1,250 LBS TEST STRENGTH AND FOOTAGE MARKINGS IN ALL EMPTY CONDUITS.

7. ALL CONDUIT, LANDSCAPE RESTORATION, ASPHALT RESTORATION, AND CONCRETE RESTORATION MUST BE INSTALLED IN ACCORDANCE WITH THE CITY OF BOULDER DESIGN & CONSTRUCTION STANDARD TRANSPORTATION DETAILS, CHAPTER 11.2.

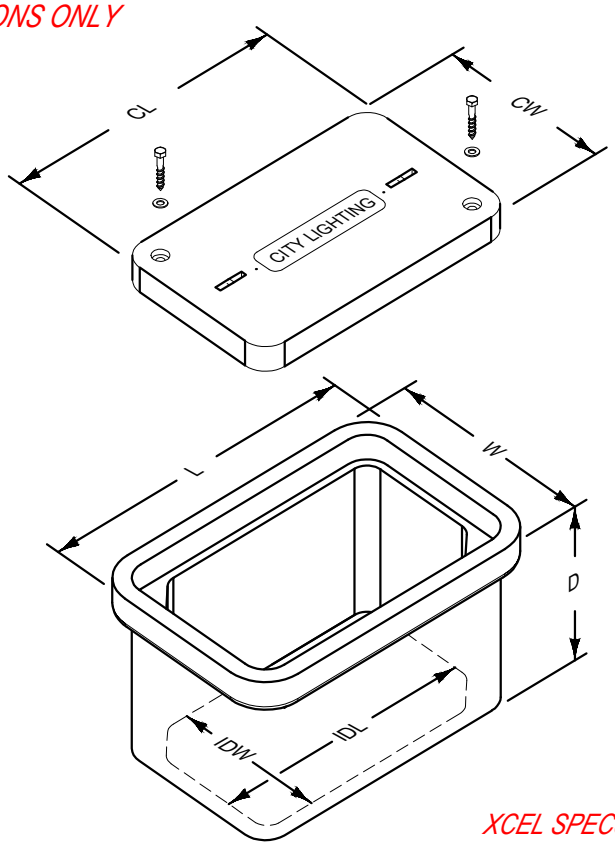
8. REFER TO NEC 300.7(D)(3) FOR WARNING TAPE REQUIREMENTS.

9. REFER TO NEC TABLE 300.7(A) FOR MINIMUM BURIAL DEPTHS BY CONDUIT MATERIAL.

10. TERMINATE BORES UNDER ROADWAYS WITH SPLICE BOX AT EACH END OF THE CONDUIT.

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	TYPICAL CONDUIT BURIAL DETAIL	STD. DWG. NO. EL-01

XCEL ENERGY SPECIFICATIONS SHOWN IN RED
SHOWN FOR EXISTING CONDITIONS ONLY



PULL BOX / SPLICE BOX - COMPOSITE MINIMUM DIMENSIONS								
TYPE 1			TYPE 2			TYPE 3		
L	=	20"	L	=	25"	L	=	32"
W	=	13"	W	=	15"	W	=	19"
D	=	12"	D	=	12"	D	=	12"
I D L	=	17"	I D L	=	21"	I D L	=	28"
I D W	=	10"	I D W	=	11"	I D W	=	15"
C L	=	18"	C L	=	23"	C L	=	30"
C W	=	11"	C W	=	13"	C W	=	17"

XCEL SPECS SHOWN FOR HISTORICAL REFERENCE ONLY

XCEL PULL BOX / SPLICE BOX - COMPOSITE MINIMUM DIMENSIONS					
SMALL			LARGE		
CAT ID #: *LID - 215570* *BODY - 215571*			*CAT ID #:* *LID - 215567* *BODY - 215569*		
L	=	*18"*	*L*	=	*24"*
W	=	*11"*	*W*	=	*13"*
D	=	*12"*	*D*	=	*12"*

DETAIL NOTES

1. BOX COVERS MUST BE POLYMER CONCRETE WITH FIBERGLASS REINFORCEMENT, INCIDENTAL TRAFFIC RATED TO TIER 22 AND BOLTED.

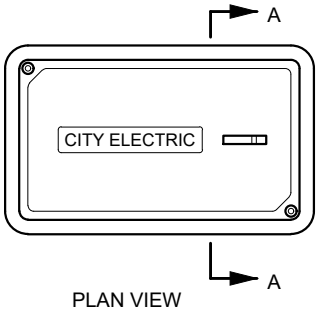
2. BOX COVERS SHALL BE LABELED AS FOLLOWS:
   - "CITY ELECTRIC" ON ALL PULL BOXES CONTAINING CITY ELECTRICAL POWER.
   - "UTILITY ELECTRIC" ON ALL PULL BOXES CONTAINING UTILITY ELECTRICAL POWER.
   - "CITY LIGHTING" ON ALL PULL BOXES CONTAINING CITY OWNED STREET LIGHTING CONDUCTORS.

   LABELING MUST BE CAST INTO THE COVER AND NOT A SEPARATE INDEPENDENT TAG.

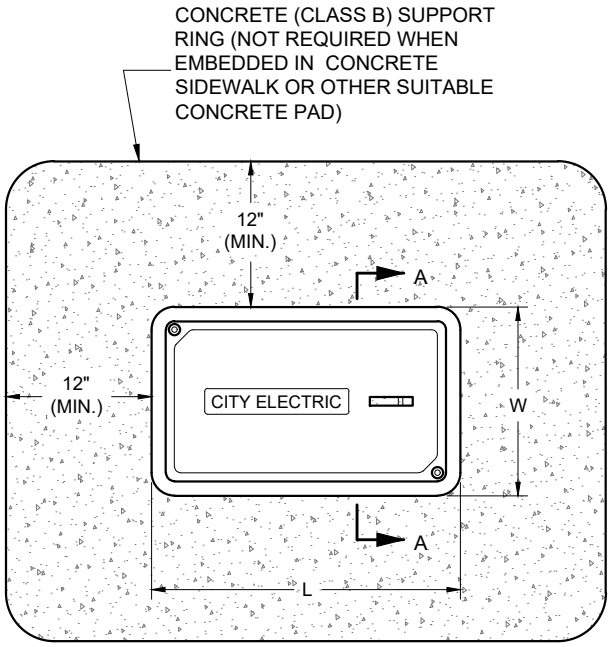
3. REFER TO N.E.C. ARTICLE 314 "PULL AND JUNCTION BOXES AND CONDUIT BODIES MINIMUM SIZE" FOR BOX SIZE REQUIREMENTS BASED ON CONDUIT AND WIRE SIZES

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	PULL BOX / SPLICE BOX TYPICAL DIMENSIONS	STD. DWG. NO. EL-02

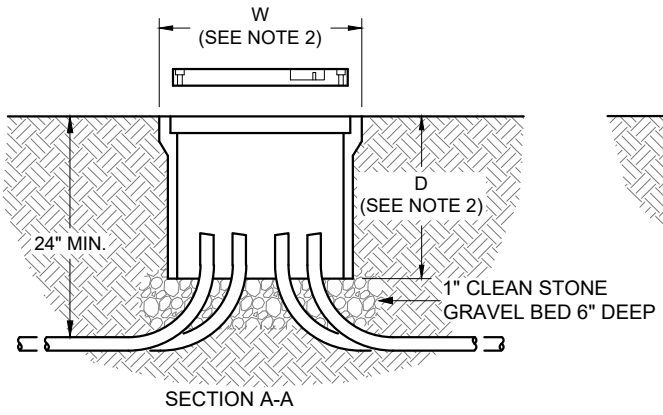
XCEL ENERGY SPECIFICATIONS SHOWN IN RED
SHOWN FOR EXISTING CONDITIONS ONLY



PLAN VIEW

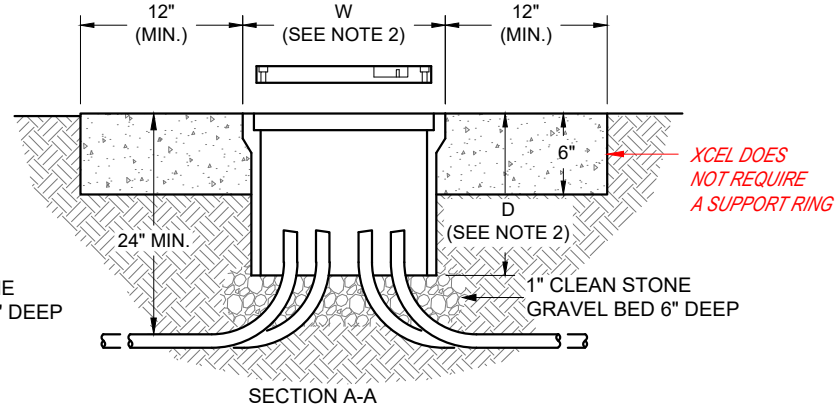


PLAN VIEW



SECTION A-A

TYPICAL PULL BOX / SPLICE BOX



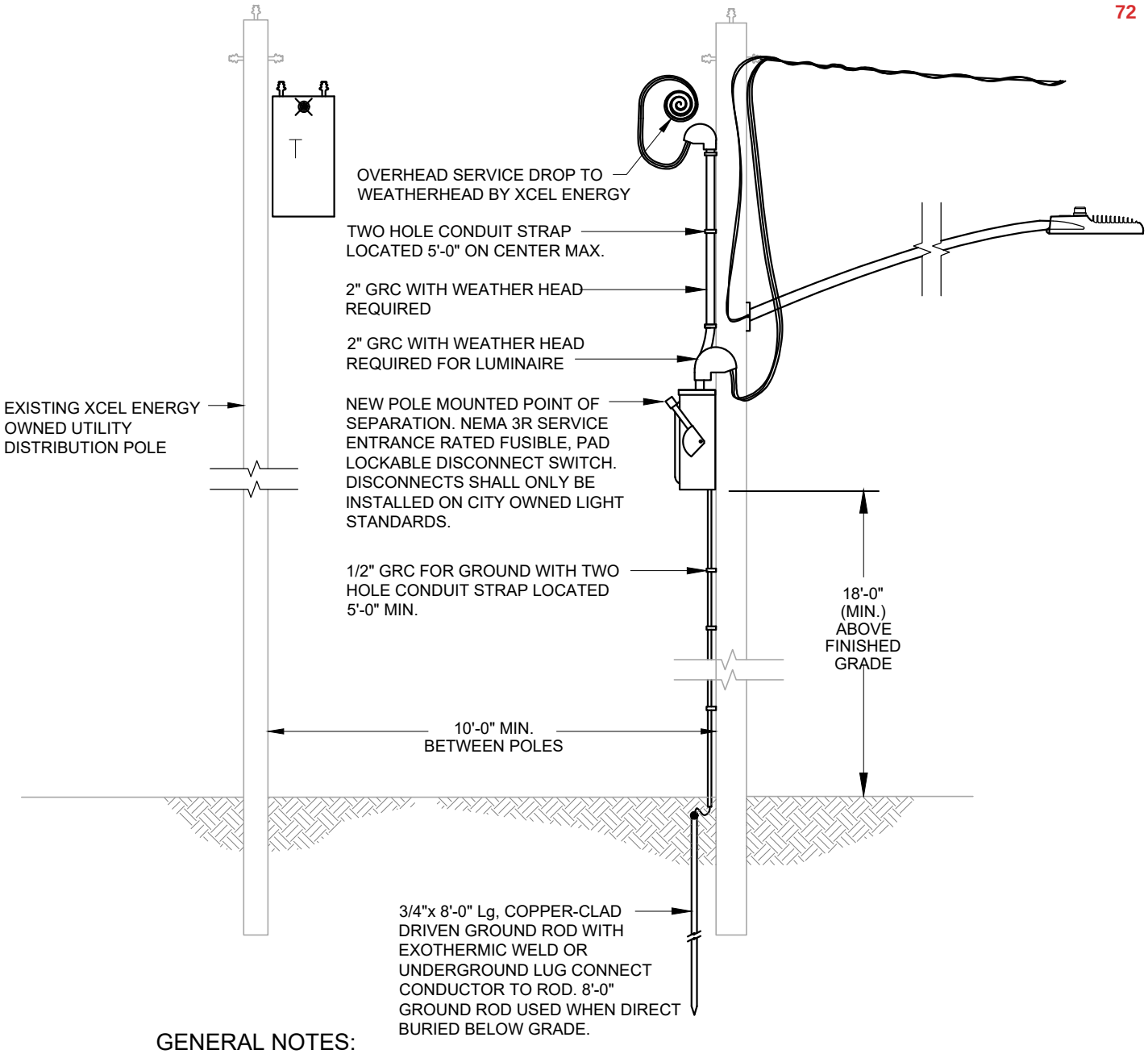
SECTION A-A

TYPICAL PULL BOX / SPLICE BOX WITH SUPPORT RING

DETAIL NOTES

1. ALL PULL OR SPLICE BOXES SHALL BE INCIDENTAL TRAFFIC RATED TO TIER 22 MINIMUM.
2. REFER TO EL-02 FOR PULL BOX SIZE REQUIREMENTS.

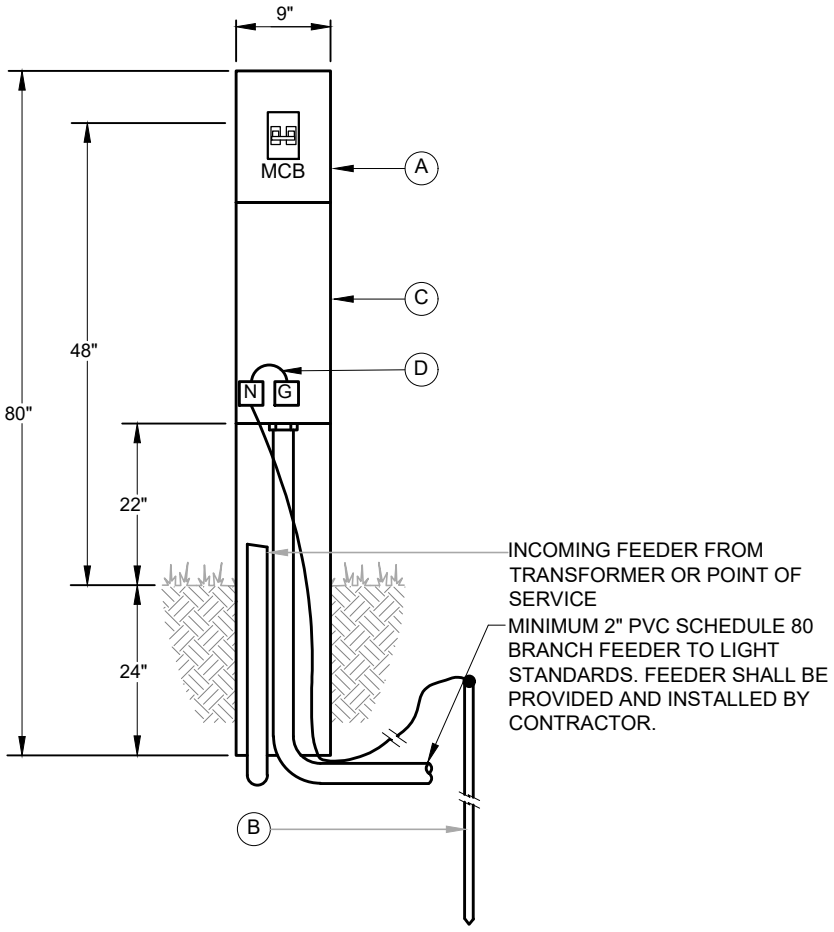
CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	IN–GRADE POLYMER CONCRETE PULL BOX / SPLICE BOX DETAIL	STD. DWG. NO. EL-03



GENERAL NOTES:

1. DISCONNECT TO BE PER XCEL ENERGY SPECIFICATIONS.
2. 30-AMP FUSIBLE DISCONNECT (FOR ONE TO TWO LIGHT STANDARDS ONLY) SHALL BE IN CONJUNCTION WITH A MIN. OF #10 CU OR #8 AL DUPLEX WIRE WITH FRN-R-30 AMP FUSES.
3. 60-AMP FUSIBLE DISCONNECT SHALL BE USED IN CONJUNCTION WITH A MIN. OF #4 AWG TRIPLEX ALUMINUM WIRE WITH FRN-R-60 AMP FUSES.
4. 100-AMP FUSIBLE DISCONNECT SHALL BE USED IN CONJUNCTION WITH A MIN. OF 1/0 TRIPLEX ALUMINUM WIRE WITH FRN-R-100 AMP FUSES.
5. PROVIDE NEUTRAL TO GROUND SERVICE ENTRANCE BONDING.
6. NEMA 3R ENCLOSURE SHALL BE LABELED WITH 'CITY OF BOULDER, ESL SERVICE'. LABEL SHALL BE PER ASTM D4956, CLASS 4.

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	POINT OF SEPARATION – TIMBER POLE MOUNTED (OVERHEAD SERVICE)	STD. DWG. NO. PS-01



COMPONENT LIST

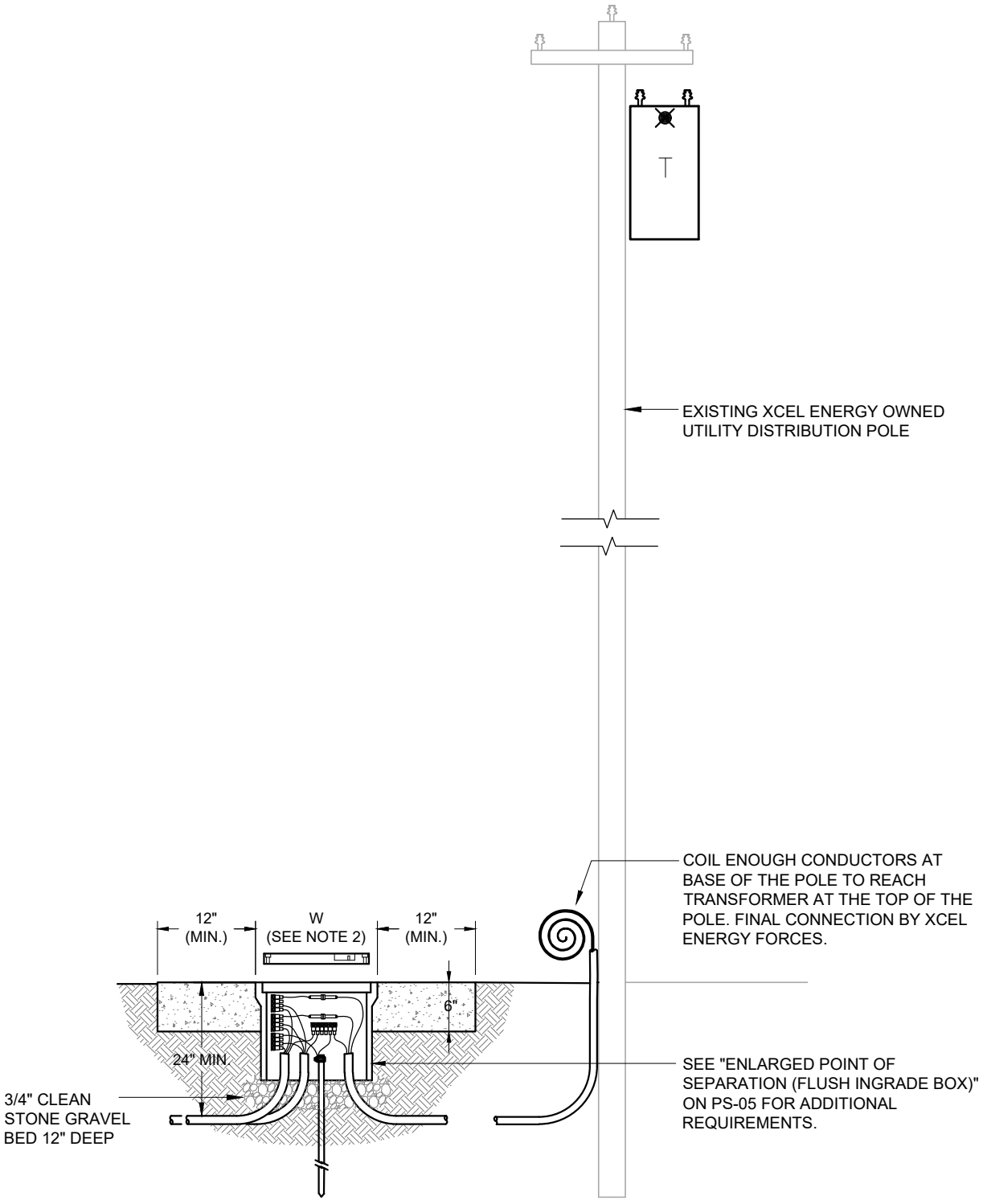
Ⓐ NEMA 3R POWER PEDESTAL WITH PAD LOCKABLE SERVICE ENTRANCE RATED BREAKER: EITHER 60A-20, 250V MAIN CIRCUIT BREAKER (MCB) OR 100A-2P, 250V MCB.

Ⓑ 3/4" by 8'-0" Lg, COPPER-CLAD DRIVEN GROUND ROD WITH EXOTHERMIC WELD OR UNDERGROUND LUG CONNECT CONDUCTOR TO ROD. 8'-0" GROUND ROD USED WHEN DIRECT BURIED BELOW GRADE.

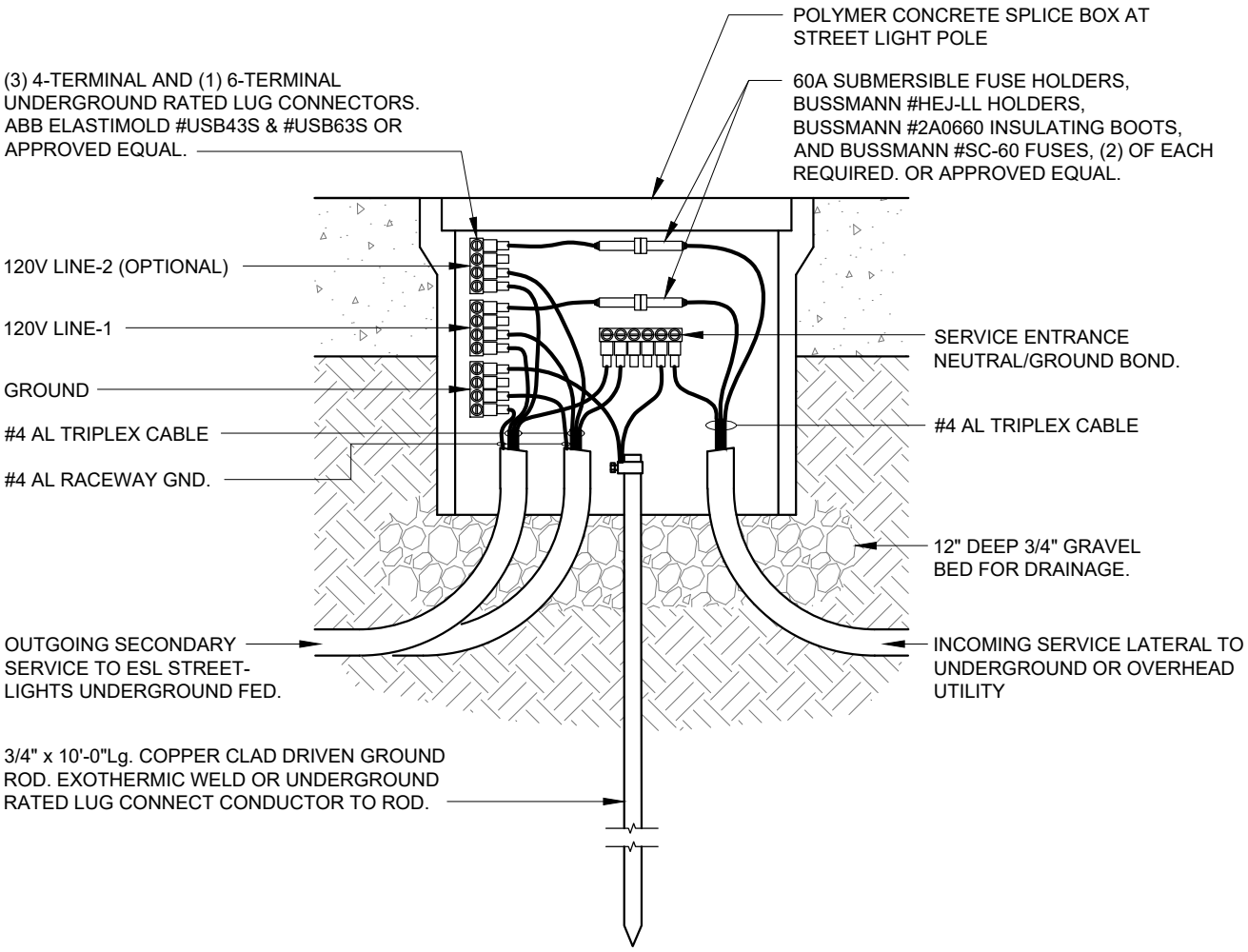
Ⓒ PEDESTAL SHALL BE LABELED WITH 'CITY OF BOULDER, ESL SERVICE'. LABEL SHALL BE PER ASTM D4956, CLASS 4.

Ⓓ PROVIDE NEUTRAL TO GROUND SERVICE ENTRANCE BONDING.

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	ENLARGED POINT OF SEPARATION – ABOVE GRADE PEDESTAL (UG/OH UTILITY)	STD. DWG. NO. PS-02



CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	POINT OF SEPARATION – FLUSH IN–GRADE BOX (OH UTILITY)	STD. DWG. NO. PS-03



GENERAL NOTES

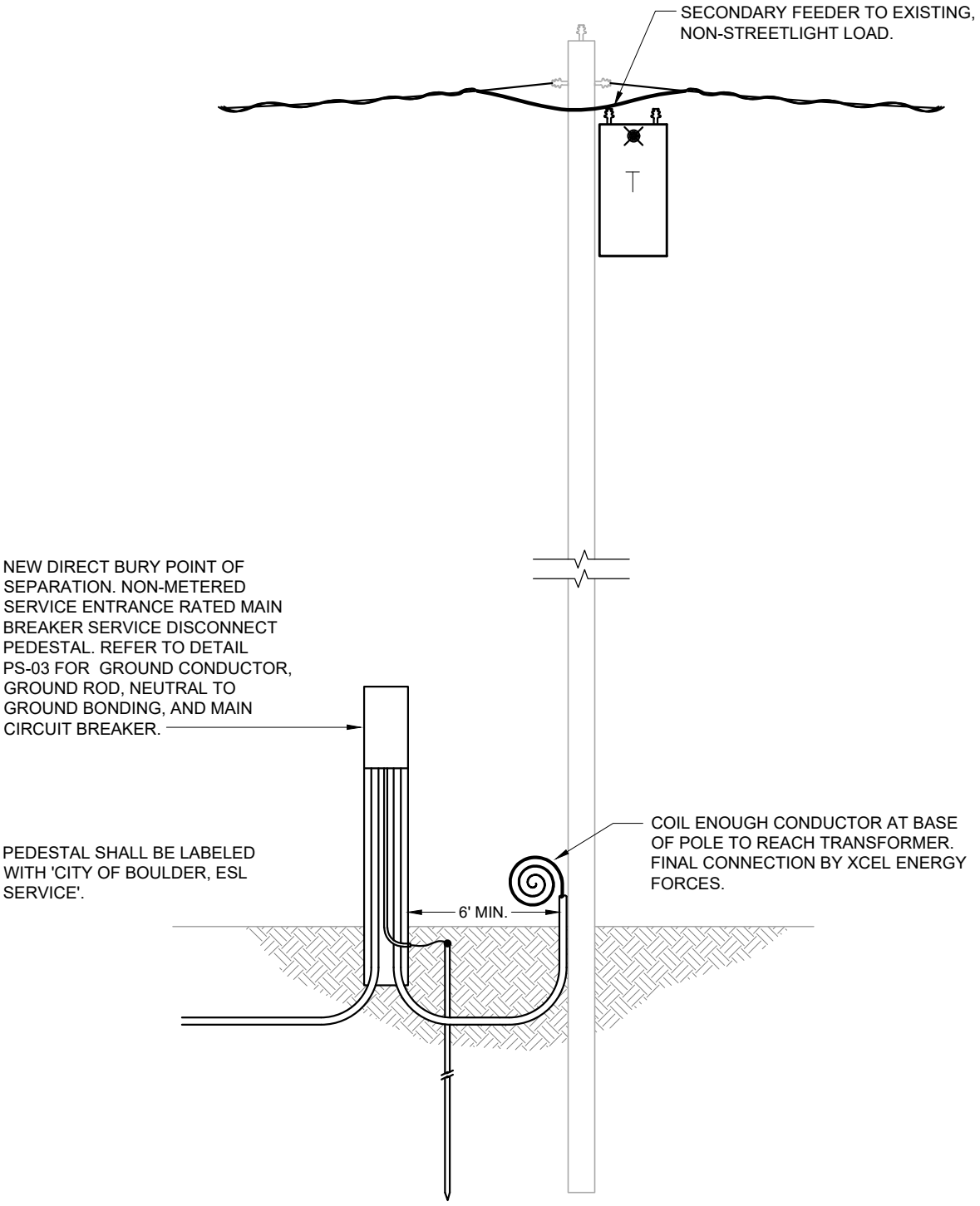
1. BOX COVERS MUST BE POLYMER CONCRETE WITH FIBERGLASS REINFORCEMENT, INCIDENTAL TRAFFIC RATED TO TIER 22.

2. BOX COVERS SHALL BE LABELED AS FOLLOWS: "CITY OF BOULDER, ESL SERVICE" LABELING MUST BE CAST INTO THE COVER AND NOT A SEPARATE INDEPENDENT TAG.

3. REFER TO N.E.C. ARTICLE 314 "PULL AND JUNCTION BOXES AND CONDUIT BODIES MINIMUM SIZE" FOR BOX SIZE REQUIREMENTS BASED ON CONDUIT AND WIRE SIZES.

4. THE WIRE TERMINATIONS IN PULL BOXES SHALL BE MADE USING URG, SUBMERSIBLE INSULATED PEDESTAL LUG CONNECTIONS WITH SUBMERSIBLE FUSES.

CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	ENLARGED POINT OF SEPARATION FLUSH IN–GRADE BOX (UG/OH UTILITY)	STD. DWG. NO. PS-04



CITY OF BOULDER PUBLIC WORKS DEPARTMENT	CITY OF BOULDER DEPARTMENT OF PUBLIC WORKS	DATE: August 24, 2026
	POINT OF SEPARATION – ABOVE GRADE PEDESTAL TO OH UTILITY	STD. DWG. NO. PS-05

APPENDIX A - NATIONAL ELECTRIC CODE (NEC) REFERENCES

NEC Article 210.19 Conductors – Minimum Ampacity and Size

(A) General.

Branch-circuit conductors shall have an ampacity not less than the larger of the following and comply with 110.14(C) for equipment terminations:

(2) Where a branch circuit supplies continuous loads or any combination of continuous and noncontinuous loads, the minimum branch-circuit conductor size shall have an ampacity not less than the noncontinuous load plus 125 percent of the continuous load in accordance with 310.14.

Exception to (1): If the assembly, including the branch-circuit OCPDs, is listed for operation at 100 percent of its rating, the ampacity of the branch-circuit conductors shall be permitted to be not less than the sum of the continuous load plus the noncontinuous load in accordance with 110.14(C).

(3) The minimum branch-circuit conductor size shall have an ampacity not less than the maximum load to be served after the application of any adjustment or correction factors in accordance with 310.15.

Exception to (1) and (2): Where a portion of a branch circuit is connected at both its supply and load ends to separately installed pressure connections as covered in 110.14(C)(2), an allowable ampacity in accordance with 310.15 not less than the sum of the continuous load plus the noncontinuous load shall be permitted. No portion of a branch circuit installed under this exception shall extend into an enclosure containing either the branch-circuit supply or the branch-circuit load terminations.

NEC Article 215.4 (A)(2) Feeders – Minimum Rating and Size - Ampacity Adjustment or Correction Factors

The minimum feeder conductor size shall have an ampacity not less than the maximum load to be served after the application of any adjustment or correction factors in accordance with 310.14.

NEC Article 250.66(A) Guidance for Grounding Electrode Conductors

“If the grounding electrode conductor or bonding jumper connected to a single or multiple rod, pipe, or plate electrode(s), or any combination thereof, as described in 250.52(A)(5) or 250.52(A)(7), does not extend on to other types of electrodes that require a larger size conductor, the grounding electrode conductor shall not be required to be larger than 6 AWG copper wire or 4 AWG aluminum or copper-clad aluminum wire.”

NEC Article 250.118(A) Types of Equipment Ground Conductors - Permitted

Each equipment grounding conductor run with or enclosing circuit conductors shall be one or more, or a combination, of the following as part of an effective ground-fault current path:

(1) A copper, aluminum, or copper-clad aluminum conductor that is solid or stranded; insulated, covered, or bare; and in the form of a wire or a busbar of any shape
(2) Rigid metal conduit
(3) Intermediate metal conduit
(4) Electrical metallic tubing
(5) Listed flexible metal conduit meeting all of the following conditions:
    a. The conduit is terminated in listed fittings.
    b. The circuit conductors contained in the conduit are protected by OCPDs rated at 20 amperes or less.
    c. The size of the conduit does not exceed metric designator 35 (trade size 1¼).
    d. The combined length of flexible metal conduit, flexible metallic tubing, and liquid-tight flexible metal conduit in the same effective ground-fault current path does not exceed 1.8 m (6 ft).
    e. If flexibility is necessary to minimize the transmission of vibration from equipment or to provide flexibility for equipment that requires movement after installation, a wire-type equipment grounding conductor or a bonding jumper in accordance with 250.102(E)(2) is installed.
    f. If flexible metal conduit is constructed of stainless steel, a wire-type equipment grounding conductor or bonding jumper in accordance with 250.102(E)(2) is installed.
(6) Listed liquid-tight flexible metal conduit meeting all of the following conditions:
    a. The conduit is terminated in listed fittings.
    b. For metric designators 12 through 16 (trade sizes ⅜ through ½), the circuit conductors contained in the conduit are protected by OCPDs rated at 20 amperes or less.
    c. For metric designators 21 through 35 (trade sizes ¾ through 1¼), the circuit conductors contained in the conduit are protected by OCPDs rated not more than 60 amperes and there is no flexible metal conduit, flexible metallic tubing, or liquid-tight flexible metal conduit in metric designators 12 through 16 (trade sizes ⅜ through ½) in the effective ground-fault current path.
    d. The combined length of flexible metal conduit, flexible metallic tubing, and liquid-tight flexible metal conduit in the same effective ground-fault current path does not exceed 1.8 m (6 ft).
    e. If flexibility is necessary to minimize the transmission of vibration from equipment or to provide flexibility for equipment that requires movement after installation, a wire-type equipment grounding conductor or a bonding jumper in accordance with 250.102(E)(2) is installed.
    f. If liquid-tight flexible metal conduit contains a stainless steel core, a wire-type equipment grounding conductor or a bonding jumper in accordance with 250.102(E)(2) is installed.
(7) Flexible metallic tubing, if the tubing is terminated in listed fittings and meets the following conditions:
    a. The circuit conductors contained in the tubing are protected by OCPDs rated at 20 amperes or less.
    b. The combined length of flexible metal conduit, flexible metallic tubing, and liquid-tight flexible metal conduit in the same effective ground-fault current path does not exceed 1.8 m (6 ft).

c. The tubing contains circuit conductors not exceeding 1000 volts, nominal.

(8) Armor of Type AC cable as provided in 320.108 containing circuit conductors not exceeding 600 volts, nominal

(9) Copper sheath of mineral-insulated, metal-sheathed cable Type MI containing circuit conductors not exceeding 600 volts, nominal

(10) Type MC cable that provides an effective ground-fault current path in accordance with one or more of the following:

a. It contains an insulated or uninsulated equipment grounding conductor complying with 250.118(A)(1).

b. The combined metallic sheath and uninsulated equipment grounding/bonding conductor of interlocked metal-tape-type MC cable is listed and identified as an equipment grounding conductor.

c. The metallic sheath or the combined metallic sheath and equipment grounding conductors of the smooth or corrugated tube-type MC cable is listed and identified as an equipment grounding conductor.

(11) Cable trays as permitted in 392.10 and 392.6

(12) Cable bus framework as permitted in 370.60(1)

(13) Other listed electrically continuous metal raceways and listed auxiliary gutters.

(14) Surface metal raceways listed for grounding and containing circuit conductors not exceeding 600 volts, nominal.

Informational Note: See Article 100 for a definition of effective ground-fault current path.

NEC Article 250.122(D) Size of Equipment Ground Conductors (EGCs) – Increased in Size

If ungrounded conductors are increased in size for any reason other than as required in 310.15(B) or 310.15(C), wire-type EGCs, if installed, shall be increased in size proportionately to the increase in circular mil area of the ungrounded conductors.

Exception: EGCs shall be permitted to be sized by a qualified person to provide an effective ground-fault current path in accordance with 250.4(A)(5) or 250.4(B)(4).

Table 300.7(A) Wiring Methods and Materials – Underground Installations – Minimum Cover Requirements

Direct-buried cable, conduit, or other raceways shall be installed to meet the minimum cover requirements of Table 300.7(A).

Table 300.7(A) Minimum Cover Requirements, 0 to 1000 Volts ac, 1500 Volts dc, Nominal, Burial in Millimeters (Inches)

Location of Wiring Method or Circuit	Type of Wiring Method or Circuit									
	Column 1 Direct Burial Cables or Conductors		Column 2 Rigid Metal Conduit or Intermediate Medal Conduit		Column 3 Electrical Metallic Tubing, Nonmetallic Raceways Listed for Direct Burial Without Concrete Encasement, or Other Approved Raceways		Column 4 Residential Branch Circuits Rated 120 Volts or Less with GFCI Protection and Maximum Overcurrent Protection of 20 Amperes		Column 5 Circuits for Control of Irrigation and Landscape Lighting Limited to Not More Than 30 Volts and Installed with Type UF or in Other Identified Cable or Raceway	
	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.
All locations not specified below	600	24	150	6	450	18	300	12	150	6
In trench below 50 mm (2 in.) thick concrete or equivalent	450	18	150	6	300	12	150	6	150	6
Under a building	0 (in raceway or Type MC or Type MI cable identified for direct burial)	0	0	0	0	0	0 (in raceway or Type MC or Type MI cable identified for direct burial)	0	0 (in raceway or Type MC or Type MI cable identified for direct burial)	0
Under minimum of 102 mm (4 in.) thick concrete exterior slab with no vehicular traffic and the slab extending not less than 152 mm (6 in.) beyond the underground installation	450	18	100	4	100	4	150 (direct burial) 100 (in raceway)	6 4	150 (direct burial) 100 (in raceway)	6 4

[1] A lesser depth shall be permitted where specified in the installation instructions of a listed low-voltage lighting system.

[2] A depth of 150 mm (6 in.) shall be permitted for pool, spa, and fountain lighting, installed in a nonmetallic raceway, limited to not more than 30 volts where part of a listed low-voltage lighting system.

Notes:

1. Cover shall be defined as the shortest distance in mm (in.) measured between a point on the top surface of any direct-buried conductor, cable, conduit, or other raceway and the top surface of finished grade, concrete, or similar cover.
2. Raceways approved for burial only where concrete encased shall require a concrete envelope not less than 50 mm (2 in.) thick.
3. Lesser depths shall be permitted where cables and conductors rise for terminations or splices or where access is otherwise required.
4. Where one of the wiring method types listed in Columns 1 through 3 is used for one of the circuit types in Columns 4 and 5, the shallowest depth of burial shall be permitted.

Table 300.7(A) Minimum Cover Requirements, 0 to 1000 Volts ac, 1500 Volts dc, Nominal, Burial in Millimeters (Inches)

Location of Wiring Method or Circuit	Type of Wiring Method or Circuit									
	Column 1 Direct Burial Cables or Conductors		Column 2 Rigid Metal Conduit or Intermediate Medal Conduit		Column 3 Electrical Metallic Tubing, Nonmetallic Raceways Listed for Direct Burial Without Concrete Encasement, or Other Approved Raceways		Column 4 Residential Branch Circuits Rated 120 Volts or Less with GFCI Protection and Maximum Overcurrent Protection of 20 Amperes		Column 5 Circuits for Control of Irrigation and Landscape Lighting Limited to Not More Than 30 Volts and Installed with Type UF or in Other Identified Cable or Raceway	
	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.
Under streets, highways, roads, alleys, driveways, and parking lots	600	24	600	24	600	24	600	24	600	24
One- and two-family dwelling driveways and outdoor parking areas, and used only for dwelling-related purposes	450	18	450	18	450	18	300	12	450	18
In or under airport runways, including adjacent areas where trespassing is prohibited	450	18	450	18	450	18	450	18	450	18

[1] A lesser depth shall be permitted where specified in the installation instructions of a listed low-voltage lighting system.

[2] A depth of 150 mm (6 in.) shall be permitted for pool, spa, and fountain lighting, installed in a nonmetallic raceway, limited to not more than 30 volts where part of a listed low-voltage lighting system.

Notes:

1. Cover shall be defined as the shortest distance in mm (in.) measured between a point on the top surface of any direct-buried conductor, cable, conduit, or other raceway and the top surface of finished grade, concrete, or similar cover.
2. Raceways approved for burial only where concrete encased shall require a concrete envelope not less than 50 mm (2 in.) thick.
3. Lesser depths shall be permitted where cables and conductors rise for terminations or splices or where access is otherwise required.
4. Where one of the wiring method types listed in Columns 1 through 3 is used for one of the circuit types in Columns 4 and 5, the shallowest depth of burial shall be permitted.

NEC Article 300.7(D)(3) Wiring Methods and Materials – Underground Installations – Protection from Damage

Conductors and cables shall be protected from damage in accordance with 300.7(D)(1) through 300.7(D)(4).

(1) Emerging from Grade.
Direct-buried conductors and cables emerging from grade and specified in Columns 1 and 4 of Table 300.7(A) shall be protected by enclosures or raceways extending from the minimum cover distance below grade required by 300.7(A) to a point at least 2.5 m (8 ft) above finished grade. In no case shall the protection be required to exceed 450 mm (18 in.) below finished grade.

(2) Conductors Entering Buildings.
Conductors entering a building shall be protected to the point of entrance.

(3) Service Conductors and Raceways.
Direct buried service conductors and raceways containing service conductors that are not encased in concrete and that are buried 450 mm (18 in.) or more below grade shall have their location identified by a warning ribbon that is placed in the trench at least 300 mm (12 in.) above the underground installation.

(4) Enclosure or Raceway Damage.
Where the enclosure or raceway is subject to physical damage, the conductors shall be installed in electrical metallic tubing, rigid metal conduit, intermediate metal conduit, RTRC-XW, Schedule 80 PVC conduit, or equivalent.

NEC Table 310.15(C)(1) Adjustment Factors for More Than Three Current-Carrying Conductors

The ampacity of each conductor shall be reduced as shown in Table 310.15(C)(1) where the number of current-carrying conductors in a raceway or cable exceeds three, installed without maintaining spacing for a continuous length longer than 600 mm (24 in.).

The ampacity of each conductor shall be reduced as shown in Table 310.15(C)(1) where single conductors or multiconductor cables not installed in raceways are installed without maintaining spacing for a continuous length longer than 600 mm (24 in.).

Each current-carrying conductor of a paralleled set of conductors shall be counted as a current-carrying conductor.

Where conductors of different systems, as provided in 300.5, are installed in a common raceway or cable, the adjustment factors shown in Table 310.15(C)(1) shall apply only to the number of power and lighting conductors.

Table 310.15(C)(1) Adjustment Factors for More Than Three Current-Carrying Conductors

Number of Conductors*	Percent of Values in Table 310.16, 310.18, and 310.20 as Adjusted for Ambient Temperature if Necessary
4-6	80
7-9	70
10-20	50
21-30	45
31-40	40
41 and above	35

*Number of conductors is the total number of conductors in the raceway or cable, including spare conductors. The count shall be adjusted in accordance with 310.15(E) and 310.15(F). The count shall not include conductors that are connected to electrical components that cannot be simultaneously energized.

NEC Table 310.16 Ampacities of Insulated Conductors with Not More Than Three Current-Carrying Conductors in Raceway, Cable, or Earth (Directly Buried)

The ampacities shall be as specified in Table 310.16 where all of the following conditions apply:

(1) Conductors are rated 0 volts through 2000 volts.
(2) Conductors are rated 60°C (140°F), 75°C (167°F), or 90°C (194°F).
(3) Wiring is installed in a 30°C (86°F) ambient temperature.
(4) There are not more than three current-carrying conductors.

Table 310.16 Ampacities of Insulated Conductors with Not More Than Three Current-Carrying Conductors in Raceway, Cable, or Earth (Directly Buried)

Size AWG or kcmil	Temperature Rating of Conductor (See Table 310.4(1))						Size AWG or kcmil
	60°C (140°F)	75°C (167°F)	90°C (194°F)	60°C (140°F)	75°C (167°F)	90°C (194°F)	
	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE, ZW	Types TBS, SA, SIS, FEP, FEPB, MI, PFA, RHH, RHW-2, THHN, THHW, THW-2, THWN-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN, Z, ZW-2	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE	Types TBS, SA, SIS, THHN, THHW, THW-2, THWN-2, RHH, RHW-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN	
	Copper			Aluminum or Copper-clad Aluminum			
16	10	15	20	-	-	-	-
14	15	20	25	10	15	20	14
12	20	25	30	15	20	25	12
10	30	35	40	25	30	35	10
8	40	50	55	35	40	45	8
6	55	65	75	40	50	55	6
4	70	85	95	55	65	75	4
3	85	100	115	65	75	85	3
2	95	115	130	75	90	100	2
1	110	130	145	86	100	115	1
1/0	125	150	170	100	120	135	1/0
2/0	145	175	195	115	135	150	2/0
3/0	165	200	225	130	155	175	3/0
4/0	195	230	260	150	180	205	4/0

Notes:

1. Section 310.15(B) shall be referenced for ampacity correction factors where the ambient temperature is other than 30°C (86°F).
2. Section 310.15(C) shall be referenced for more than three current-carrying conductors.
3. Section 310.16 shall be referenced for conditions of use.
4. Ampacity for 18 AWG copper Type MI and Type MTW shall be 14 amperes at 90°C (194°F).
5. Ampacity for 16 AWG copper Type MI and Type MTW shall be 18 amperes at 90°C (194°F).

[1] Section 240.4(D) shall be referenced for conductor overcurrent protection limitations, except as modified elsewhere in the code.

[2] Ampacity shall be used for adjustment or correction only.

[3] Ampacity shall be applicable only to copper-clad aluminum conductors.

Table 310.16 Ampacities of Insulated Conductors with Not More Than Three Current-Carrying Conductors in Raceway, Cable, or Earth (Directly Buried)

Size AWG or kcmil	Temperature Rating of Conductor (See Table 310.4(1))						Size AWG or kcmil
	60°C (140°F)	75°C (167°F)	90°C (194°F)	60°C (140°F)	75°C (167°F)	90°C (194°F)	
	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE, ZW	Types TBS, SA, SIS, FEP, FEPB, MI, PFA, RHH, RHW-2, THHN, THHW, THW-2, THWN-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN, Z, ZW-2	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE	Types TBS, SA, SIS, THHN, THHW, THW-2, THWN-2, RHH, RHW-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN	
	Copper			Aluminum or Copper-clad Aluminum			
250	215	255	290	170	205	230	250
300	240	285	320	195	230	260	300
350	260	310	350	210	250	280	350
400	280	335	380	225	270	305	400
500	320	380	430	260	310	350	500
600	350	420	475	285	340	385	600
700	385	460	520	315	375	425	700
750	400	475	535	320	385	435	750
800	410	490	555	330	395	445	800
900	435	520	585	355	425	480	900
1,000	455	545	615	375	445	500	1,000
1,250	495	590	665	405	485	545	1,250
1,500	525	625	705	435	520	585	1,500
1,750	545	650	735	455	545	615	1,750
2,000	555	665	750	470	560	630	2,000

Notes:

1. Section 310.15(B) shall be referenced for ampacity correction factors where the ambient temperature is other than 30°C (86°F).
2. Section 310.15(C) shall be referenced for more than three current-carrying conductors.
3. Section 310.16 shall be referenced for conditions of use.
4. Ampacity for 18 AWG copper Type MI and Type MTW shall be 14 amperes at 90°C (194°F).
5. Ampacity for 16 AWG copper Type MI and Type MTW shall be 18 amperes at 90°C (194°F).

[1] Section 240.4(D) shall be referenced for conductor overcurrent protection limitations, except as modified elsewhere in the code.

[2] Ampacity shall be used for adjustment or correction only.

[3] Ampacity shall be applicable only to copper-clad aluminum conductors.

NEC Table 310.17 Ampacities of Single-Insulated Conductors in Free Air

The ampacities shall be as specified in Table 310.17 where all of the following conditions apply:

(1) Conductors are rated 0 volts through 2000 volts.
(2) Conductors are rated 60°C (140°F), 75°C (167°F), or 90°C (194°F).
(3) Wiring is installed in a 30°C (86°F) ambient temperature.

Table 310.17 Ampacities of Single-Insulated Conductors in Free Air

Size AWG or kcmil	Temperature Rating of Conductor (See Table 310.4(1))						Size AWG or kcmil
	60°C (140°F)	75°C (167°F)	90°C (194°F)	60°C (140°F)	75°C (167°F)	90°C (194°F)	
	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, ZW	Types TBS, SA, SIS, FEP, FEPB, MI, PFA, RHH, RHW-2, THHN, THHW, THW-2, THWN-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN, Z, ZW-2	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN	Types TBS, SA, SIS, THHN, THHW, THW-2, THWN-2, RHH, RHW-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN	
	Copper			Aluminum or Copper-clad Aluminum			
18	-	-	18	-	-	-	-
16	-	-	24	-	-	-	-
14*	25	30	35	-	-	-	-
12*	30	35	40	25	30	35	12*
10*	40	50	55	35	40	45	10*
8	60	70	80	45	55	60	8
6	80	95	105	60	75	85	6
4	105	125	140	80	100	115	4
3	120	145	165	95	115	130	3
2	140	170	190	110	135	150	2
1	165	195	220	130	155	175	1
1/0	195	230	260	150	180	205	1/0
2/0	225	265	300	175	210	235	2/0
3/0	260	310	350	200	240	270	3/0
4/0	300	360	405	235	280	315	4/0

Notes:
1. Section 310.15(B) shall be referenced for ampacity correction factors where the ambient temperature is other than 30°C (86°F).
2. Section 310.17 shall be referenced for conditions of use.
*Section 240.4(D) shall be referenced for conductor overcurrent protection limitations, except as modified elsewhere in the code.

Table 310.17 Ampacities of Single-Insulated Conductors in Free Air

Size AWG or kcmil	Temperature Rating of Conductor (See Table 310.4(1))						Size AWG or kcmil
	60°C (140°F)	75°C (167°F)	90°C (194°F)	60°C (140°F)	75°C (167°F)	90°C (194°F)	
	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE, ZW	Types TBS, SA, SIS, FEP, FEPB, MI, PFA, RHH, RHW-2, THHN, THHW, THW-2, THWN-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN, Z, ZW-2	Types TW, UF	Types RHW, THHW, THW, THWN, XHHW, XHWN, USE	Types TBS, SA, SIS, THHN, THHW, THW-2, THWN-2, RHH, RHW-2, USE-2, XHH, XHHW, XHHW-2, XHWN-2, XHHN	
	Copper			Aluminum or Copper-clad Aluminum			
250	340	405	455	265	315	355	250
300	375	445	500	290	350	395	300
350	420	505	570	330	395	445	350
400	455	545	615	355	425	480	400
500	515	620	700	405	485	545	500
600	575	690	780	455	545	615	600
700	630	755	850	500	595	670	700
750	655	785	885	515	620	700	750
800	680	815	920	535	645	725	800
900	730	870	980	580	700	790	900
1,000	780	935	1,055	625	750	845	1,000
1,250	890	1,065	1,200	710	855	965	1,250
1,500	980	1,175	1,325	795	950	1,070	1,500
1,750	1,070	1,280	1,445	875	1,050	1,185	1,750
2,000	1,155	1,385	1,560	960	1,150	1,295	2,000

Notes:
1. Section 310.15(B) shall be referenced for ampacity correction factors where the ambient temperature is other than 30°C (86°F).
2. Section 310.17 shall be referenced for conditions of use.
*Section 240.4(D) shall be referenced for conductor overcurrent protection limitations, except as modified elsewhere in the code.

NEC Article 352.24 (B) Rigid Polyvinyl Chloride Conduit (PVC) – Installation – Bends – Number in One Run

The total degrees of bends in a conduit run shall not exceed 360 degrees between pull points.

Chapter 9, Table 1, Conduit Fill

Table 1 Percent of Cross Section of Conduit and Tubing for Conductors and Cables	
Number of Conductors and/or Cables	Cross-Sectional Area (%)
1	53
2	31
Over 2	40

Note (2) - Table 1 applies only to complete conduit or tubing systems and is not intended to apply to sections of conduit or tubing used to protect exposed wiring and cable from physical damage.

All references in these standards cite the following version of the NFPA:

NFPA 70®, National Electrical Code® (Quincy: National Fire Protection Association, 2023)

CONSTRUCTION CONTRACT
(Design-Bid-Build)

Department: **Department Name**

Project Name: **Project Name**
Project Summary: **Project Summary**

Contractor: **Contractor's Legal Name**
Contract Amount: **$Amount**

EXHIBITS

Exhibit A	Solicitation
Exhibit B	Response to Solicitation
Exhibit C	General Conditions
Exhibit D	Special Conditions
Exhibit E	Federal Aid Provisions

EFFECTIVE DATE: _____

This Construction Contract (this "Contract") is made and entered into as of the date appearing on the title page (above) (the "Effective Date") between the City of Boulder, a Colorado home rule municipality (the "City"), and **Contractor's legal name**, a **Contractor's state of incorporation Contractor's form of entity** ("Contractor"). The City and Contractor may hereinafter be referred to individually as a "Party" or collectively as the "Parties."

Defined terms for this Contract, including these pages comprising the main body of this Contract, are contained at **Appendix 1** of **Exhibit C**.

RECITALS

A. In accordance with the City's purchasing ordinance, the City issued a solicitation known as **RFB/RFP No. (and Title, if you like)** (the "Solicitation"), for the construction of **project name or**

describe the project , as further described herein (the "Project"). The Solicitation, including any Addendum, is attached as **Exhibit A** and incorporated by this reference.

B. Contractor responded to the Solicitation, and Contractor's response is attached as **Exhibit B**.

C. The City has evaluated proposers and has selected Contractor to perform the Work.

AGREEMENT

NOW, THEREFORE, in consideration of the foregoing and the terms, conditions and covenants included in this Contract, the receipt and sufficiency of which are hereby acknowledged, the Parties, intending to be legally bound, agree as follows:

1. The "Work." The construction work and services described in the Solicitation and herein constitute the "Work." Contractor shall perform the Work pursuant to the terms and conditions of the Contract Documents. Contractor agrees to render all services and activities that are necessary to complete the Project except the Parties agree that the "Work" does not include: (a) any architectural, engineering, or other design services that have been, or are to be rendered by, the City or a separate City consultant; or (b) any other services or activities which are expressly identified in these Contract Documents as excluded from Contractor's services.

2. Payment. Subject to the terms and conditions of this Contract, including any additions or deductions to the Contract Price that occur by Change Order, the City shall pay Contractor, for the performance of the Work, the Contract Price of **TYPE WRITTEN DOLLAR AMOUNT** DOLLARS (**$NUMBER DOLLAR AMOUNT**) (the "Contract Price"), as set forth in **Exhibit B**. This is a fixed-fee lump sum, subject to change and payment only as provided in these Contract Documents.

A. Federal-Aid Provisions. Contractor agrees to adhere to those general federal aid funding provisions designed to comply with law that are contained in **Exhibit E**, Federal Aid Provisions, as well as those unique obligations and requirements that are applicable by virtue of the City's use of funds under the following federal program: **federal program name**. Contractor agrees these unique funding obligations and requirements include, but are not limited to, the following: **Insert any specific requirements pertaining to the use of funds, such as final results that must be achieved, statute or regulations that govern the use of the funds, and/or the unique legal or regulatory requirements of the use of funds**

3. Overhead and Profit Markup. Contractor will serve as general contractor for the Work. Unless modified by **Exhibit D**, Special Conditions, all Work shall be performed at the rates and mark-up set forth in Contractor's bid. For any Work in which additional payment is warranted or required, such as Extra Work, the Parties agree such Work shall be subject to the following terms and conditions:

A. Self-Performed Work. For Work performed directly by Contractor with its own forces, Contractor shall be entitled to a markup of **[#]%** of the Direct Costs of such Work, which shall constitute full compensation for overhead and profit.

B. Subcontracted Work. For Work performed by Subcontractors under contract with Contractor, Contractor shall be entitled to a markup of **[#]%** of the Subcontractor's Direct Costs of such Work, which shall constitute full compensation for overhead and profit. Subcontractors shall be entitled to a markup that is not to exceed **[#]%** of Subcontractor's Direct Costs of such Work, which shall constitute its full compensation for overhead and profit.

C. Additional Markup Terms. The Parties further agree as follows: **Reserved.**

The Contract Price in Section 2 above, already includes all markup, overhead, profit, and administrative expenses for the Work and the Project.

The markup percentages above constitute the total allowable markup for all markups, overhead, profits, and administrative expenses, and shall apply for all of the Work. See General Conditions (**Exhibit C**), **Article VII**, *Changes*, for additional terms and conditions governing changes to the Work or Project.

There shall be no different fees or charges to the City for markups, overhead, profits, or administrative expenses unless both (i) the Parties agree to such via a Change Order signed by each Parties' project manager, and (ii) the resulting markup percentage is lower than those set forth above.

4. Substantial Completion Deadline. Contractor agrees to achieve Substantial Completion on or by the Substantial Completion Deadline. With respect to the Substantial Completion Deadline:

☐ Set Now. The Substantial Completion Deadline is hereby set as **[Date] or "Reserved"**.

☐ Set by Future Change Order. The Substantial Completion Deadline shall be set by a Change Order within sixty (60) days of the Effective Date. If no such Change Order is issued, then the date of Substantial Completion Deadline shall be one hundred and eighty (180) days from the Effective Date.

☐ Set by Notice to Proceed. The Substantial Completion Deadline shall be set by a set number of days following the City's issuance of a written Notice to Proceed. The deadline date shall be **[#]** days after the date of issuance of the Notice to Proceed. Contractor and the City agree that the City shall not issue the Notice to Proceed unless and until the following occurs: **Reserved.**

The date or deadline above shall constitute the "Substantial Completion Deadline" for all purposes under this Contract unless and except as such deadline may be extended by a Change Order or amendment.

The terms and conditions governing Substantial Completion of the Work and Project and the Substantial Completion Deadline are further detailed in the General Conditions (**Exhibit C**), including, without limitation, Section 608 of that exhibit.

5. Final Completion Date. Unless revised by a Change Order or Amendment, Contractor agrees to achieve Final Completion no later than **[# of days]** after the Substantial Completion Date.

6. Liquidated Damages. Contractor shall be liable for liquidated damages for each day that the Substantial Completion of the Work is delayed past the established time limit or limits, accounting for any extension of time agreed to by the City. Liquidated damages per day of delay are hereby set at:

☐ Standard. The City's default rate of **ONE TENTH OF A PERCENT (0.1%)** of the Contract Price, which each Party deems reasonable to compensate the City for such delay.

☐ Negotiated. The negotiated rate of **$Reserved** per day of delay.

☐ Set by Law or Funding Source. A rate set by law or funding source of **$Reserved** per day of delay.

The terms and conditions governing liquidated damages are further detailed in the General Conditions (**Exhibit C**), including, without limitation, Section 606 of that exhibit.

7. The Contract Documents. The Contract Documents consist of: (a) this Contract (main-body), including the cover page through signature page; the Solicitation (**Exhibit A**), Contractor's Response to the Solicitation (**Exhibit B**), the General Conditions (**Exhibit C**), the Special Conditions (**Exhibit D**), the Federal Aid Provisions (**Exhibit E**); (b) Amendments; (c) *upon their submission by Contractor to the City or inclusion herein as part of contract execution, whichever occurs first*, performance bond and labor and material payment bond; (d)

upon their completion and approval by the City, any Drawings and Specifications, and Forms 1 through 9 and Form 13 of **Exhibit C** (e.g., Change Orders, Field Orders, Guarantee Period Inspection Report); and (e) *upon their completion, signature by all affected parties*, and submission to the City, Forms 10 through 12 of **Exhibit C** (e.g., release and waivers; Affidavit that All Claims are Satisfied; Surety Consent to Final Payment).

8. Order of Precedence.

A. In the event of any conflict arising from errors or inconsistencies within these Contract Documents, the following order of precedence shall apply:

1. Amendments
2. Federal Aid Provisions (**Exhibit E**)
3. Change Orders, Field Orders
4. This Contract (main body)
5. Special Conditions (**Exhibit D**)
6. General Conditions (**Exhibit C**)
7. Performance Bond and Labor and Material Bond documents
8. Permitting or Licensing Conditions as interpreted by the Permitting or Licensing Authority
9. Drawings (*See* General Conditions, Section 301)
10. Technical Specifications
11. Referenced Standard Specifications
12. The Solicitation, including any Addenda thereto (**Exhibit A**)
13. Contractor's Response to the Solicitation (**Exhibit B**)

B. With reference to the Drawings, the order of precedence, as between them, shall be as follows:

1. Figures govern over scaled dimensions.
2. Detail Drawings govern over general Drawings.
3. Addenda/Change Order drawings govern over Contract Drawings.
4. Contract Drawings govern over standard drawings.

9. Notice.

A. Formal Notice – Requirements. Any "Formal Notice" (as described below) shall be in writing to the other Party shall be deemed given (1) if by hand delivery, upon receipt thereof; (2) three (3) days after deposit in the United States mails, postage prepaid, certified mail, return receipt requested; or (3) one (1) day after deposit with a nationally recognized overnight courier, specifying overnight priority delivery. Either Party may change its address for purposes of these Contract Documents at any time by giving Notice of such change to the other Party.

If to Contractor:
Individual Name
Individual Title
Street Address
City/Town, State ZIP
If to the City:
Individual Name
Individual Title
Department
Street Address
Boulder, Colorado 80306

With a copy to:

City Attorney's Office
1777 Broadway, Second Floor
P.O. Box 791
Boulder, Colorado 80306

B. Distinguishing "Formal Notice" from Other Notice. This "Notice" applies only to instances in these Contract Documents where a Party is required to give "Formal Notice." References in these Contract Documents, including, for avoidance of doubt, **Exhibit C**, General Conditions, to other forms of notice, even if required to be in writing, need not comply with this section. The purpose of distinguishing "Formal Notice" from any other form of written notice – such as "notice," "written notice," "notice in writing," or similar phrasing – is to ensure: (1) the Contract Documents clearly identify those matters of significance that require elevation according to this Formal Notice process; and, (2) to permit other notices, even if "in writing," to be given without compliance with this section.

10. Appropriations. The City uses a purchase order process to complete requisitions for payment. Work shall only proceed if a purchase order so authorizes the Work. [ALTERNATIVE #1: To be used for construction projects that are scheduled to be funded wholly within one fiscal year.]With respect to the full Contract Price, the City has appropriated funds to pay Contractor for its Work under this Contract that are equal to or greater than the Contract Price stated above. [ALTERNATIVE #2: To be used for phased or annual renewable programs construction projects occurring over more than one fiscal year where funds sufficient to pay the full Contract Price are expected to be appropriated by the City for fiscal years subsequent to the one in which the Work commenced.] With respect to the full Contract Price, the amount of funds appropriated to pay Contractor for its Work under this Contract is **$Appropriated Dollar Value**. Before Contractor commences any Work for which the City has agreed to pay a sum in excess of the appropriated amount stated above, Contractor shall request written assurance that the funds in excess of the amount in the preceding sentence have been lawfully appropriated, and the City will provide such written assurance if appropriation occurred. Bonds. [ADDITION FOR BONDS: [Include only if the construction projects is funded by bonded indebtedness; if bonds *are not* issued for this project, replace this with "RESERVED"; if bonds *are* issued for this project, consult with CAO about additional, custom terms (if any) that tie the project to any bond conditions/restrictions] The funds to be used to pay Contractor for its Work under this Contract have been raised by issuing bonds. Appropriations by the City to pay the debt on those bonds may occur subsequent to the date of this Contract. Upon request by Contractor, the City shall provide Contractor with a written assurance that each such appropriation has been lawfully made shortly after it occurs.

11. No Multi-Fiscal Year Obligation. The Parties understand and acknowledge that the City is subject to TABOR requirements. The Parties do not intend to violate the terms and requirements of TABOR by the execution of this Contract. It is understood and agreed that this Contract does not create a multiple fiscal year direct or indirect debt or obligation within the meaning of TABOR and notwithstanding anything in this Contract to the contrary, all payment obligations of the City are expressly dependent and conditioned upon the continuing availability of funds beyond the term of the City's current fiscal period ending upon the next succeeding December 31. Financial obligations of the City payable after the current fiscal year are contingent upon funds for that purpose being appropriated, budgeted, and otherwise made available in accordance with the rules, regulations, and resolutions of the City and applicable law. Any failure of a city council annually to appropriate adequate monies to finance the City's obligations under this Contract shall terminate this Contract at such time as such then-existing appropriations are to be depleted. Notice shall be given promptly to Contractor of any failure to appropriate such adequate monies.

12. Compliance with Laws. At all times during the performance of this Contract, Contractor shall strictly adhere to all Laws in effect during its time of performance under this Contract. This shall include, by way of example and not by way of limitation, the United States Department of Health, Occupational Safety and Health Administration (OSHA) standards.

13. City Manager and City Attorney. The city manager of the City of Boulder, Colorado, alone has the authority to solicit bids for a contract and must sign the Contract for it to be effective. This Contract must be approved as to form by the City Attorney's Office.

14. Effective Date. The "Effective Date" is the date appearing on the cover page of the main body of this Contract following signature of all Parties. The city manager is authorized, at the time of their signing, to affix such date to the cover page.

15. Authorization. Contractor warrants that the individual executing this Contract is properly authorized to bind Contractor to this Contract.

16. Binding Effect. This Contract shall be binding upon and inure to the benefit of the heirs, successors and assigns of the Parties.

17. Counterparts and Signatures. This Contract may be signed in multiple counterparts. Each signed counterpart will be treated as an original, and all counterparts together will form one single agreement. Signed copies may be sent by email or other electronic means. Electronic signatures – including images of handwritten signatures or other electronic representations showing a Party's intent to be bound – will have the same legal effect as an original handwritten ("wet") signature. This Contract becomes effective on the date the last Party signs it. The city manager, as the last Party to sign, is authorized to enter the Effective Date on the cover page.

[Signature page follows]

The Parties to this Contract have caused it to be executed by their authorized officers, effective as of the day and year first above written.

CONTRACTOR
[Contractor's Legal Name]

By: _____ Date: _____
Printed Name: _____
Title: _____

CITY OF BOULDER,
a Colorado home rule municipality

_____ Date: _____
Nuria Rivera-Vandermyde,
City Manager

_____ Date: _____
Purchasing Agent
Printed Name: _____

ATTEST:

_____
City Clerk

APPROVED AS TO FORM:

_____ Date: _____
City Attorney's Office